

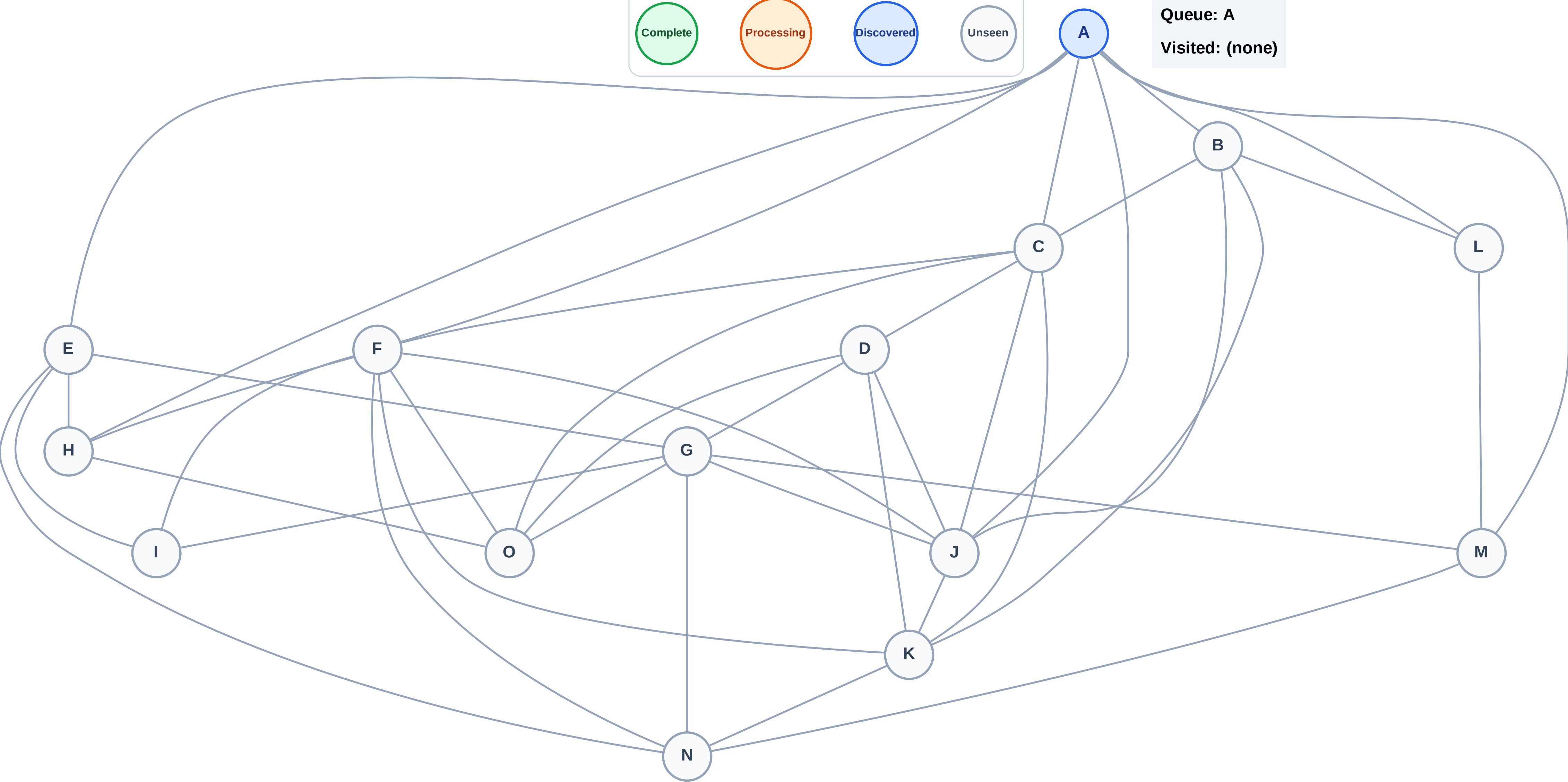
BFS Traversal

Step 1: Start at A

Legend



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

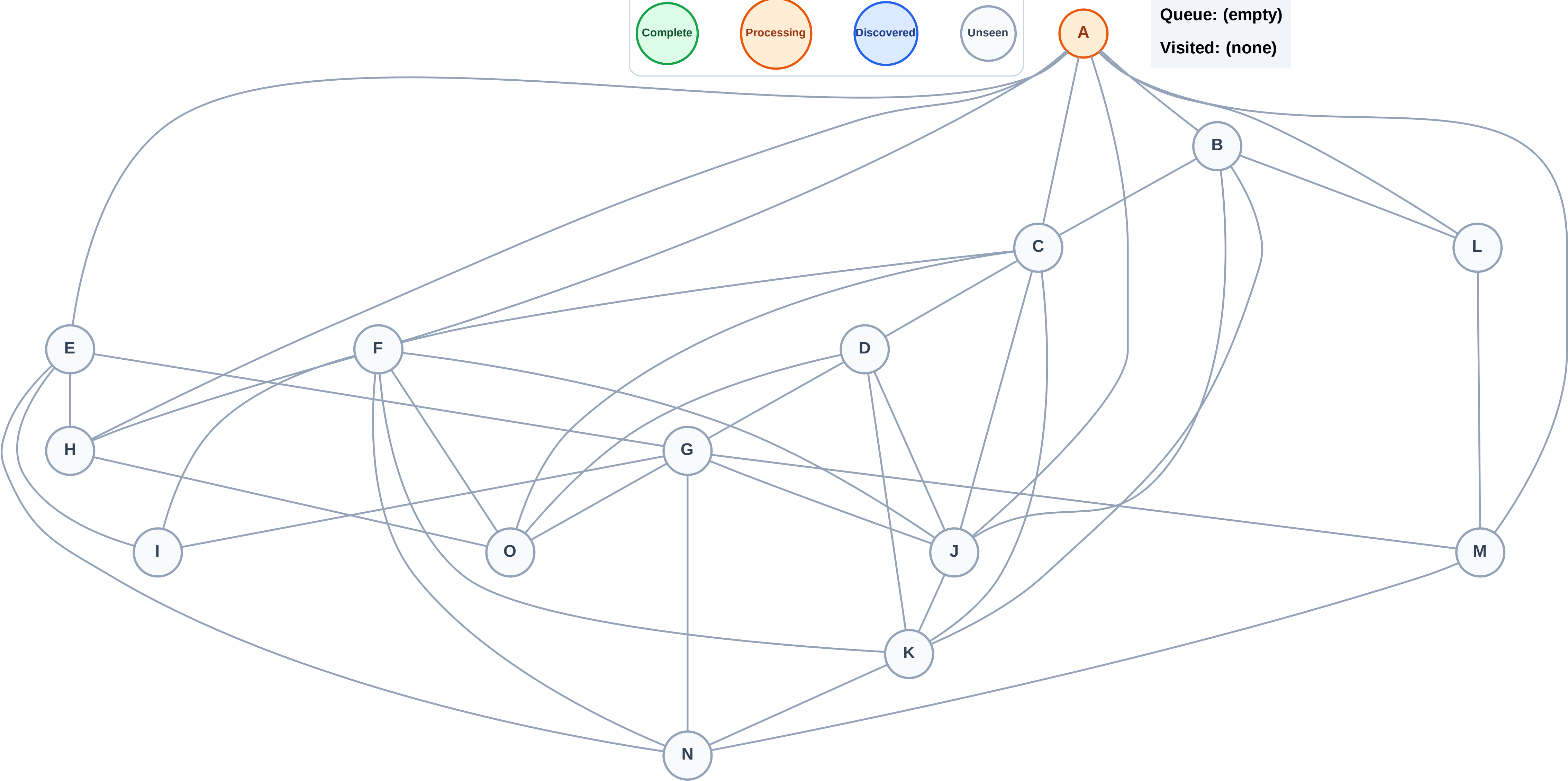
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, F, H, J, L, M from A

Legend



Complete



Processing

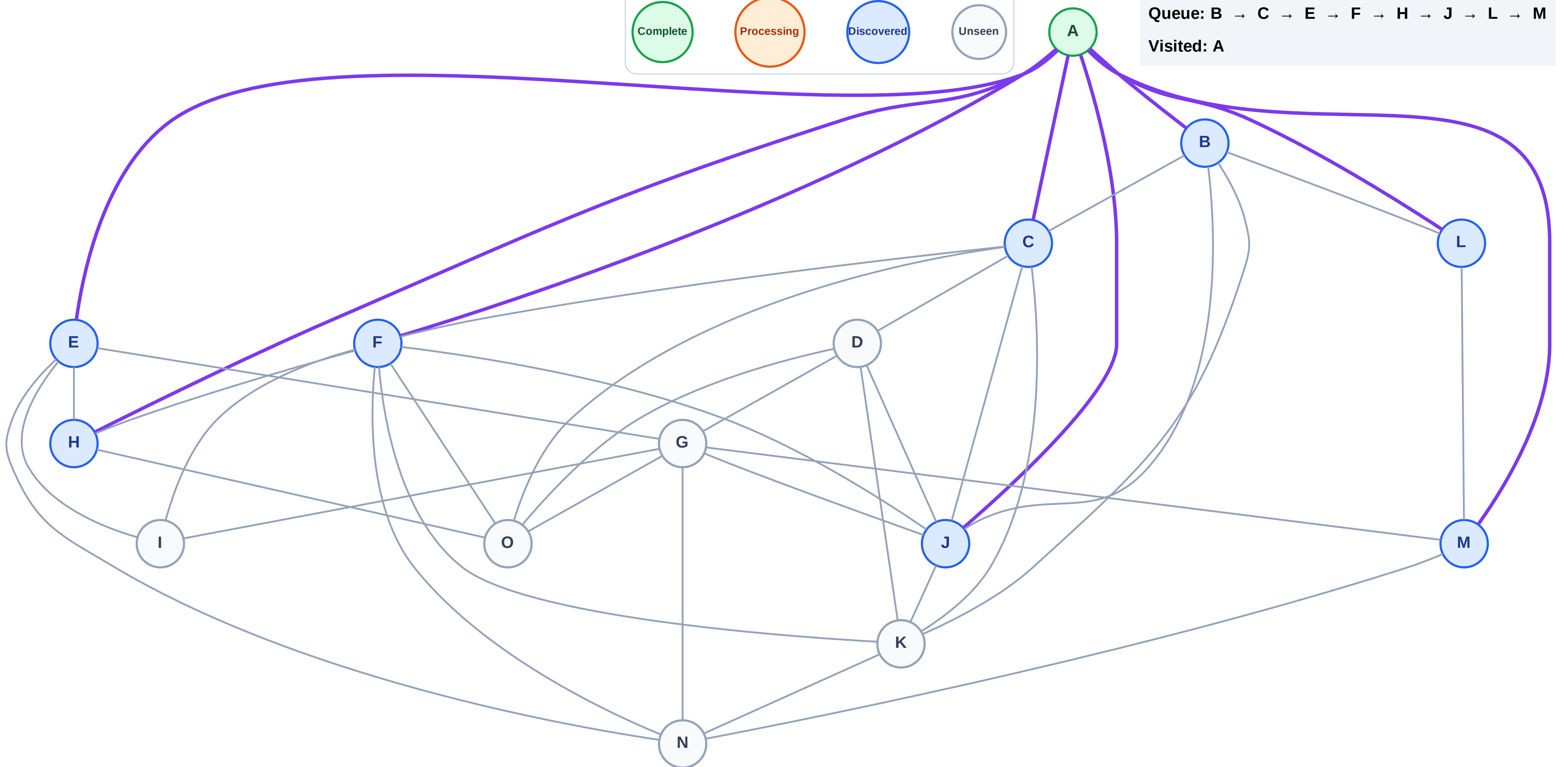


Discovered



Unseen

Queue: B → C → E → F → H → J → L → M
Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

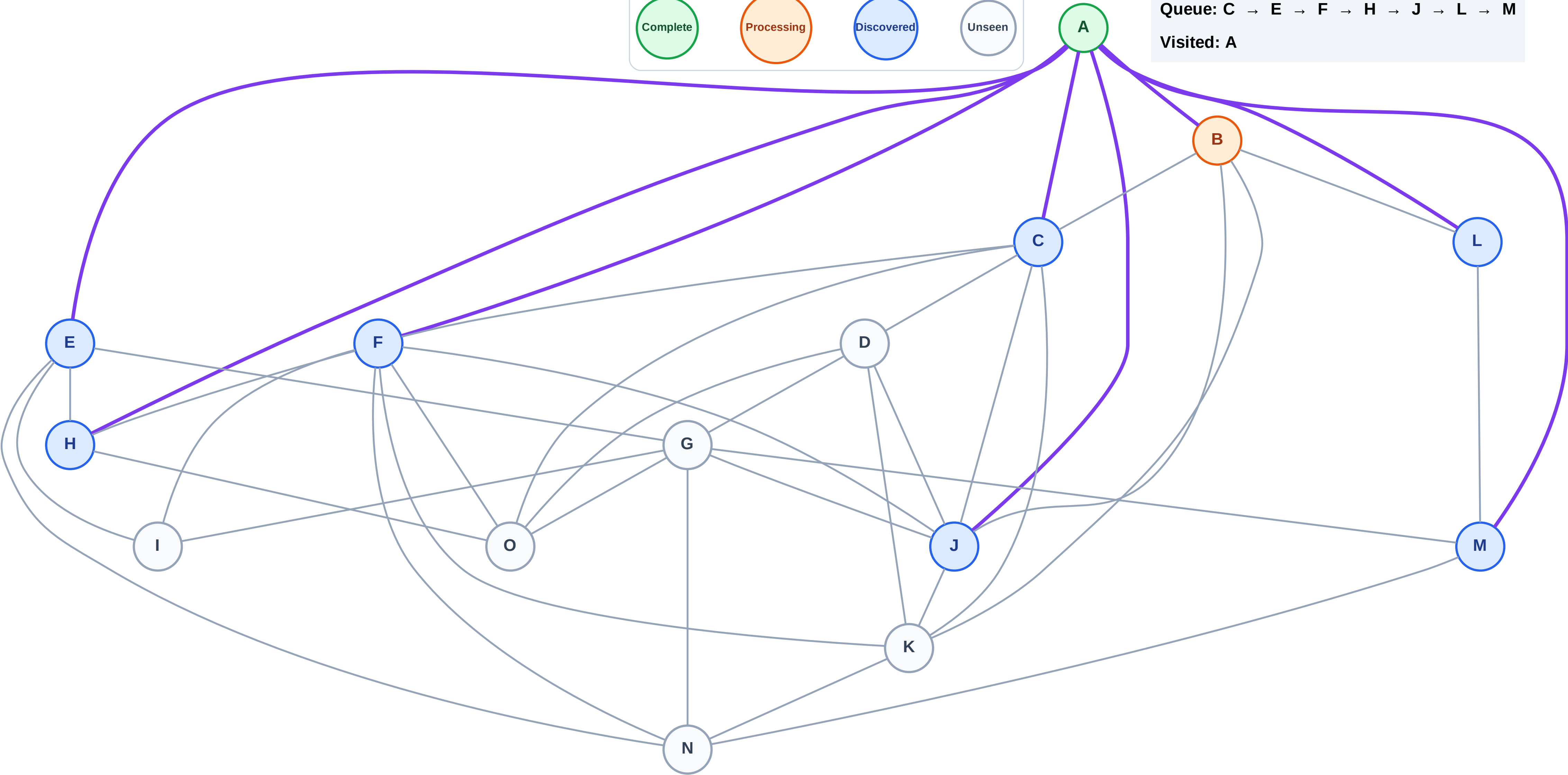
Processing

Discovered

Unseen

Queue: C → E → F → H → J → L → M

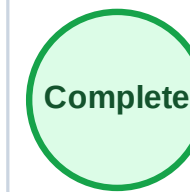
Visited: A



BFS Traversal

Step 5: Discover K from B

Legend



Complete



Processing

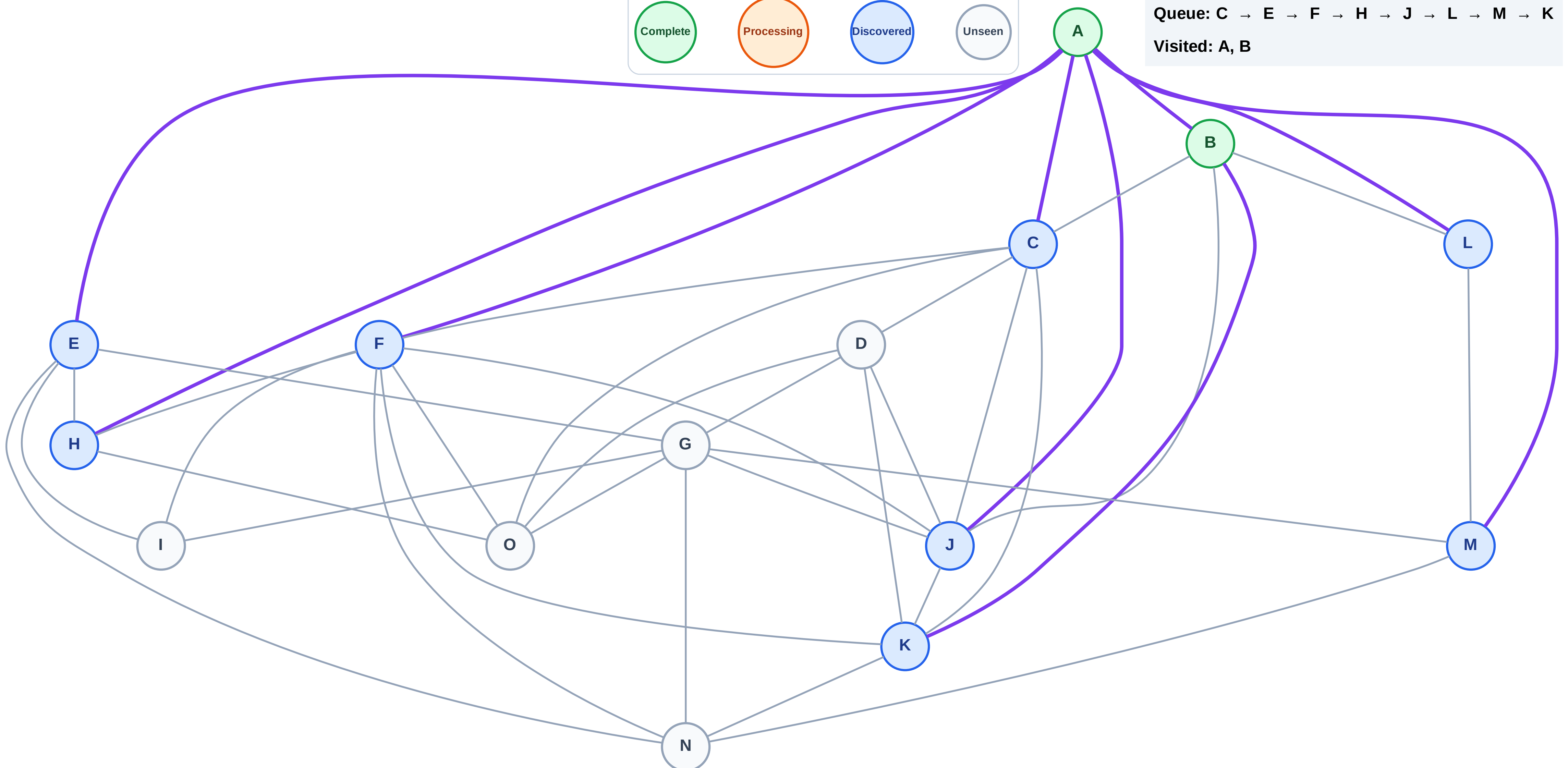


Discovered



Unseen

Queue: C → E → F → H → J → L → M → K
Visited: A, B



BFS Traversal
Step 6: Process node C

Legend

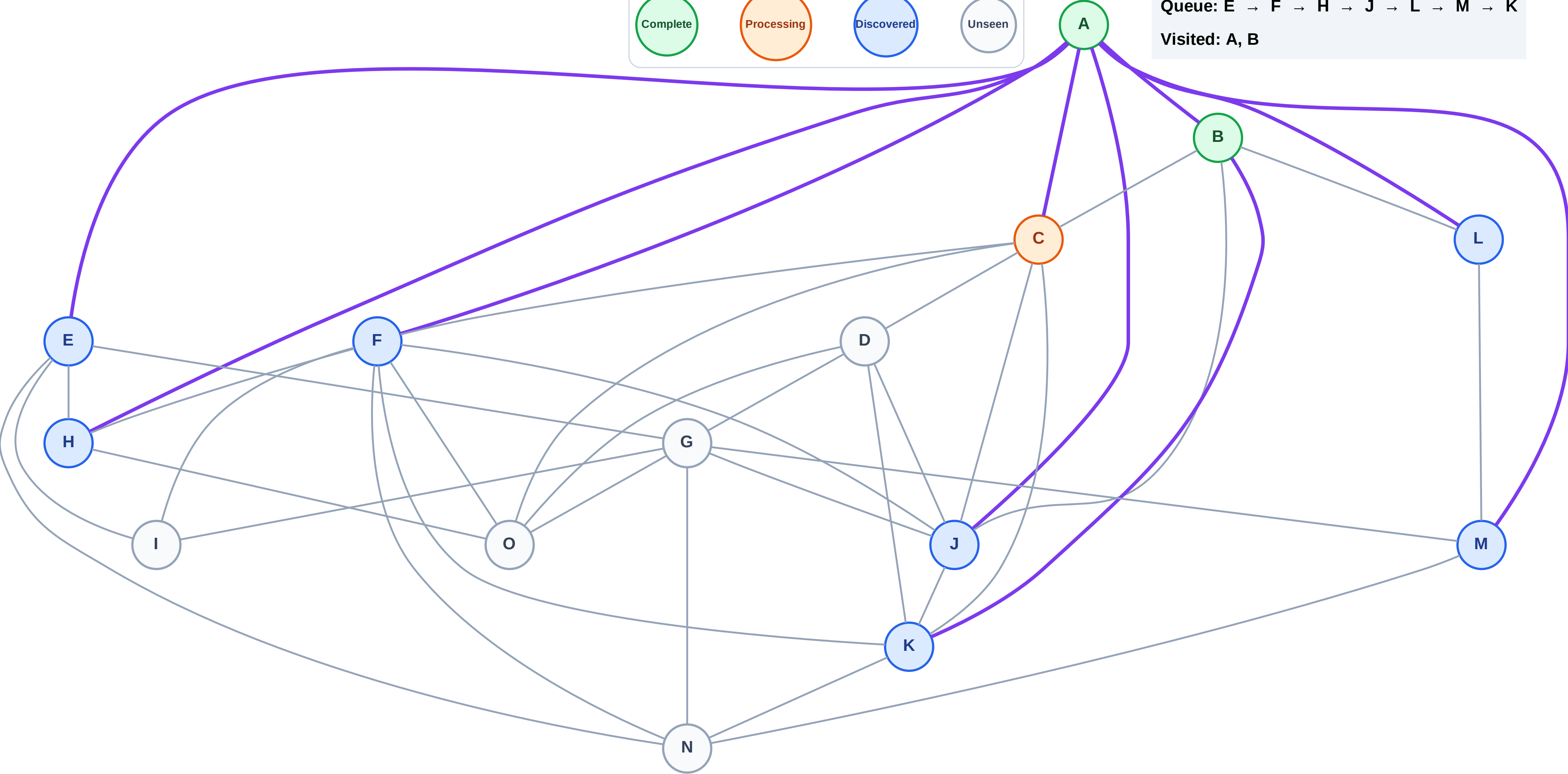
Complete

Processing

Discovered

Unseen

Queue: E → F → H → J → L → M → K
Visited: A, B



BFS Traversal

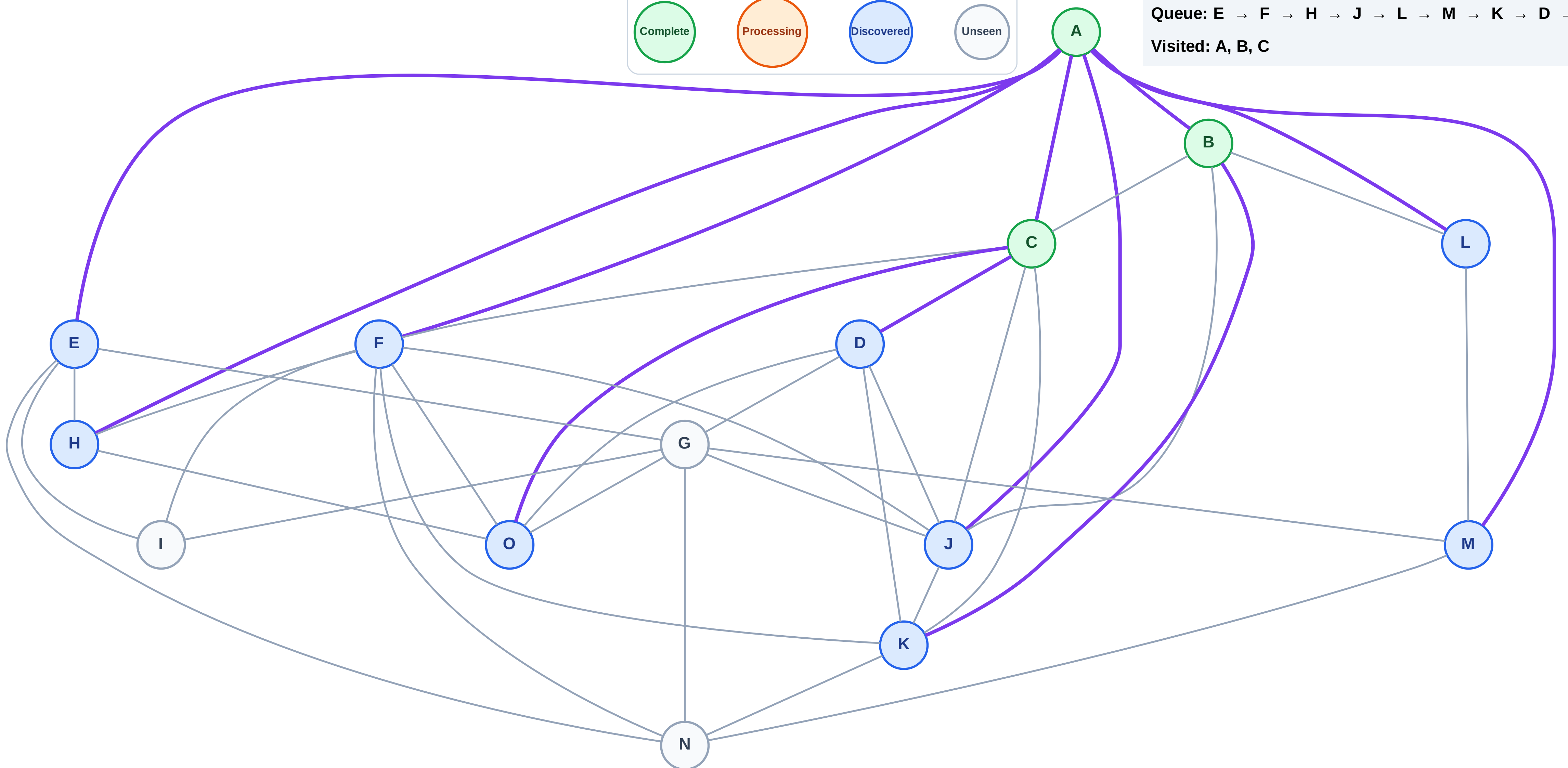
Step 7: Discover D, O from C

Legend



Queue: E → F → H → J → L → M → K → D → O

Visited: A, B, C



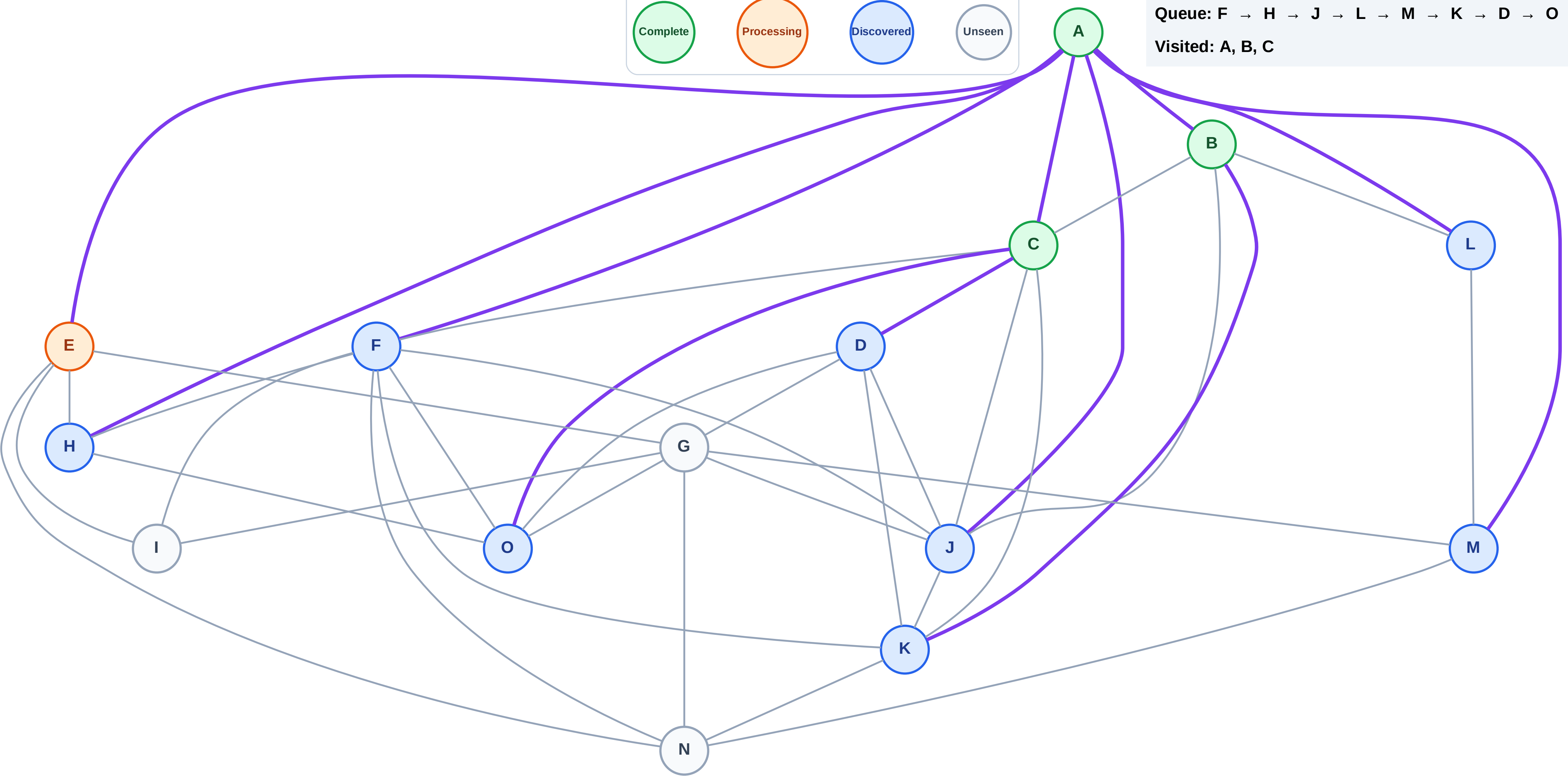
BFS Traversal

Step 8: Process node E

Legend



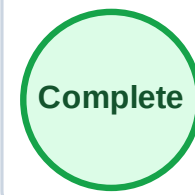
Queue: F → H → J → L → M → K → D → O
Visited: A, B, C



BFS Traversal

Step 9: Discover G, I, N from E

Legend



Complete



Processing



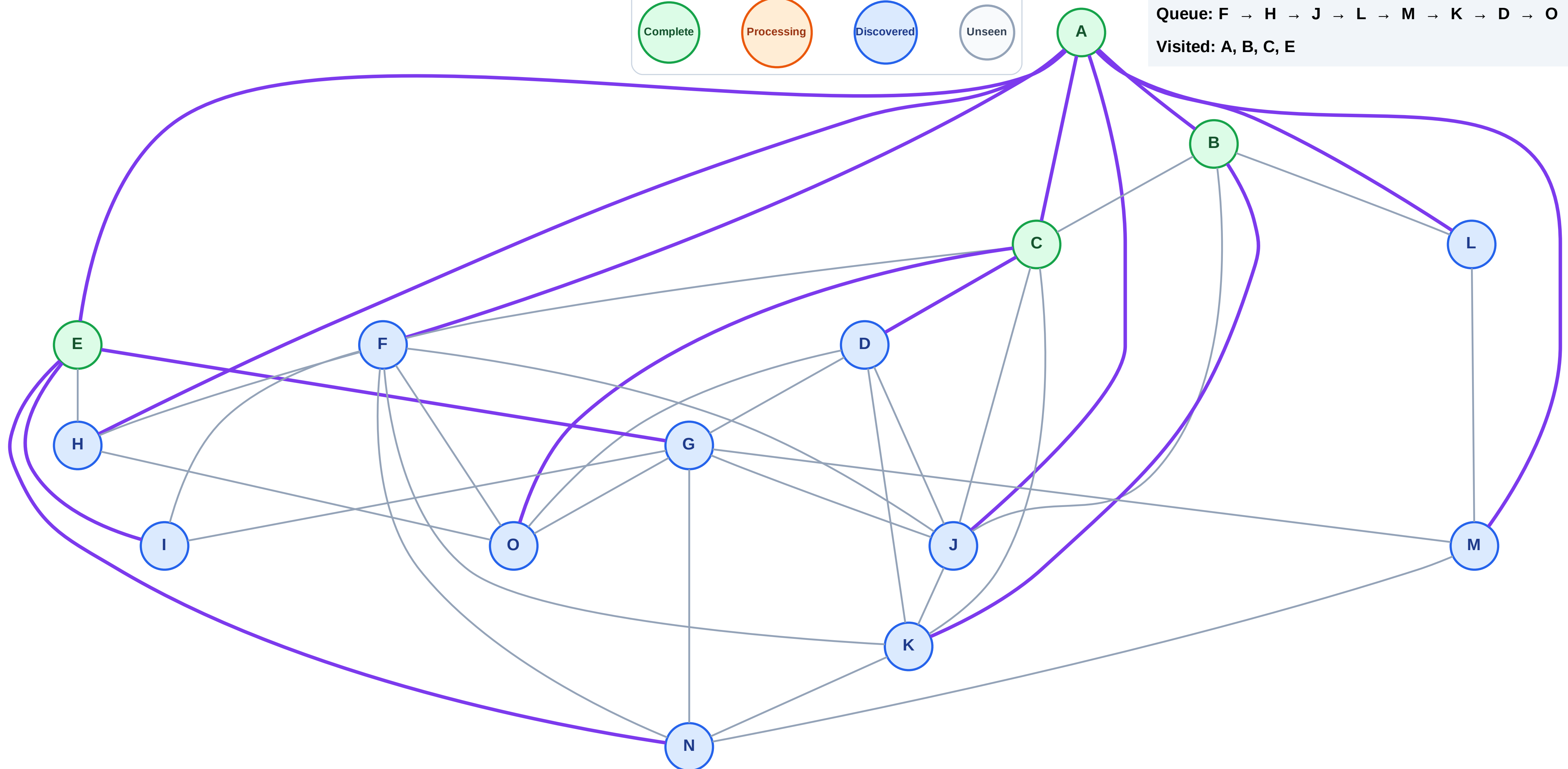
Discovered



Unseen

Queue: F → H → J → L → M → K → D → O → G → I → N

Visited: A, B, C, E



BFS Traversal

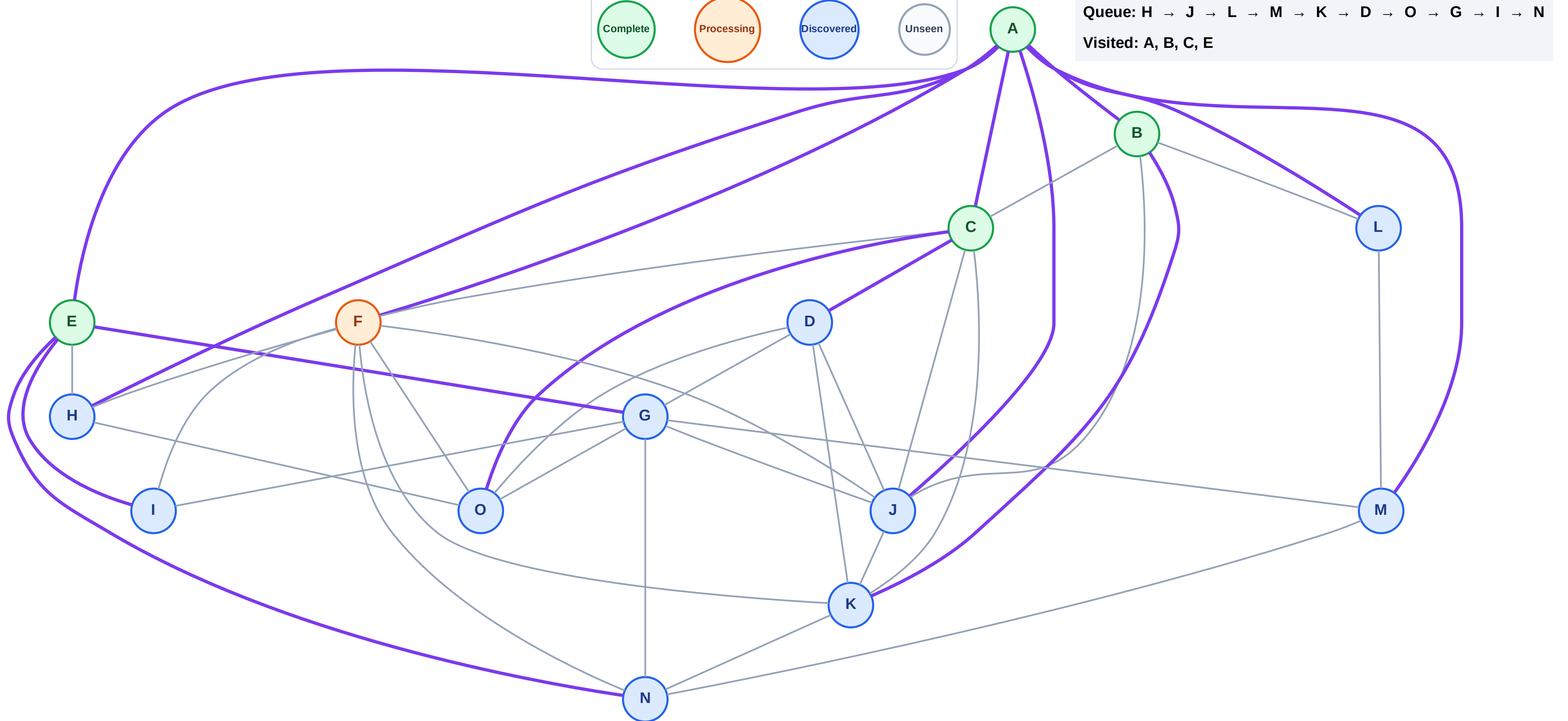
Step 10: Process node F

Legend



Queue: H → J → L → M → K → D → O → G → I → N

Visited: A, B, C, E



BFS Traversal

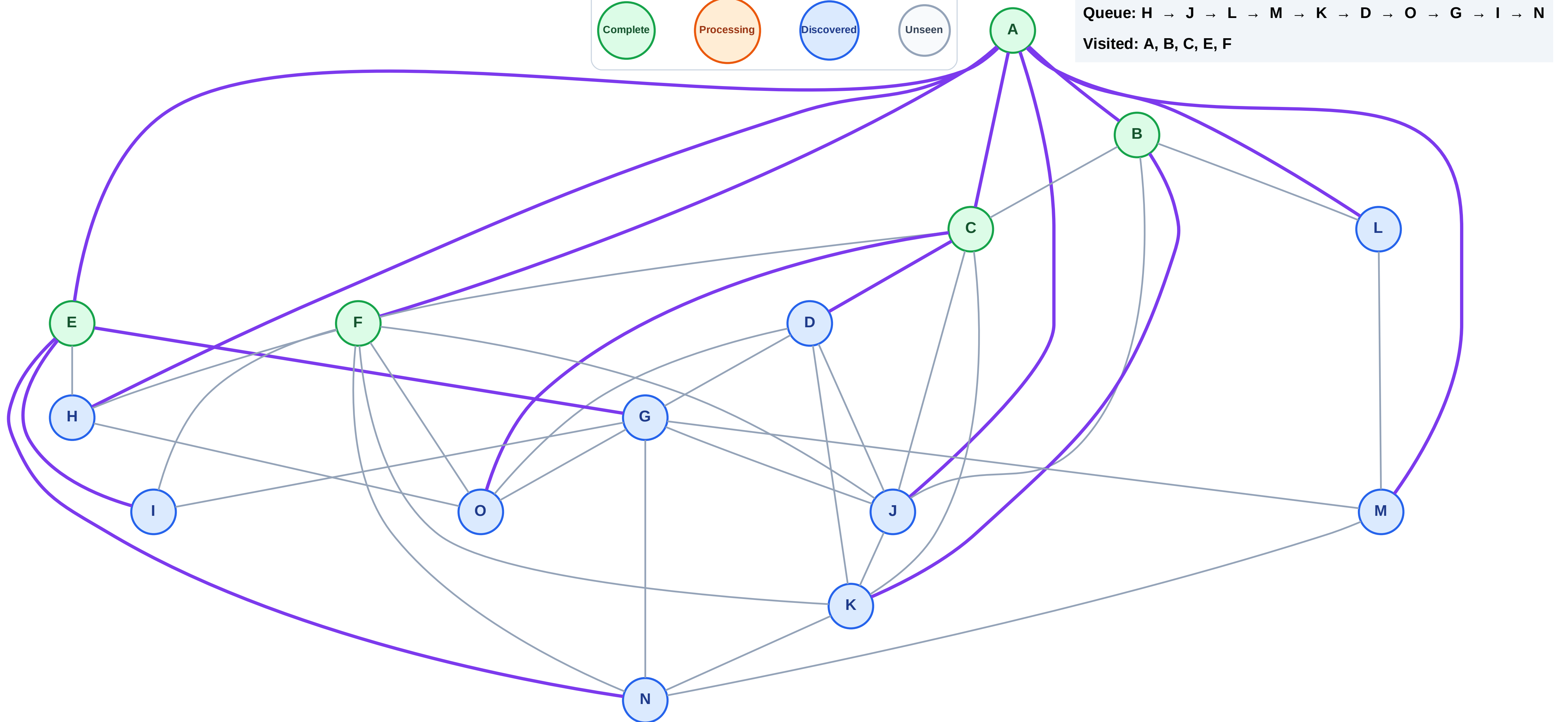
Step 11: No new neighbors from F

Legend



Queue: H → J → L → M → K → D → O → G → I → N

Visited: A, B, C, E, F



BFS Traversal

Step 12: Process node H

Legend



Complete



Processing

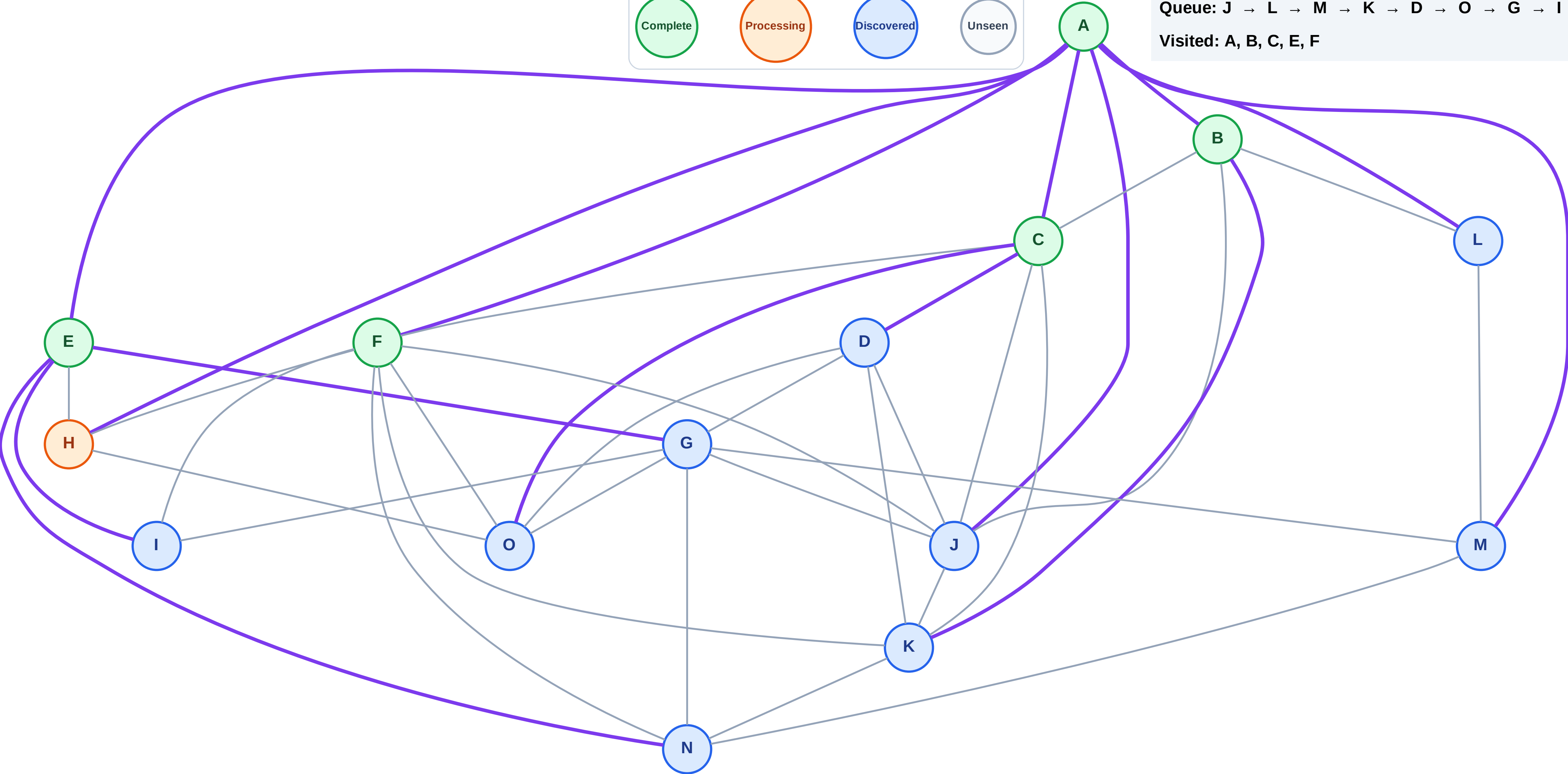


Discovered



Unseen

Queue: J → L → M → K → D → O → G → I → N
Visited: A, B, C, E, F



Step 13: No new neighbors from H

Step 13: No new neighbors from H



BFS Traversal

Step 14: Process node J

Legend



Complete



Processing



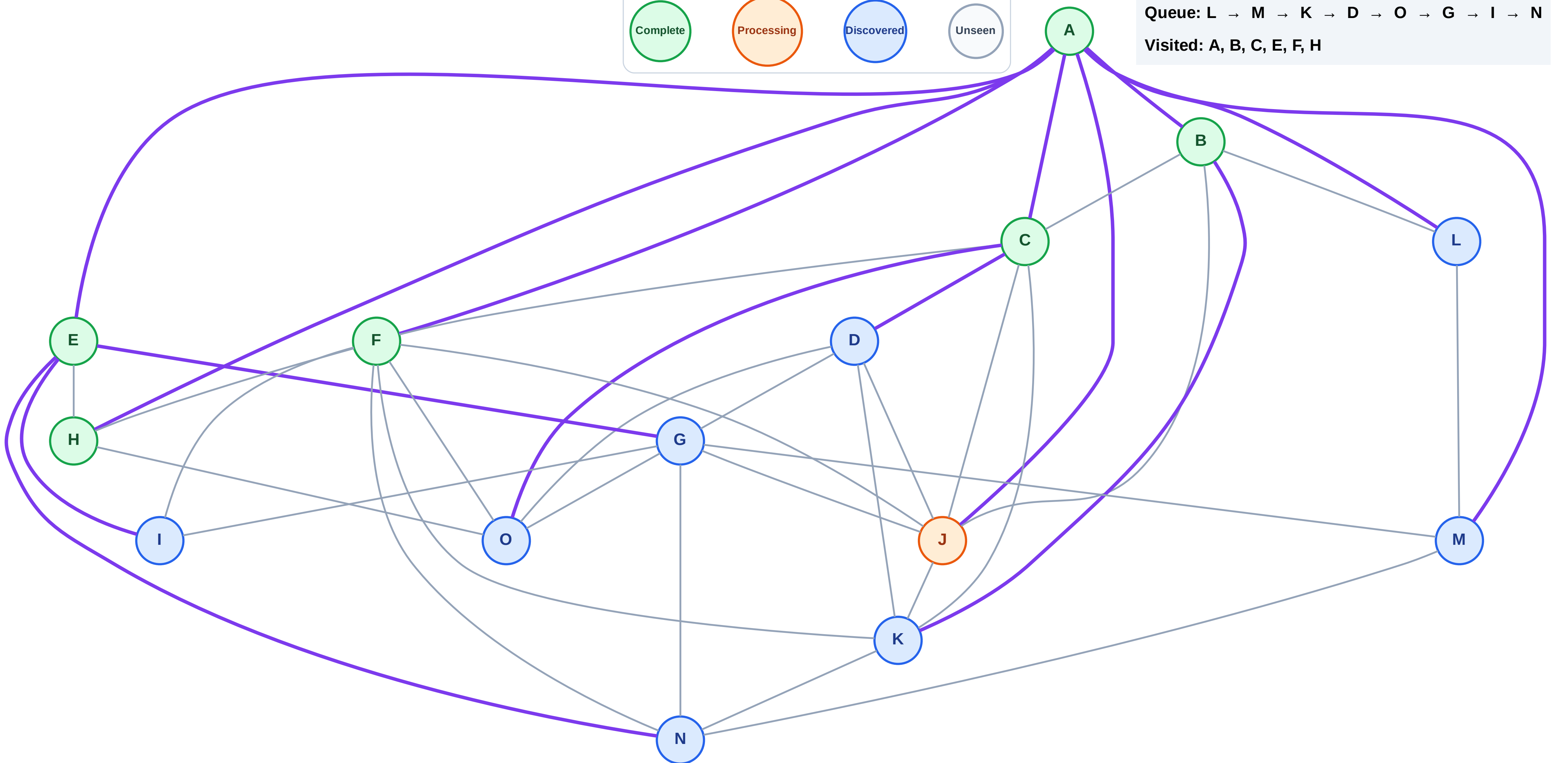
Discovered



Unseen

Queue: L → M → K → D → O → G → I → N

Visited: A, B, C, E, F, H



BFS Traversal

Step 15: No new neighbors from J

Legend



Complete



Processing

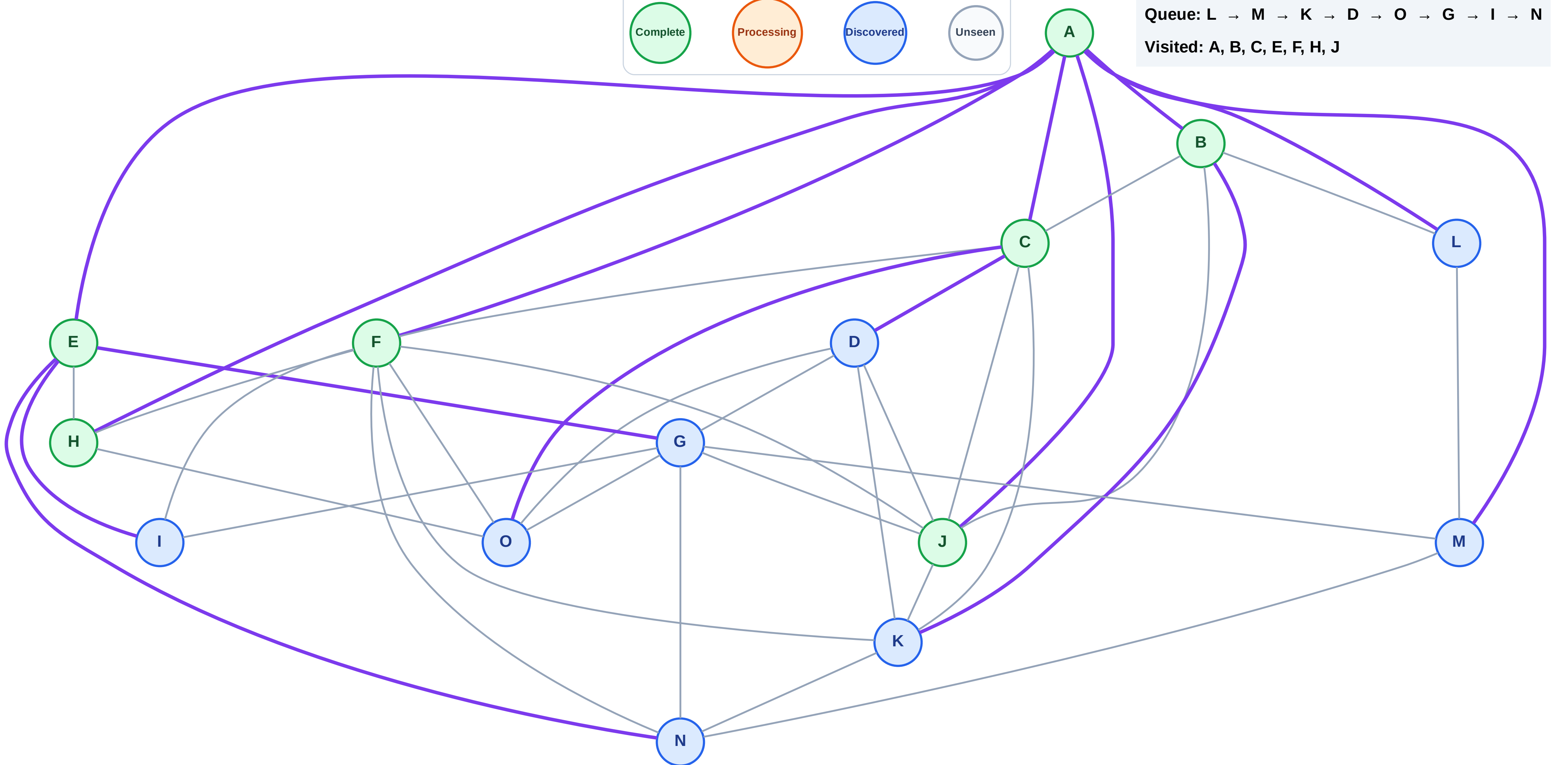


Discovered



Unseen

Queue: L → M → K → D → O → G → I → N
Visited: A, B, C, E, F, H, J



Step 16: Process node L

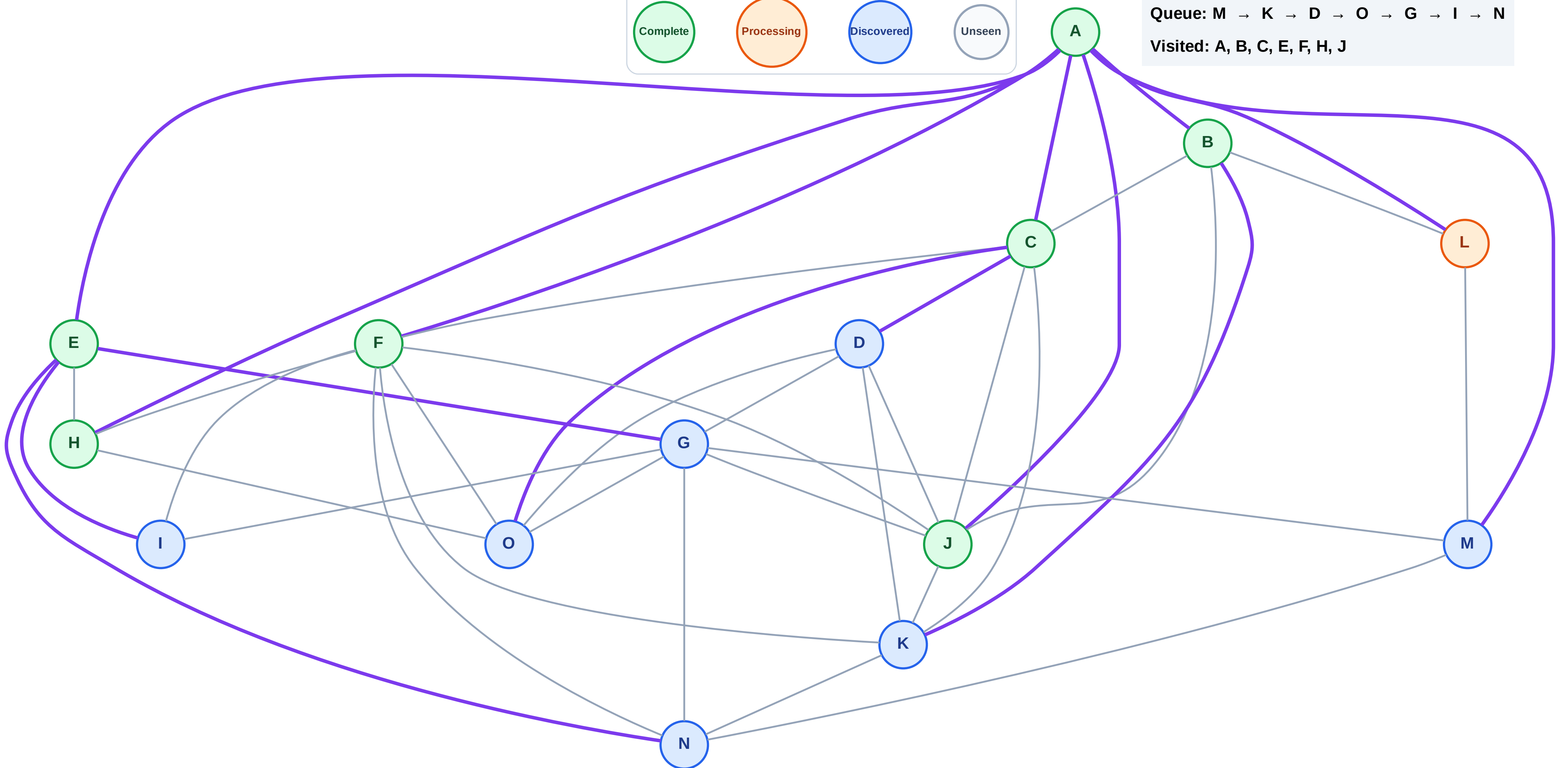
Step 16: Process node L

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

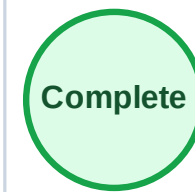
Visited: A, B, C, E, F, H, J



BFS Traversal

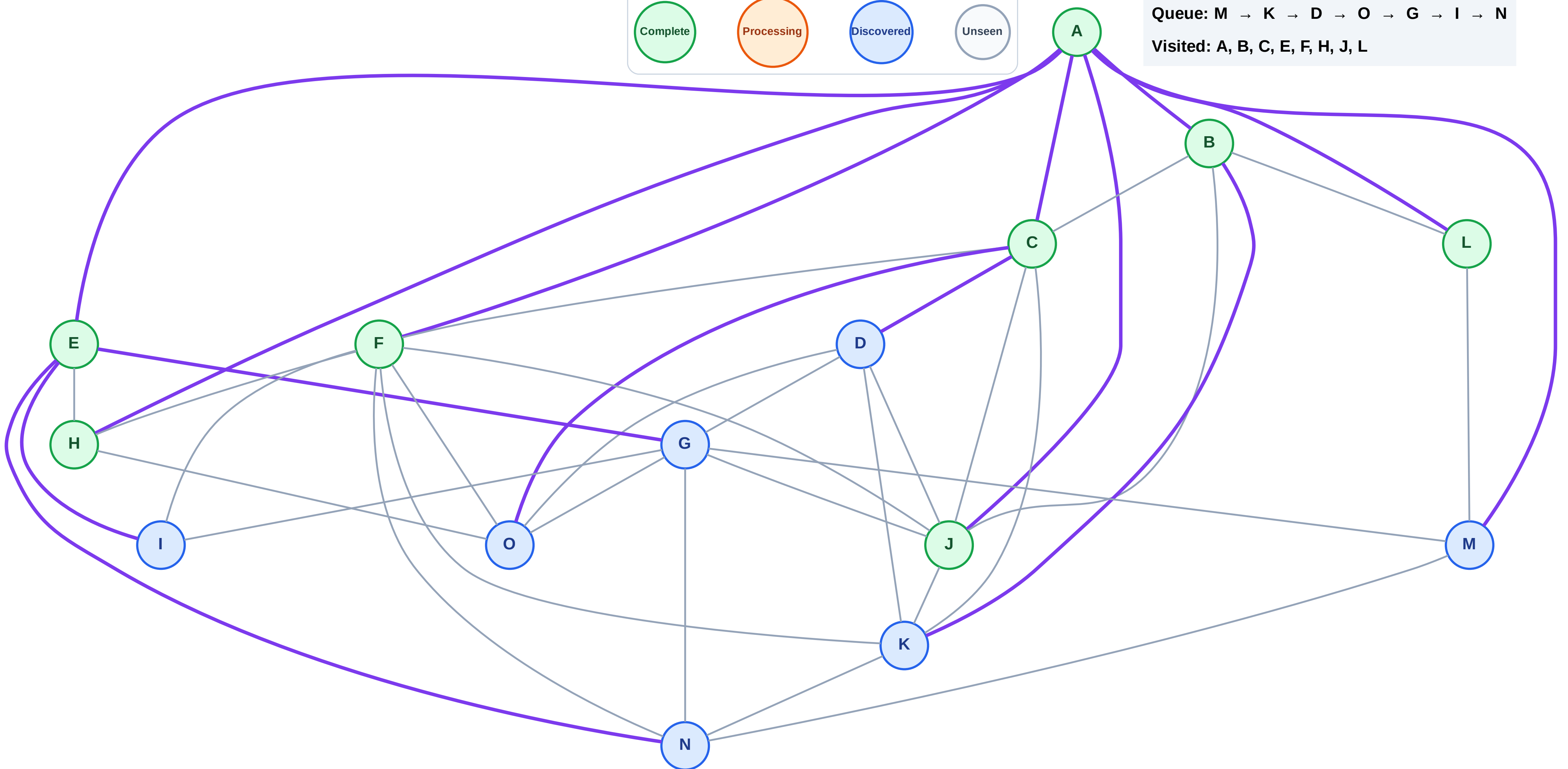
Step 17: No new neighbors from L

Legend



Queue: M → K → D → O → G → I → N

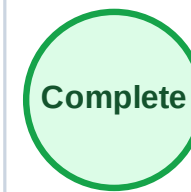
Visited: A, B, C, E, F, H, J, L



BFS Traversal

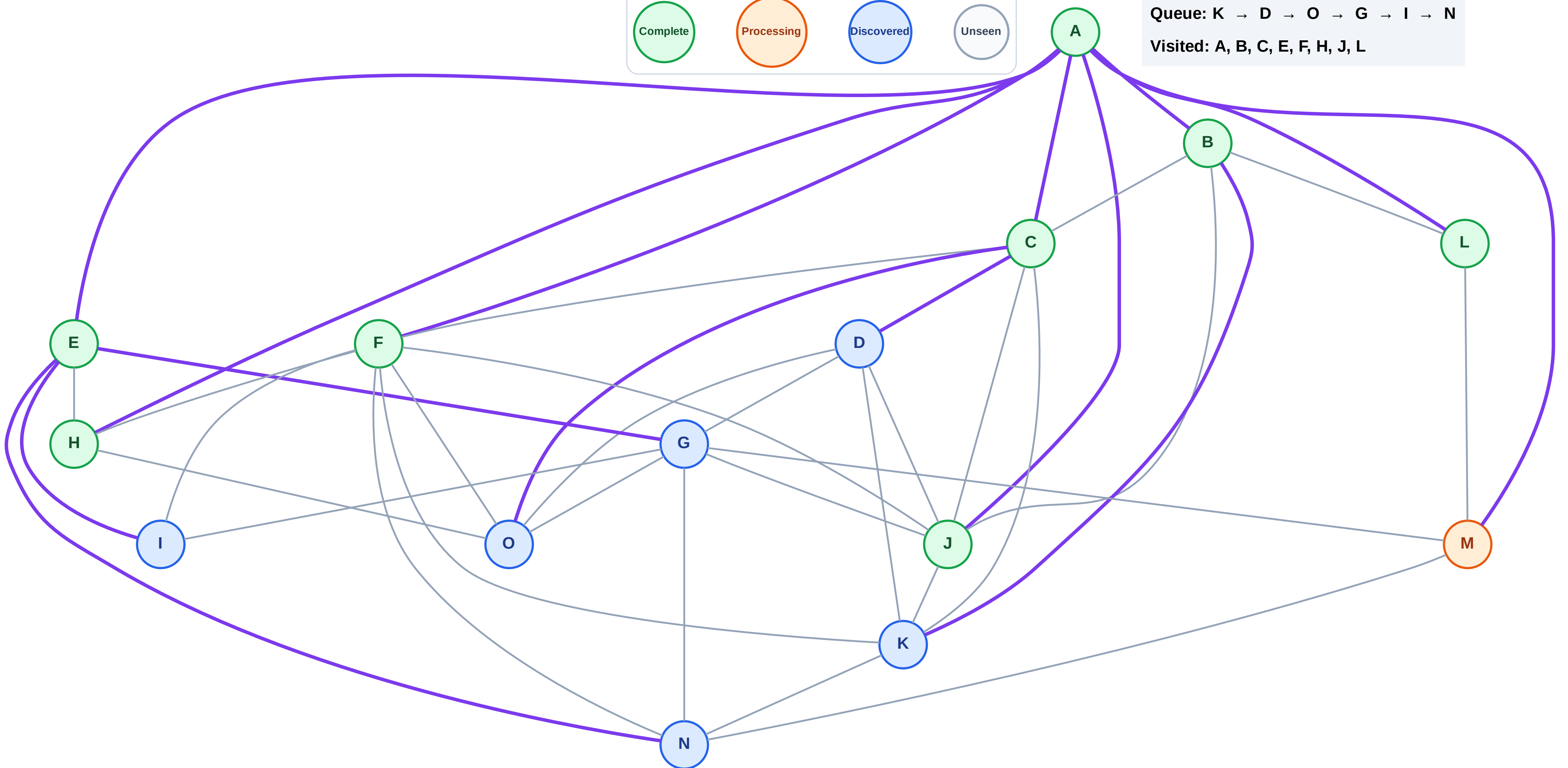
Step 18: Process node M

Legend



Queue: K → D → O → G → I → N

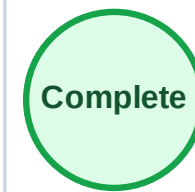
Visited: A, B, C, E, F, H, J, L



BFS Traversal

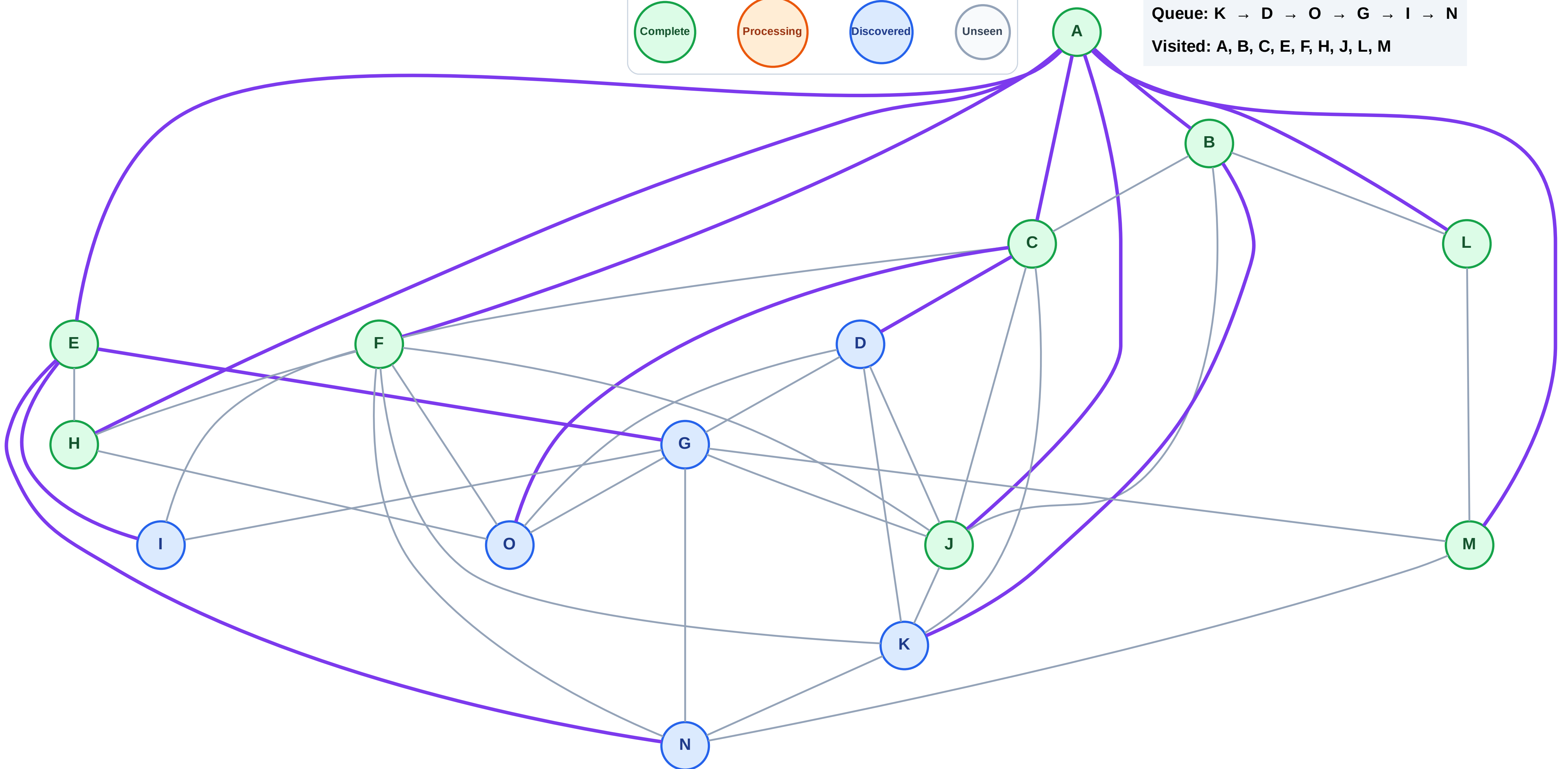
Step 19: No new neighbors from M

Legend



Queue: K → D → O → G → I → N

Visited: A, B, C, E, F, H, J, L, M



BFS Traversal

Step 20: Process node K

Legend

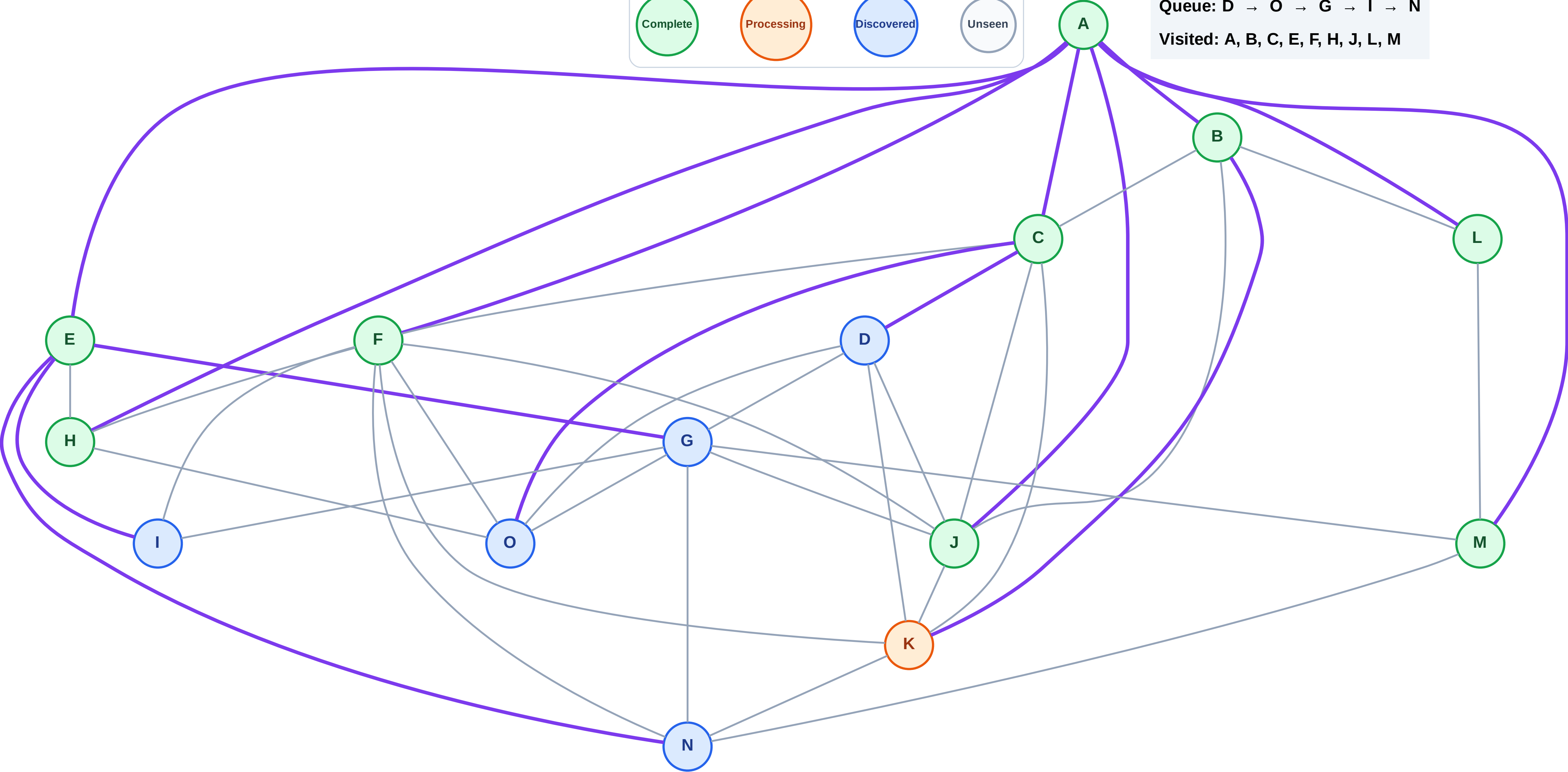
Complete

Processing

Discovered

Unseen

Queue: D → O → G → I → N
Visited: A, B, C, E, F, H, J, L, M



BFS Traversal

Step 21: No new neighbors from K

Legend



Complete



Processing

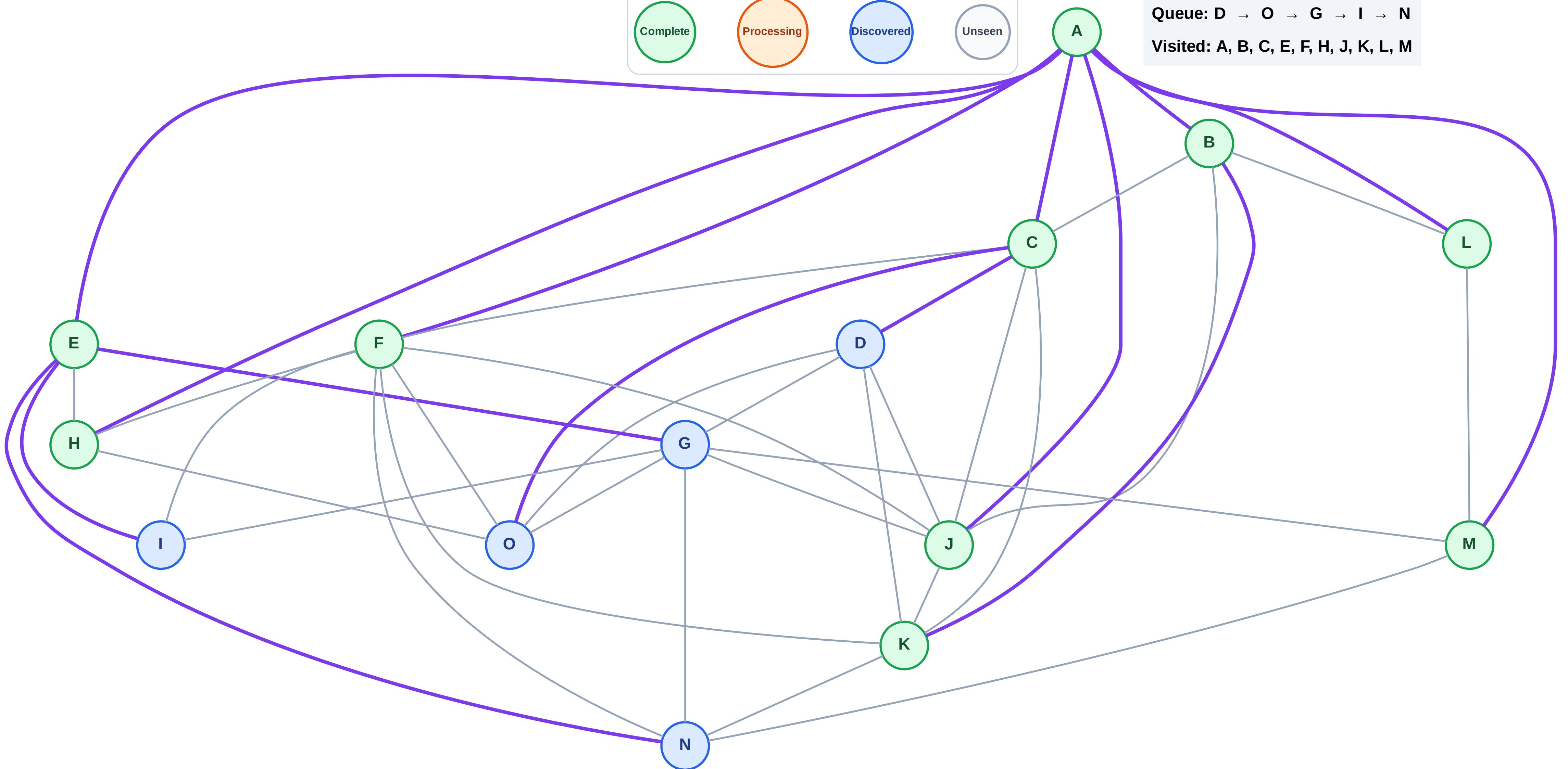


Discovered



Unseen

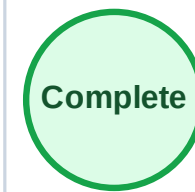
Queue: D → O → G → I → N
Visited: A, B, C, E, F, H, J, K, L, M



BFS Traversal

Step 22: Process node D

Legend



Complete



Processing

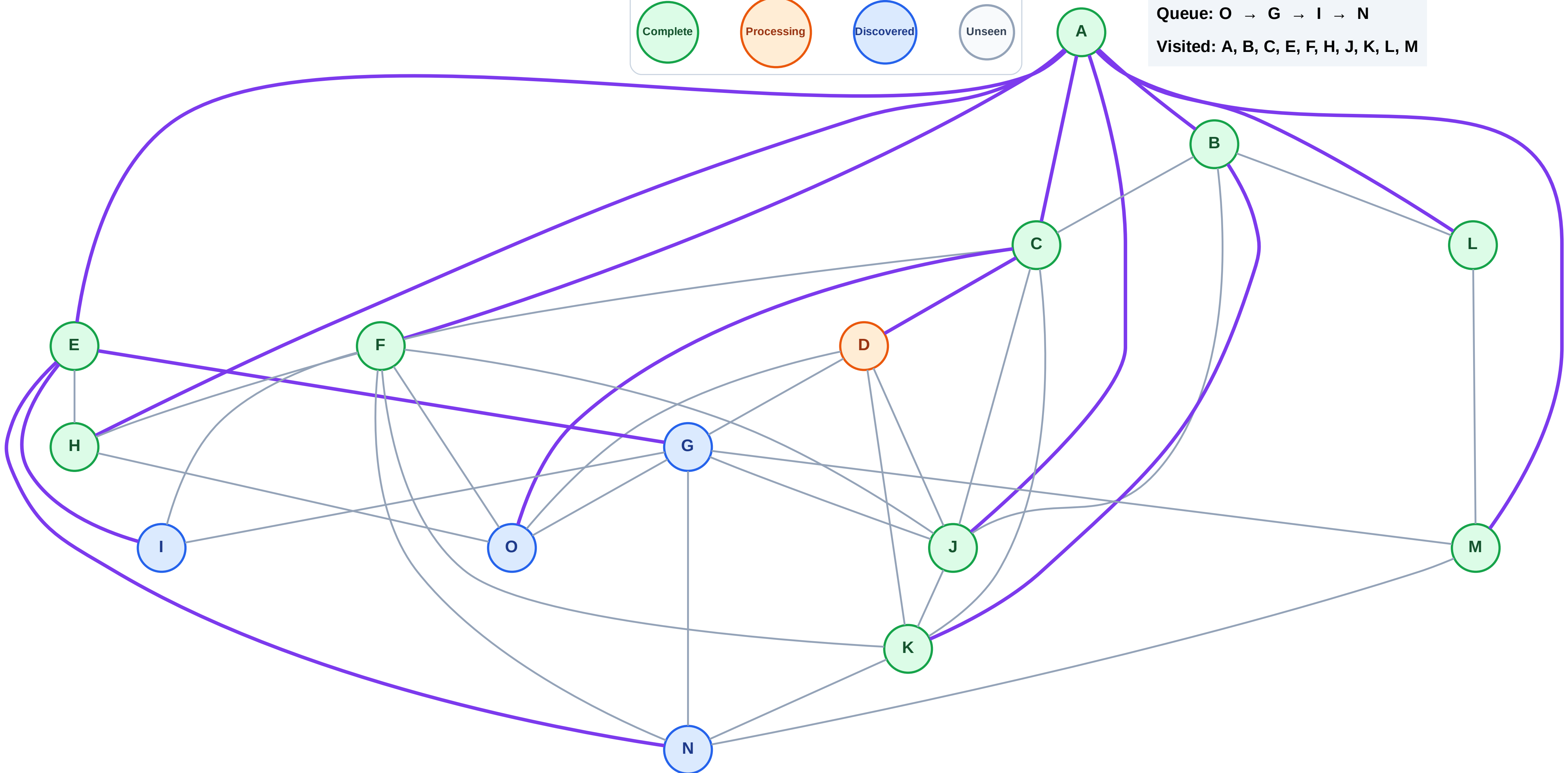


Discovered



Unseen

Queue: O → G → I → N
Visited: A, B, C, E, F, H, J, K, L, M



BFS Traversal

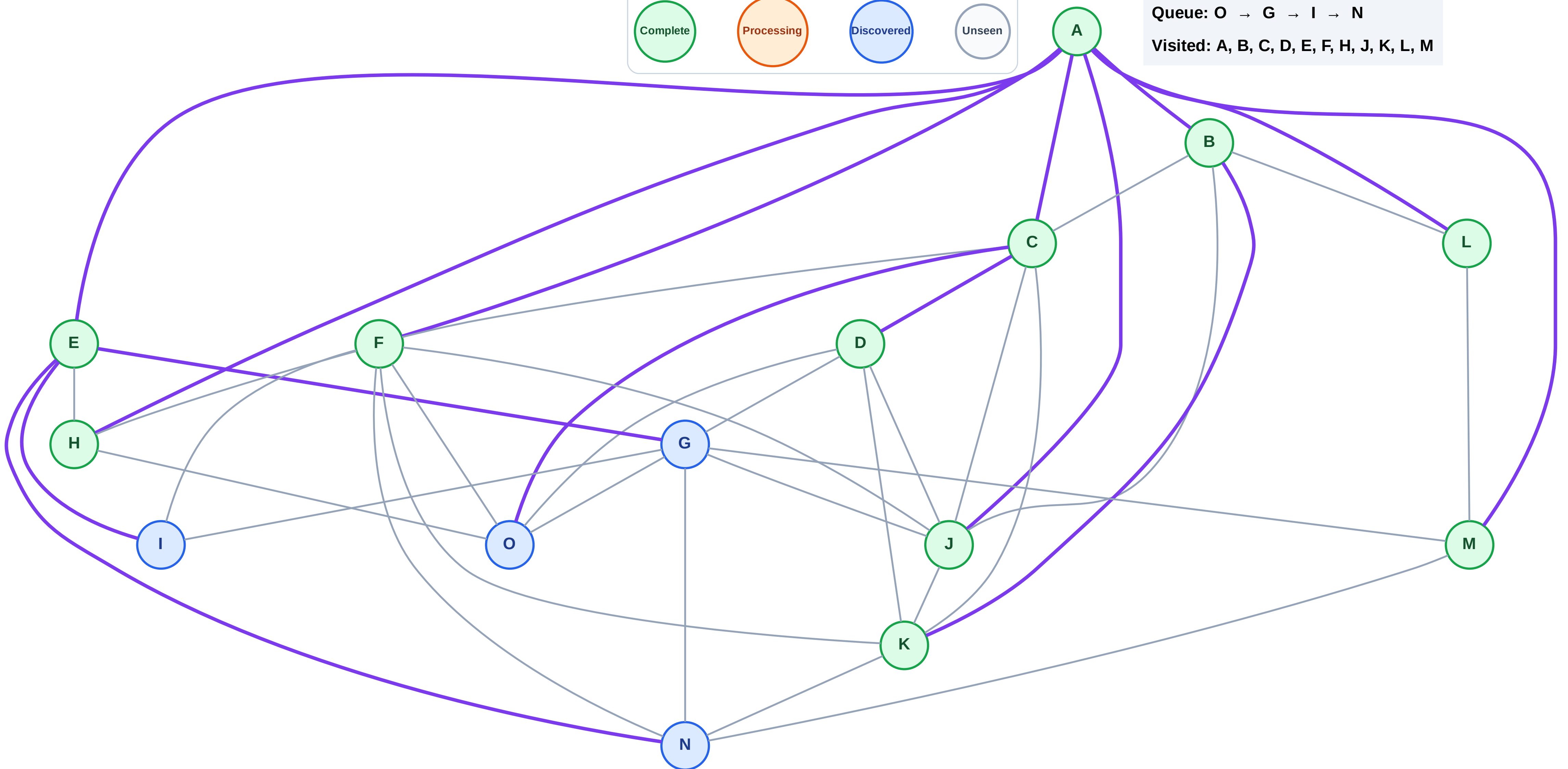
Step 23: No new neighbors from D

Legend



Queue: O → G → I → N

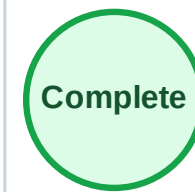
Visited: A, B, C, D, E, F, H, J, K, L, M



BFS Traversal

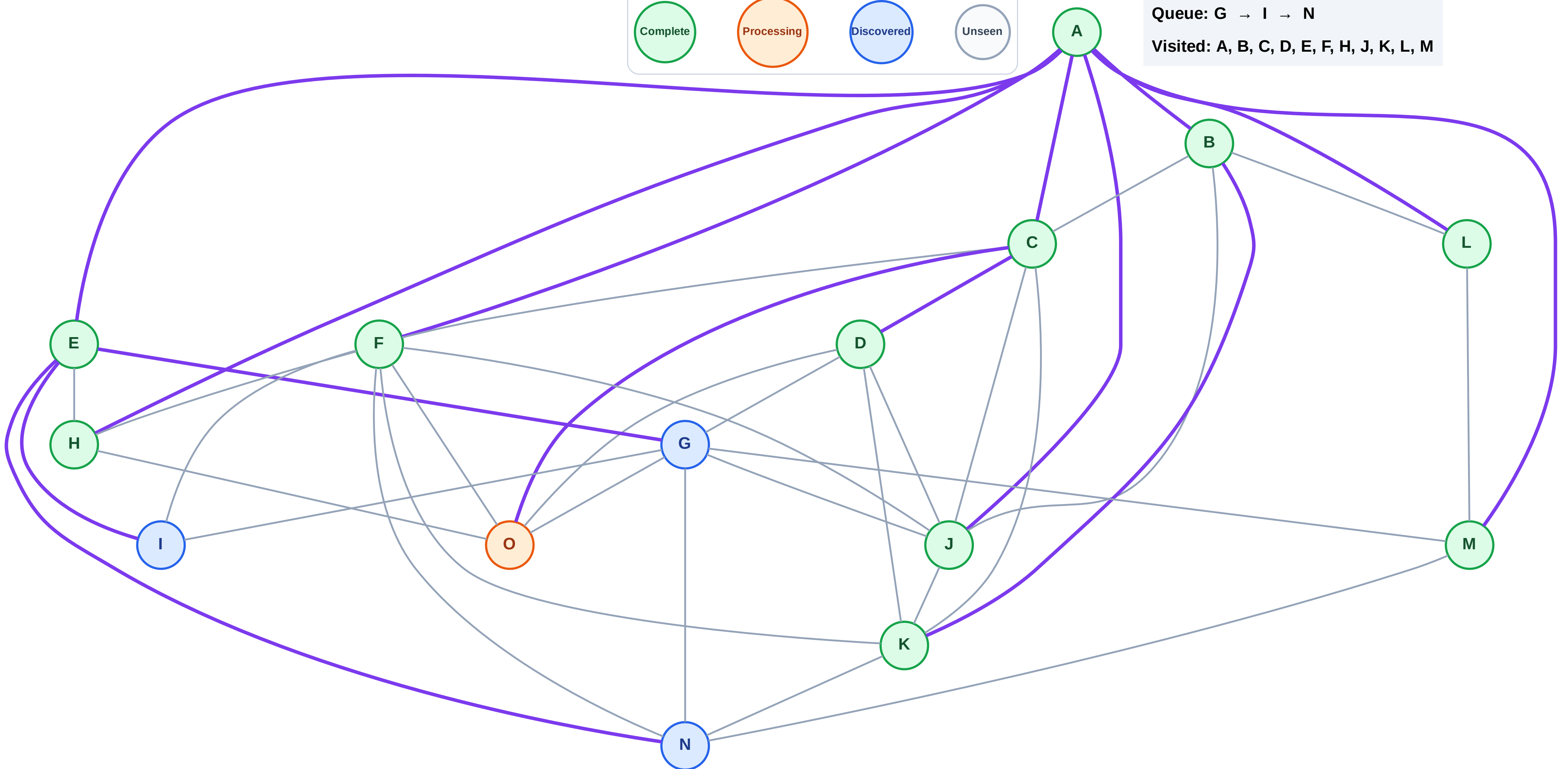
Step 24: Process node O

Legend



Queue: G → I → N

Visited: A, B, C, D, E, F, H, J, K, L, M



Legend

Complete Processing Discovered Unseen

Legend

Complete Processing Discovered Unseen

Legend

Complete Processing Discovered Unseen

Legend

Complete Processing Discovered Unseen

Legend

Complete Processing Discovered Unseen

Legend

Complete Processing Discovered Unseen

Legend

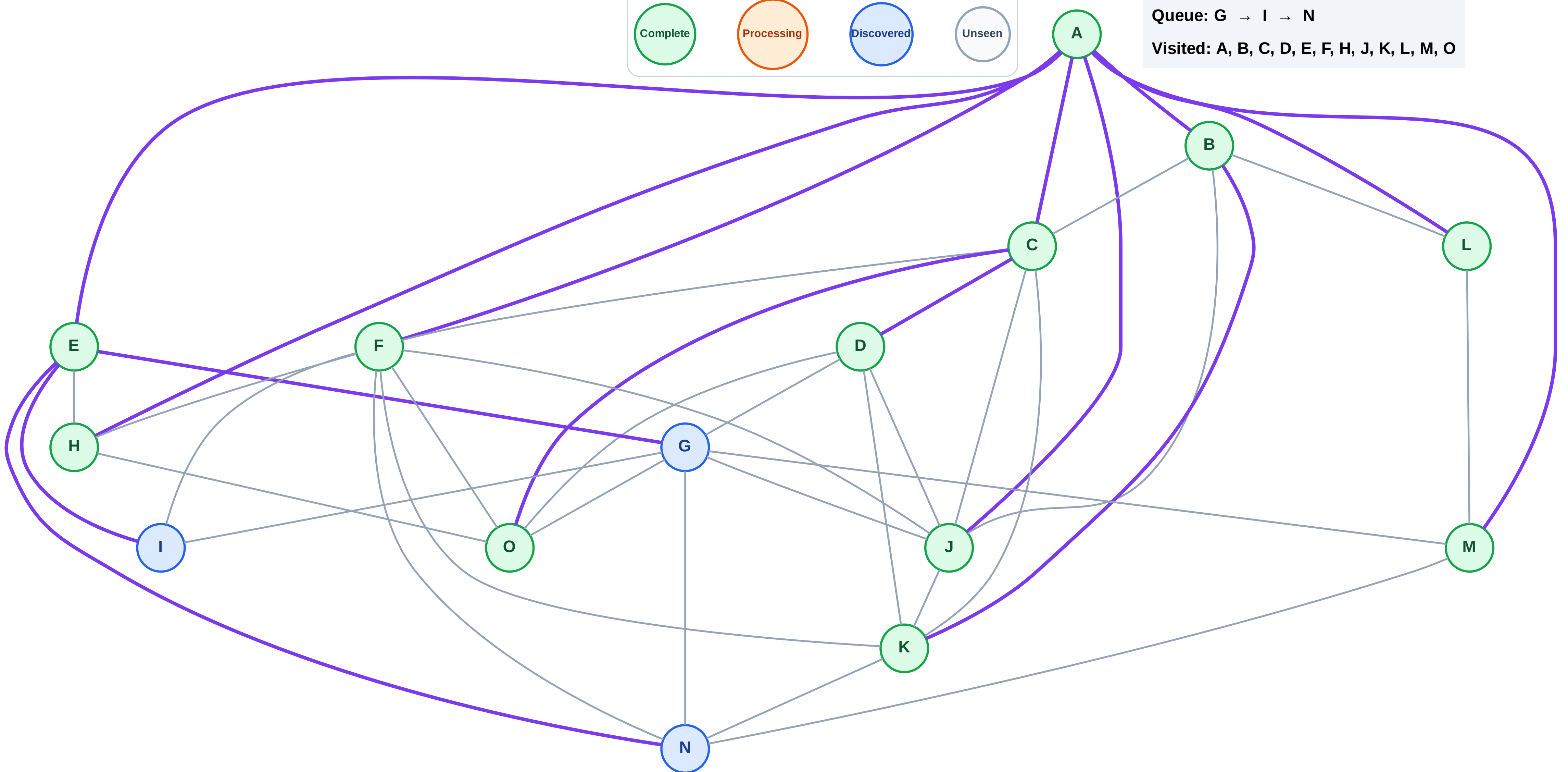
Complete Processing Discovered Unseen

Queue: G → I → N

Visited: A, B, C, D, E, F, H, J, K, L, M, O

Queue: G → I → N

Visited: A, B, C, D, E, F, H, J, K, L, M, O



BFS Traversal

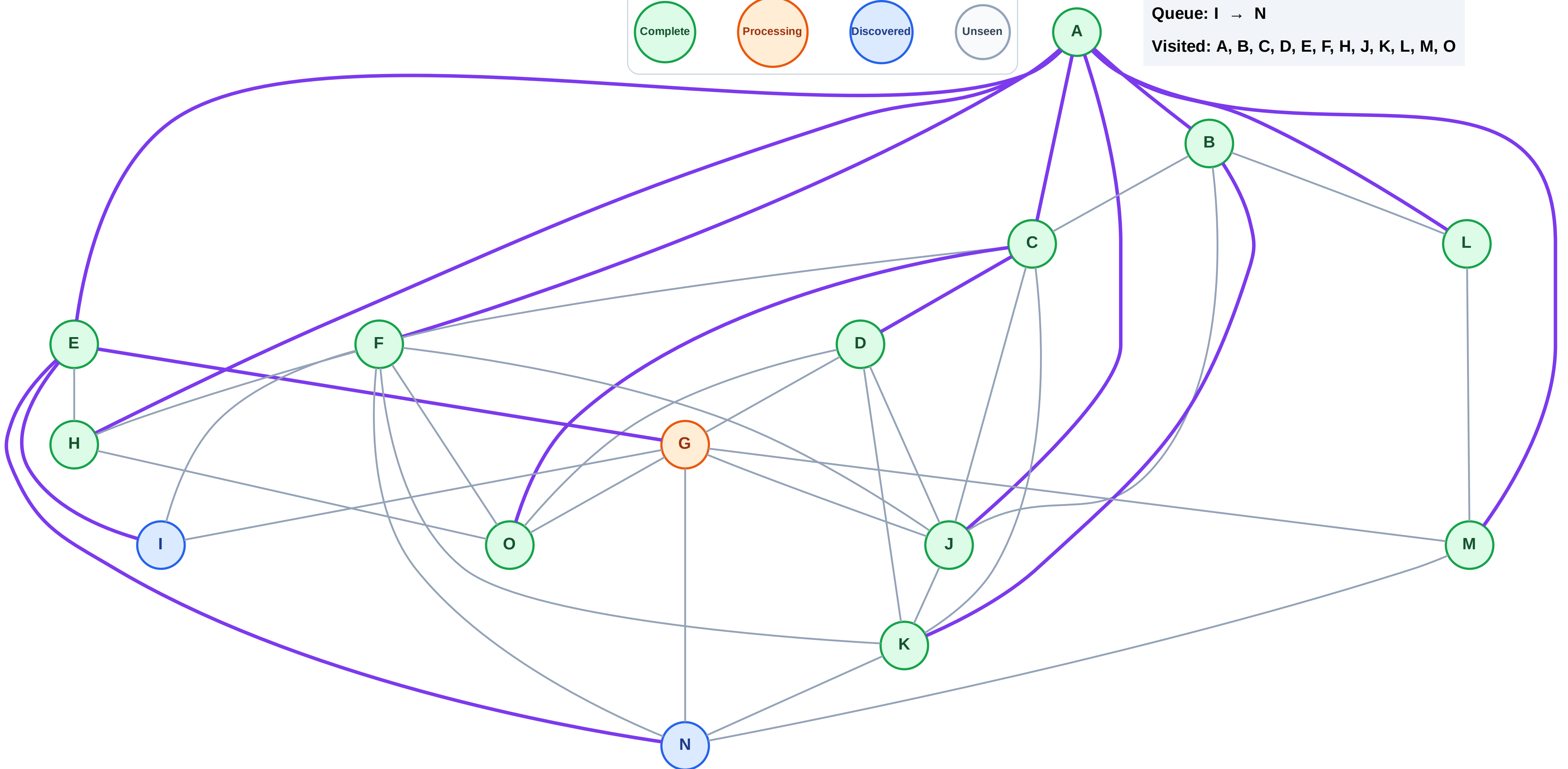
Step 26: Process node G

Legend



Queue: I → N

Visited: A, B, C, D, E, F, H, J, K, L, M, O



BFS Traversal

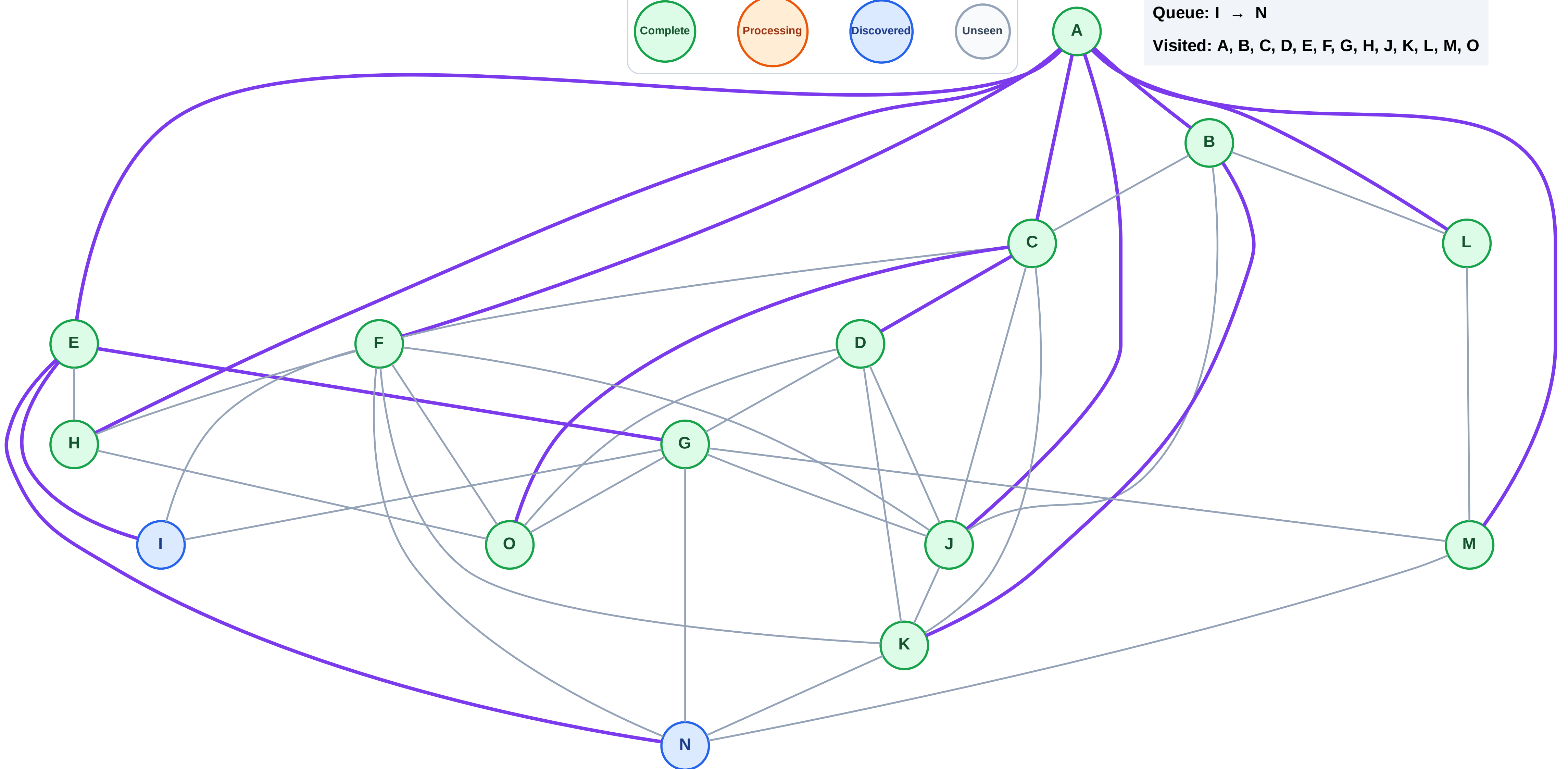
Step 27: No new neighbors from G

Legend



Queue: I → N

Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



BFS Traversal
Step 28: Process node I

Legend

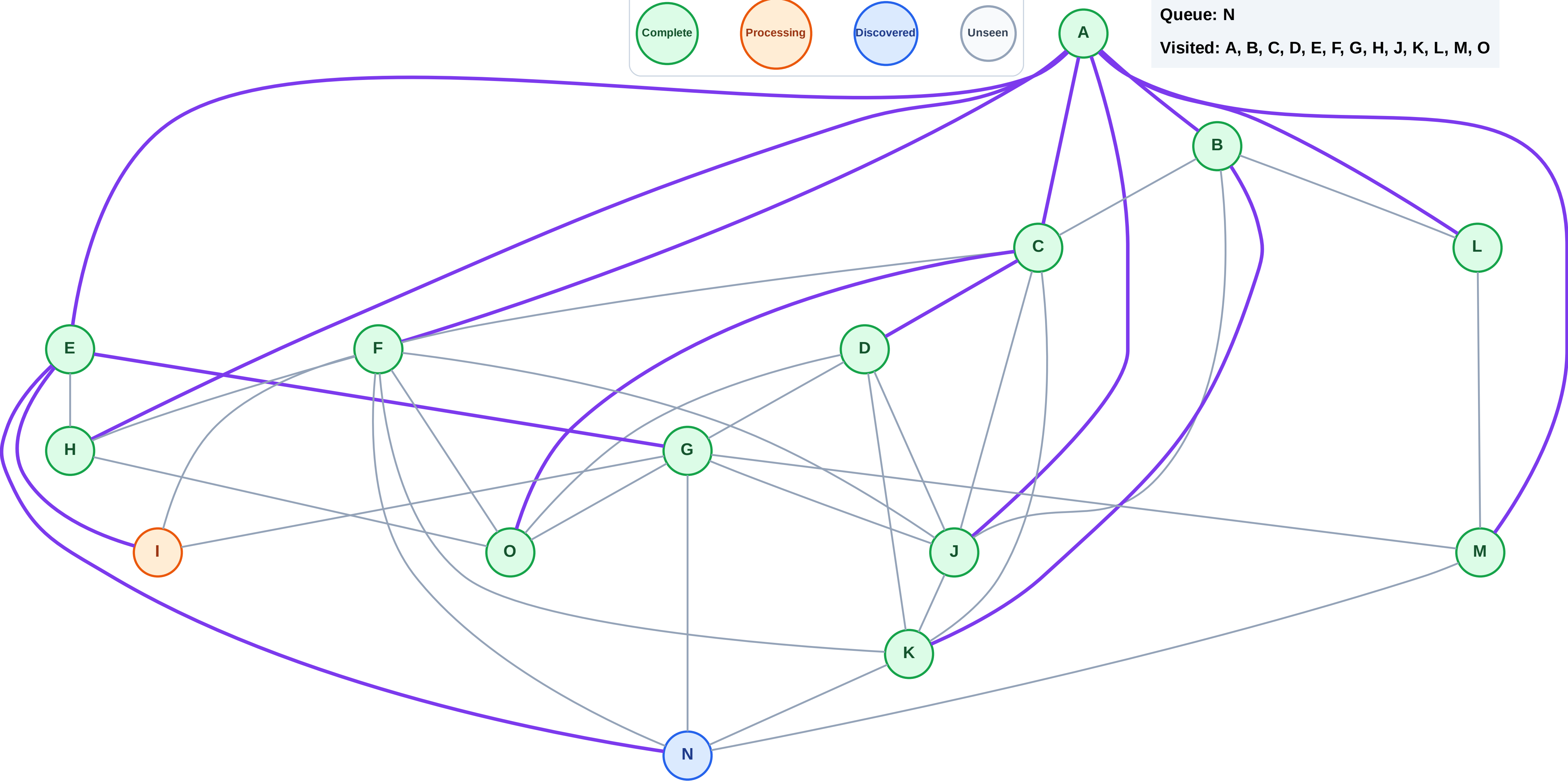
Complete

Processing

Discovered

Unseen

Queue: N
Visited: A, B, C, D, E, F, G, H, J, K, L, M, O



BFS Traversal

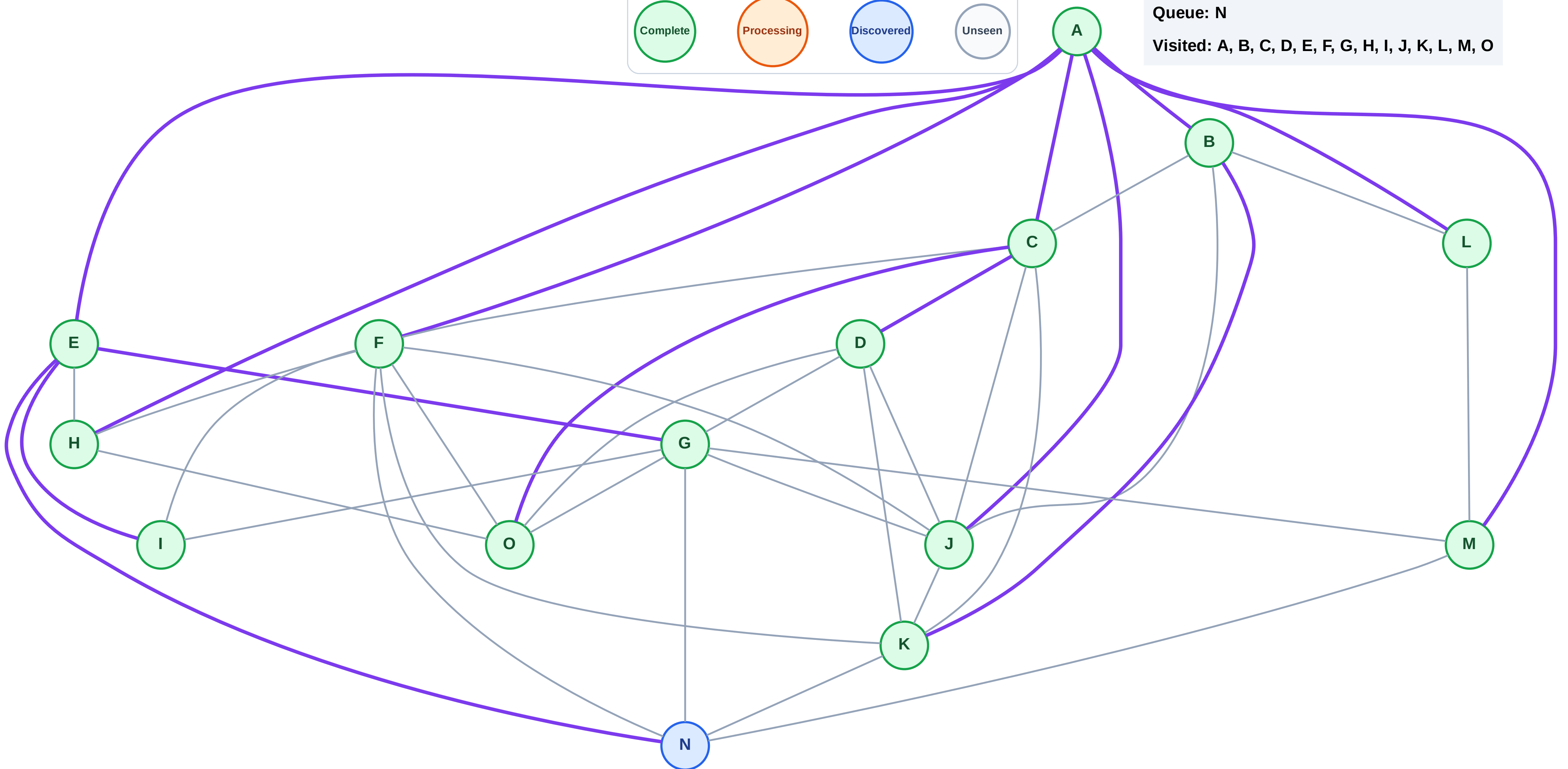
Step 29: No new neighbors from I

Legend



Queue: N

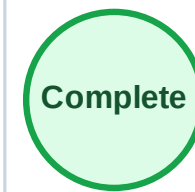
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



BFS Traversal

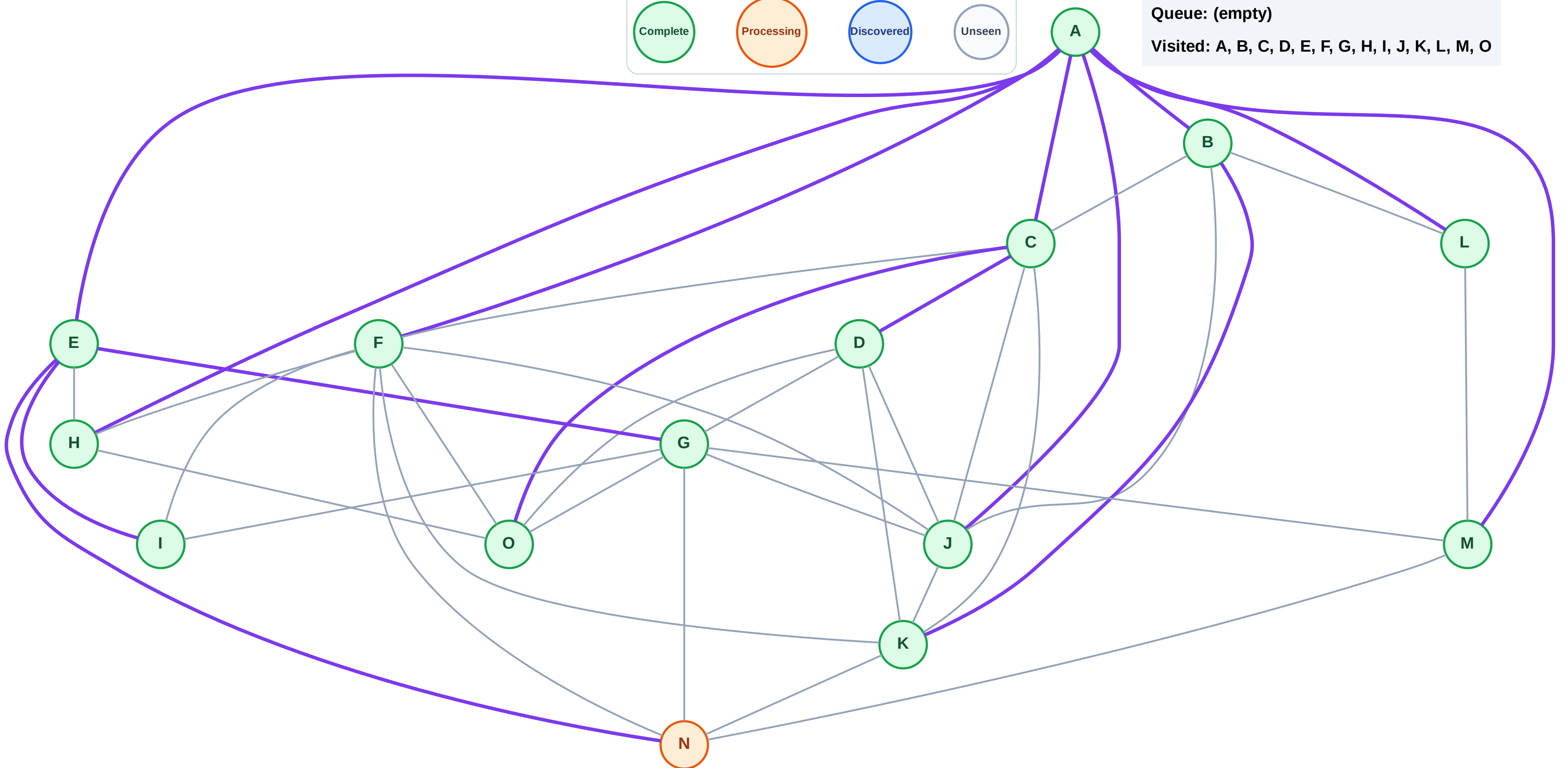
Step 30: Process node N

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, O



BFS Traversal

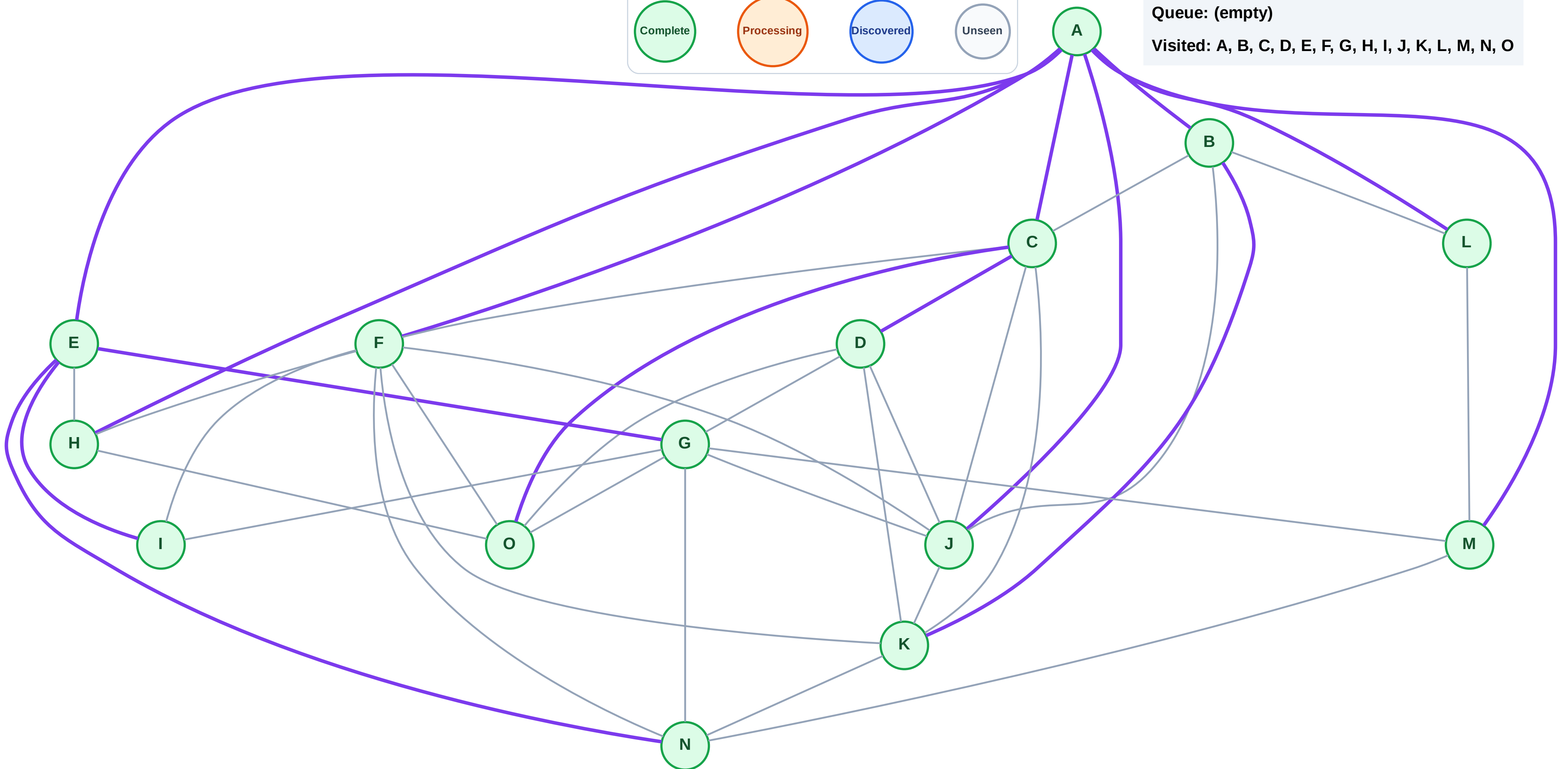
Step 31: No new neighbors from N

Legend



Queue: (empty)

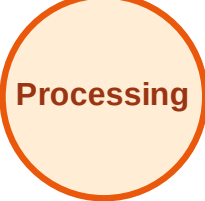
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



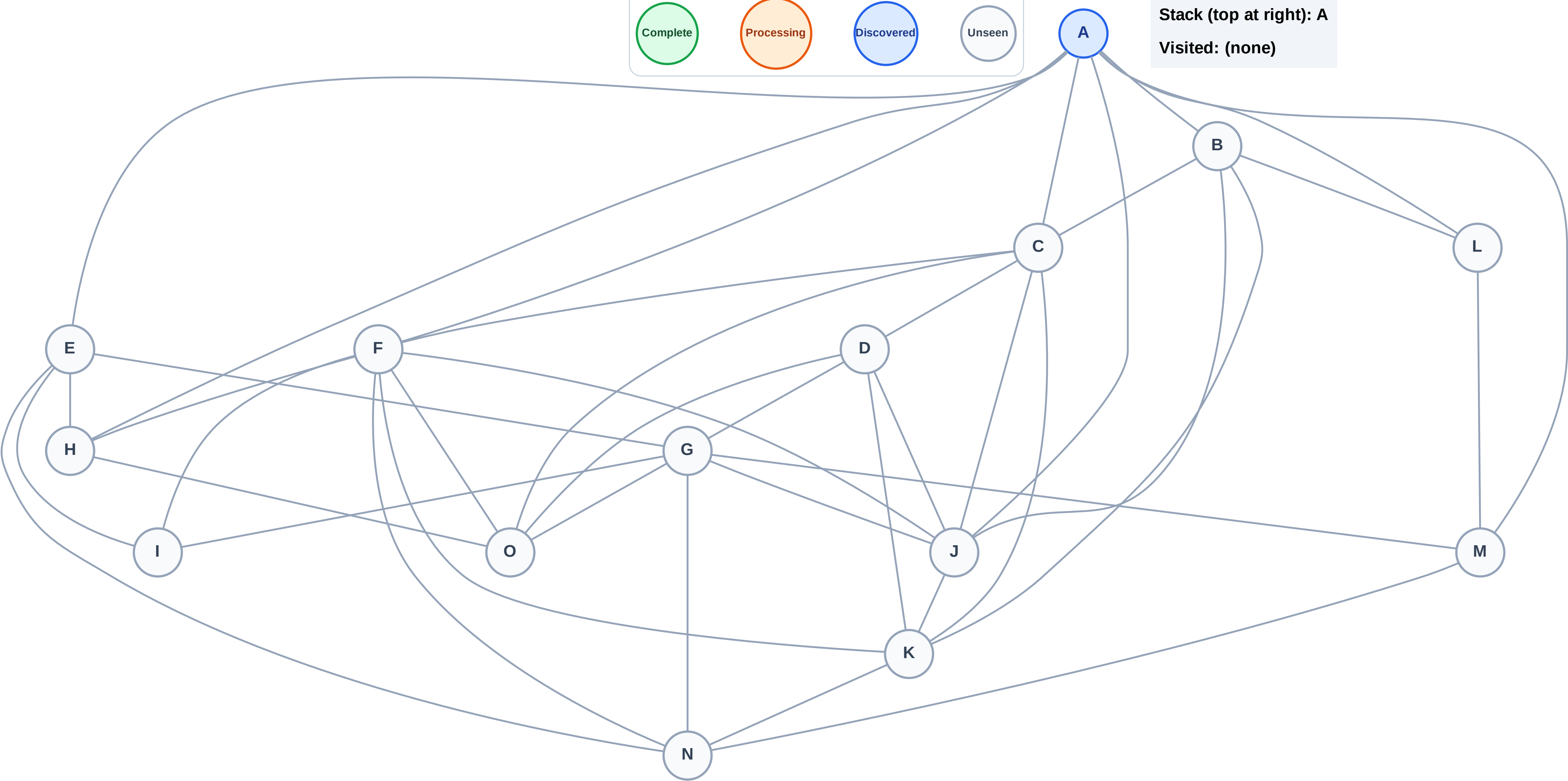
DFS Traversal

Step 1: Start at A

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

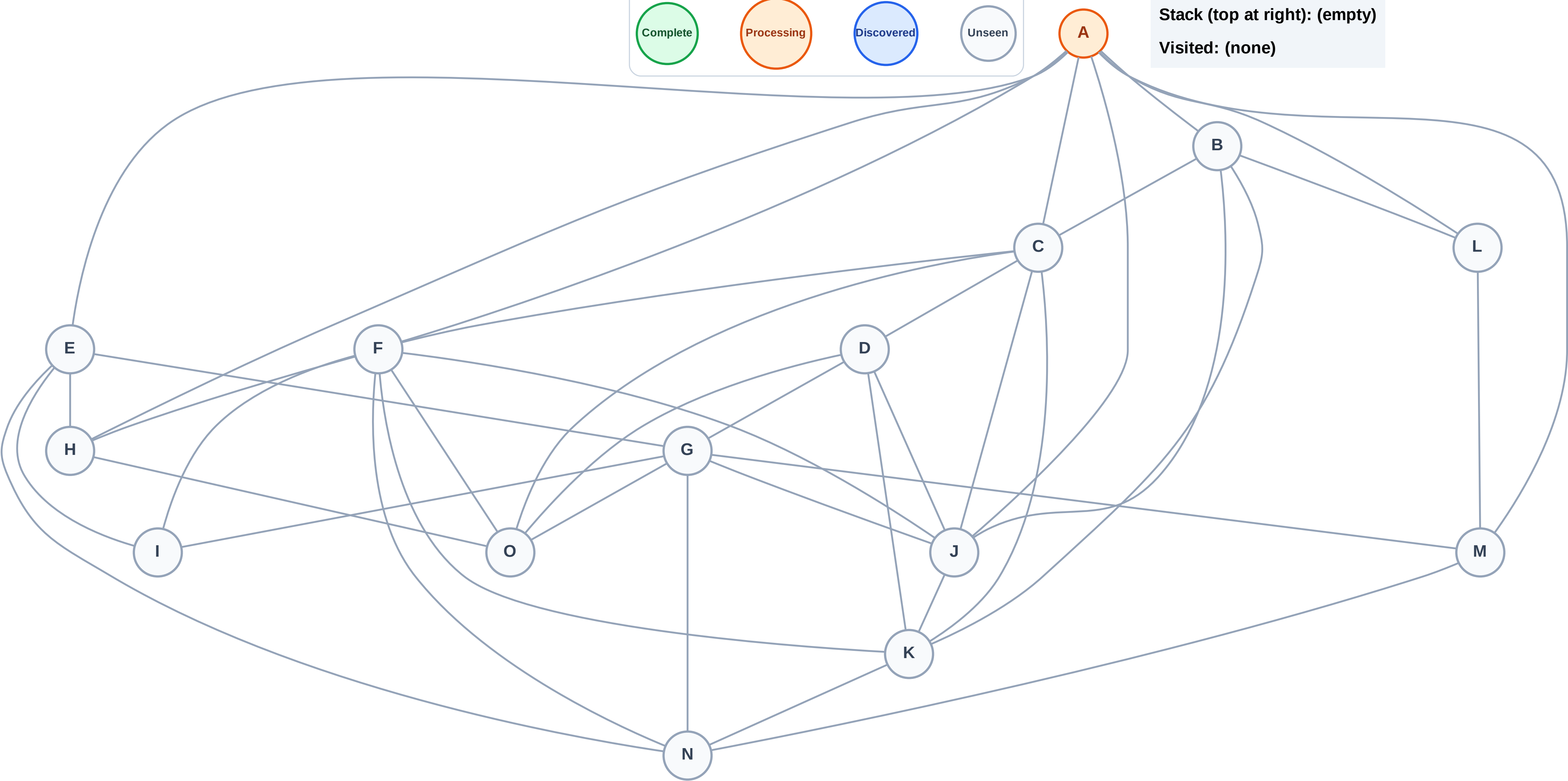
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

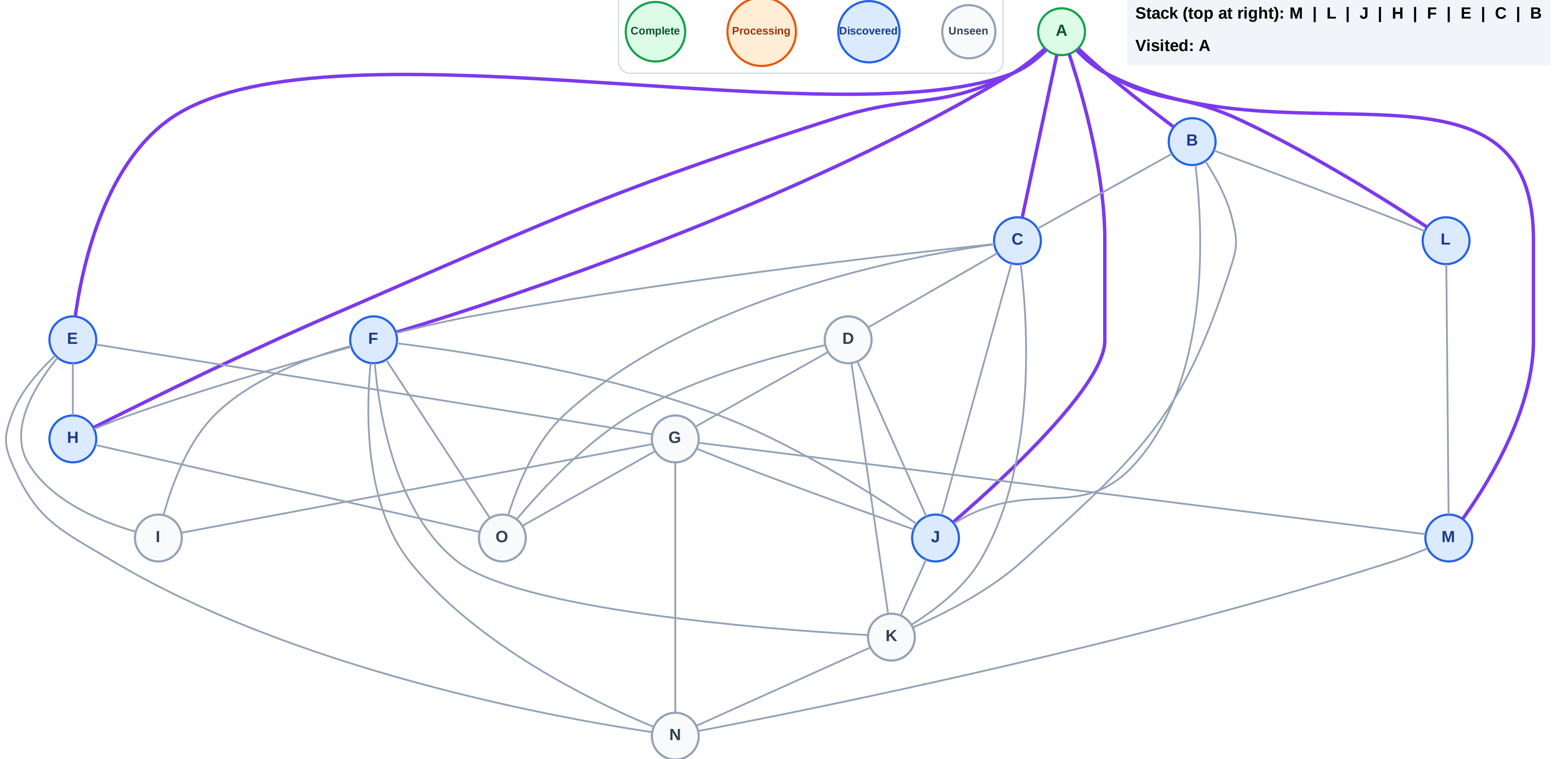
Step 3: Discover M, L, J, H, F, E, C, B from A

Legend



Stack (top at right): M | L | J | H | F | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

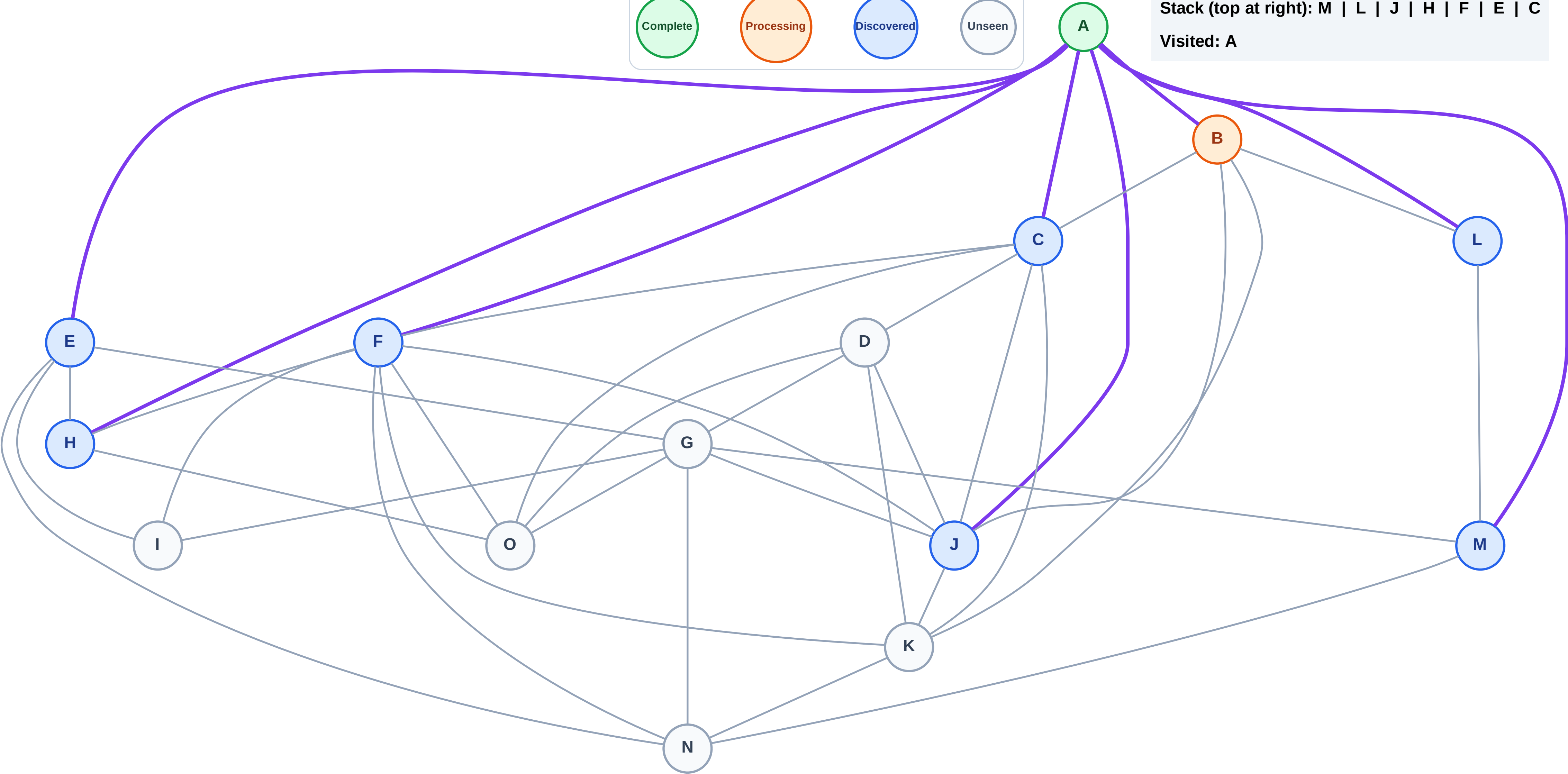
Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | J | H | F | E | C
Visited: A



Step 5: Discover K from B

Step 5: Discover K from B

Legend

Complete

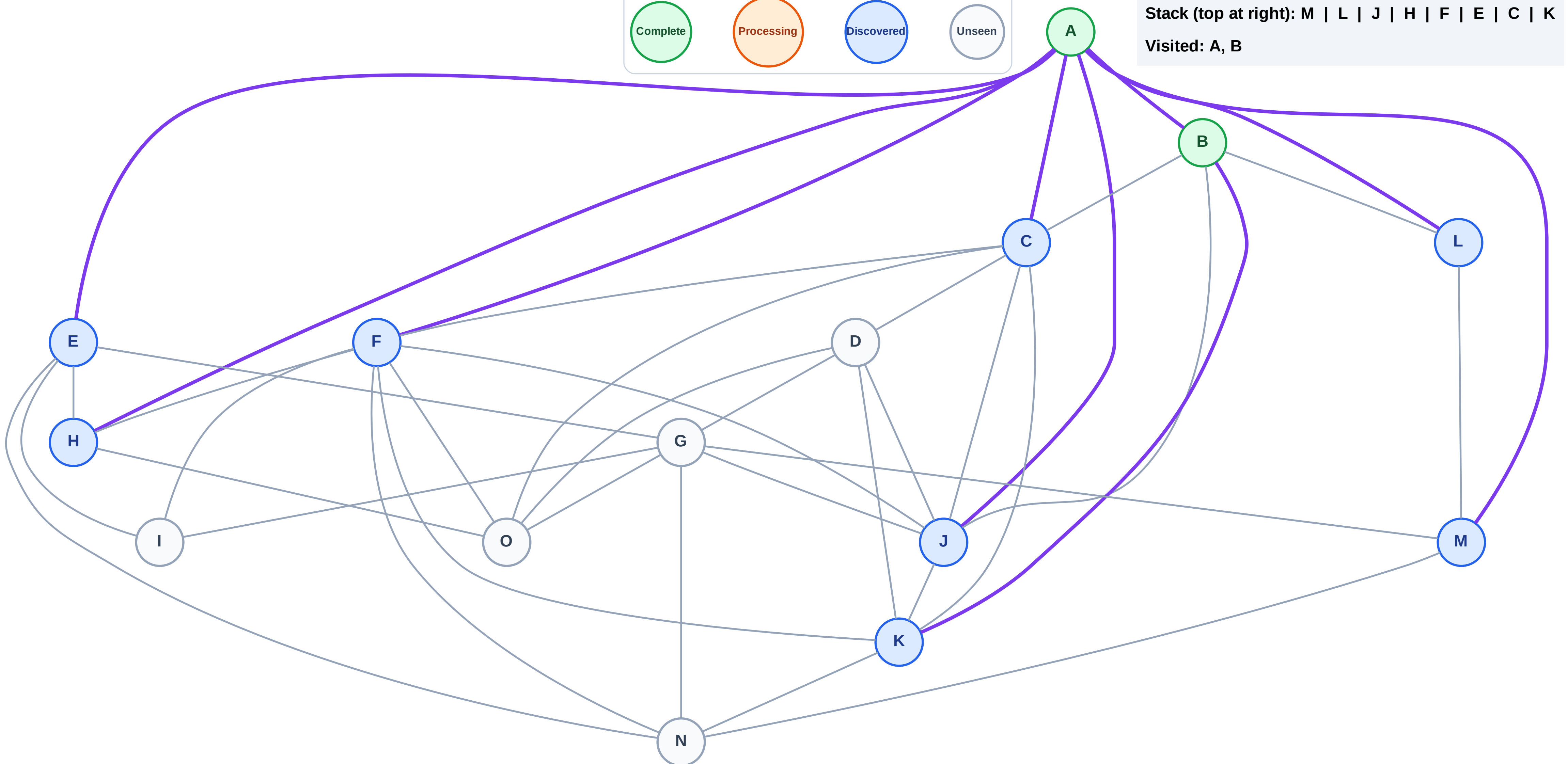
Processing

Discovered

Unseen

Stack (top at right): M | L | J | H | F | E | C | K

Visited: A, B



Step 6: Process node K

Step 6: Process node K

Legend



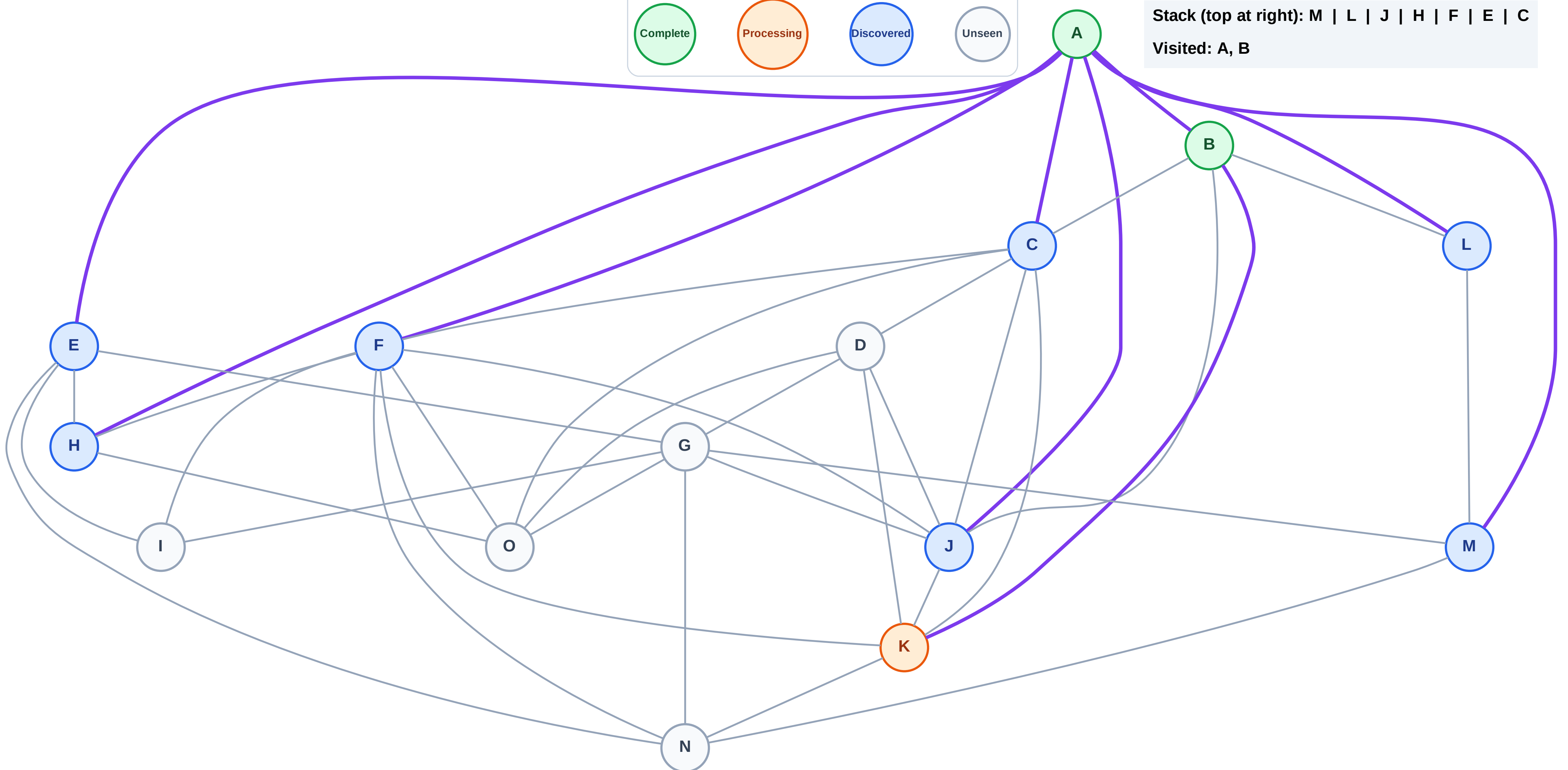
The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.



Processing

Unseen

Visited: A, B



DFS Traversal

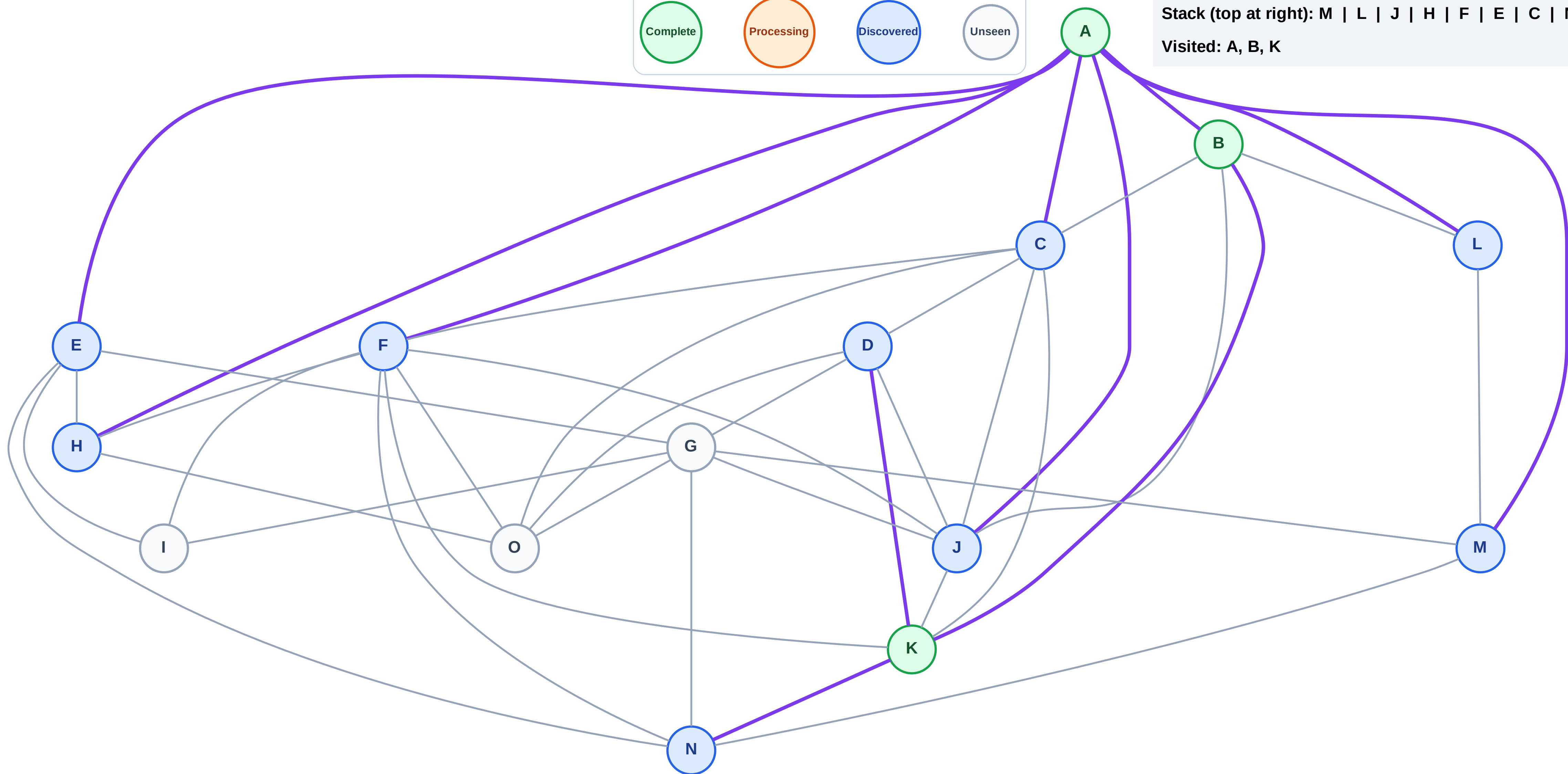
Step 7: Discover N, D from K

Legend



Stack (top at right): M | L | J | H | F | E | C | N | D

Visited: A, B, K



DFS Traversal
Step 8: Process node D

Legend

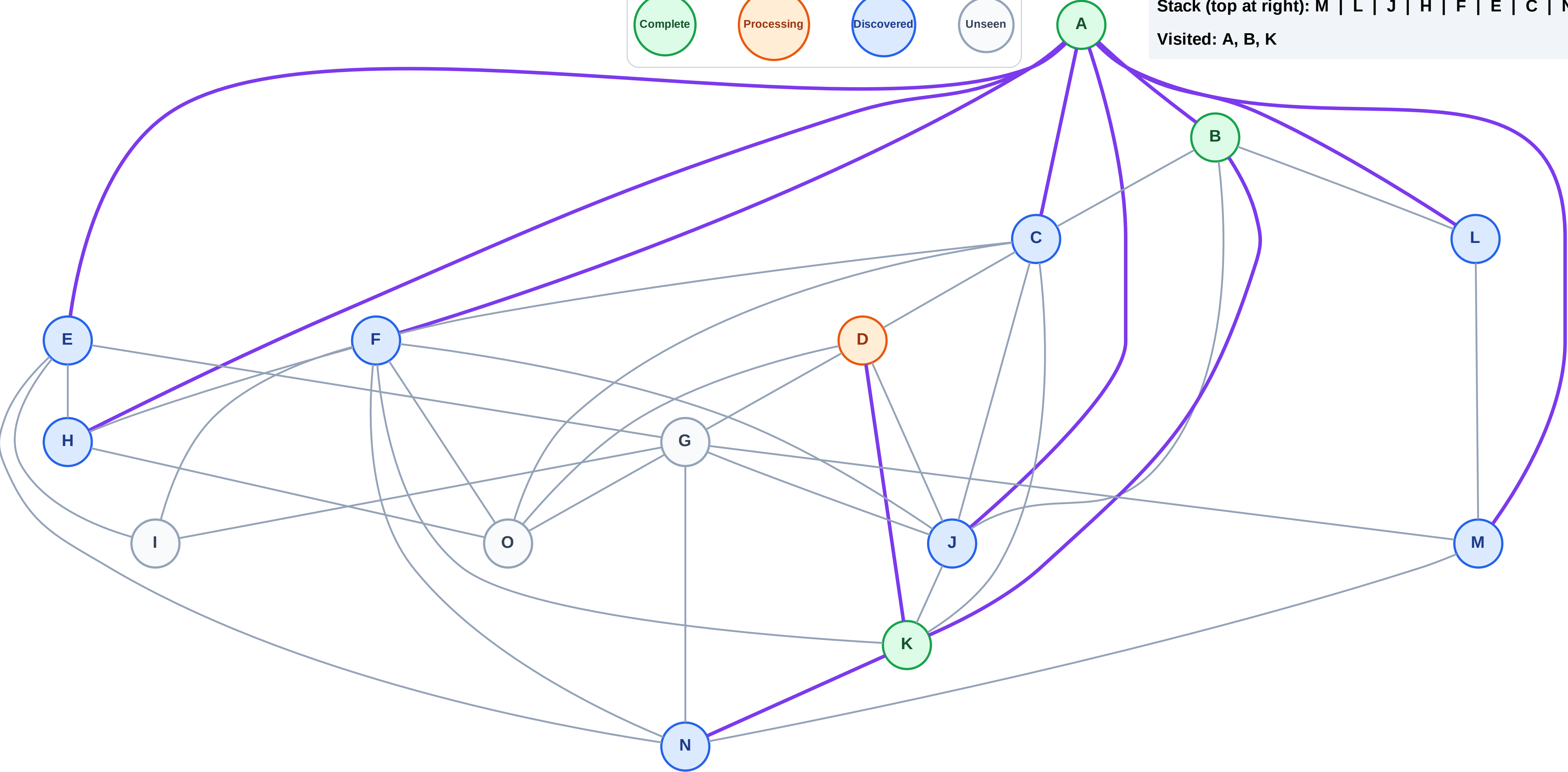
Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | J | H | F | E | C | N
Visited: A, B, K



DFS Traversal

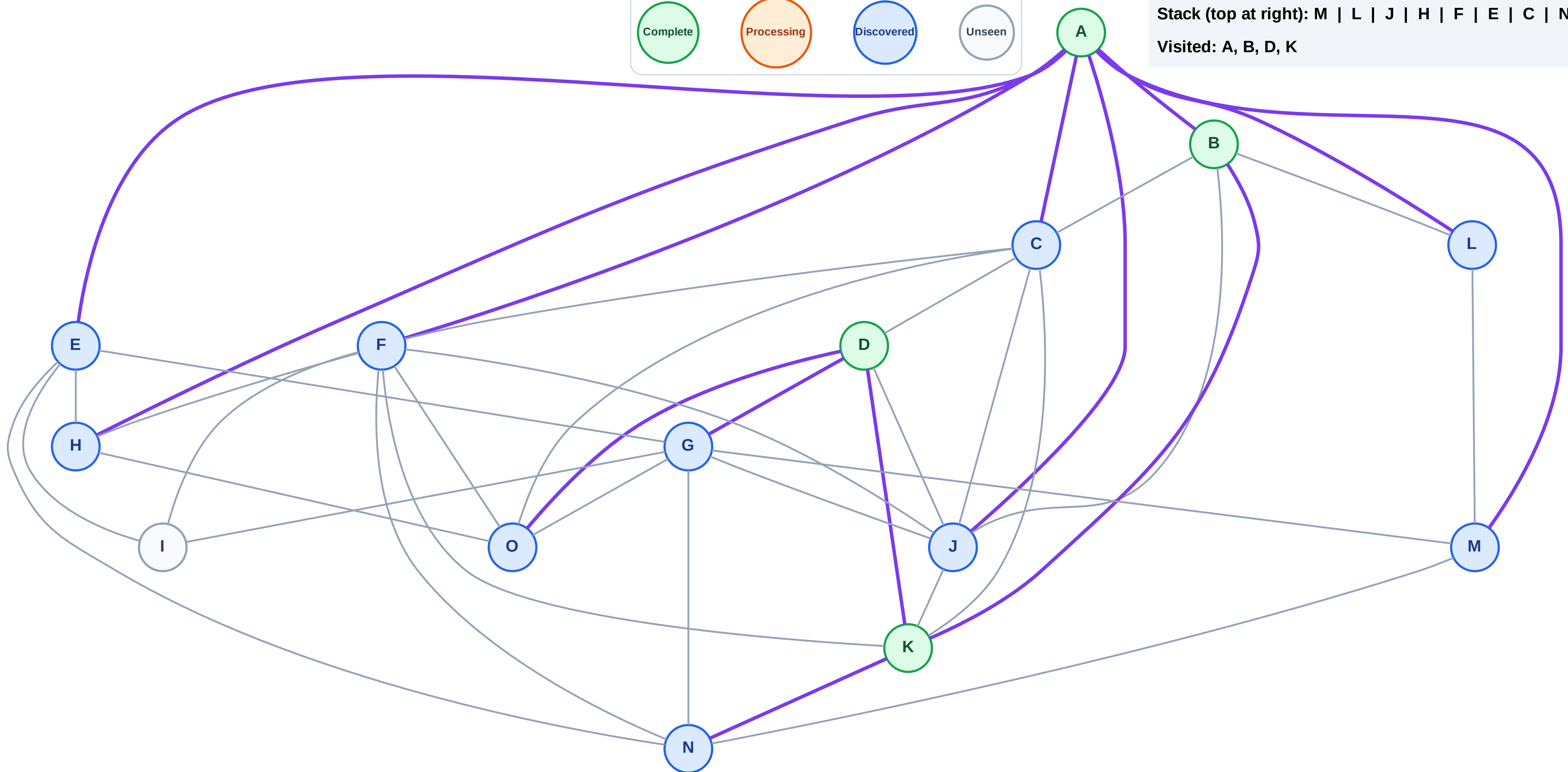
Step 9: Discover O, G from D

Legend



Stack (top at right): M | L | J | H | F | E | C | N | O | G

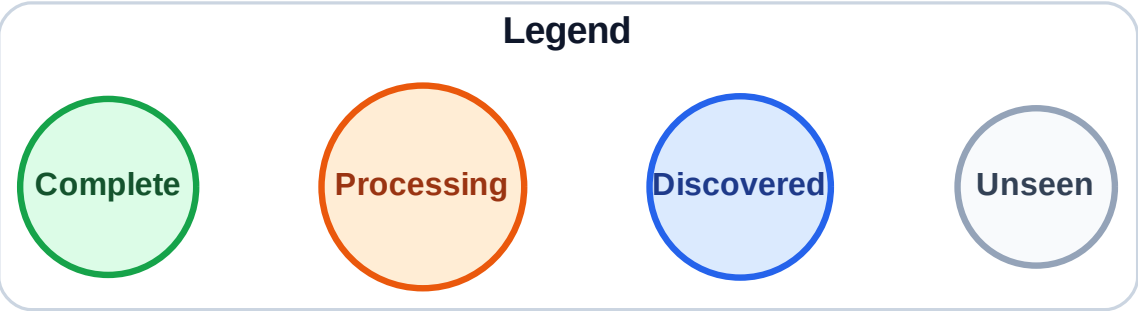
Visited: A, B, D, K



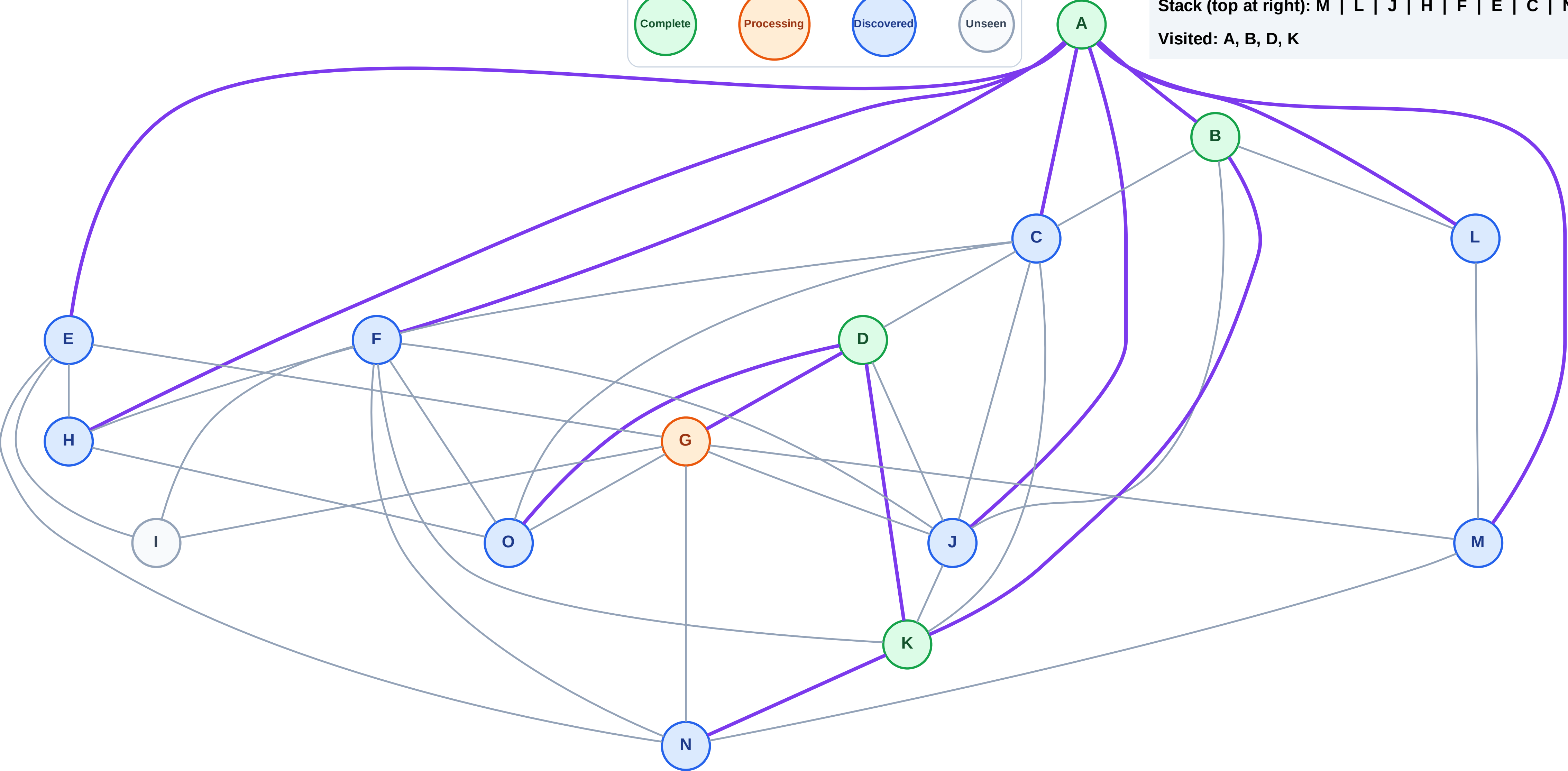
DFS Traversal

Step 10: Process node G

Legend



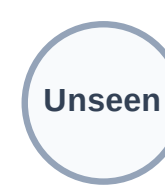
Stack (top at right): M | L | J | H | F | E | C | N | O
Visited: A, B, D, K



DFS Traversal

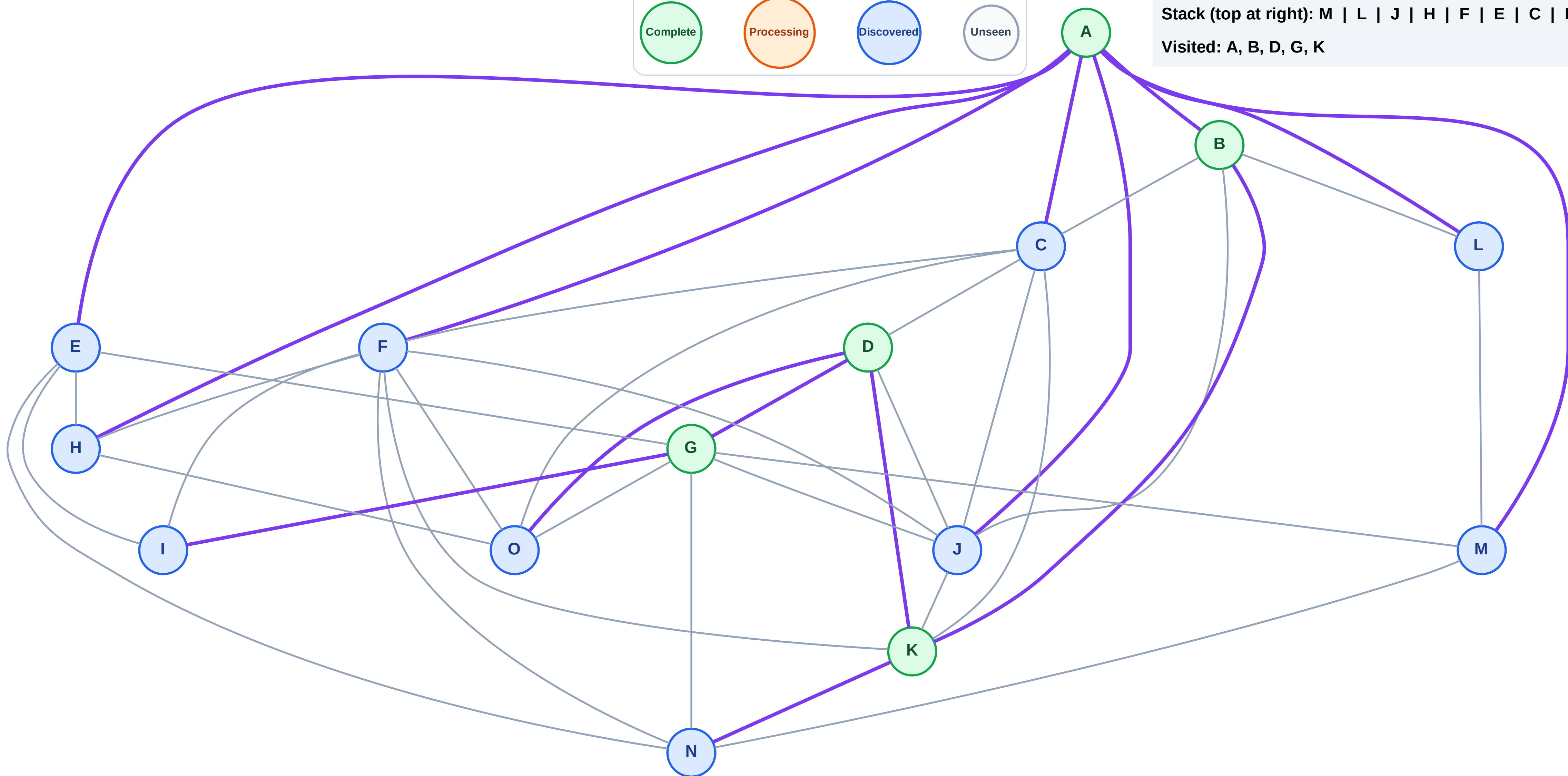
Step 11: Discover I from G

Legend



Stack (top at right): M | L | J | H | F | E | C | N | O | I

Visited: A, B, D, G, K



DFS Traversal
Step 12: Process node I

Legend

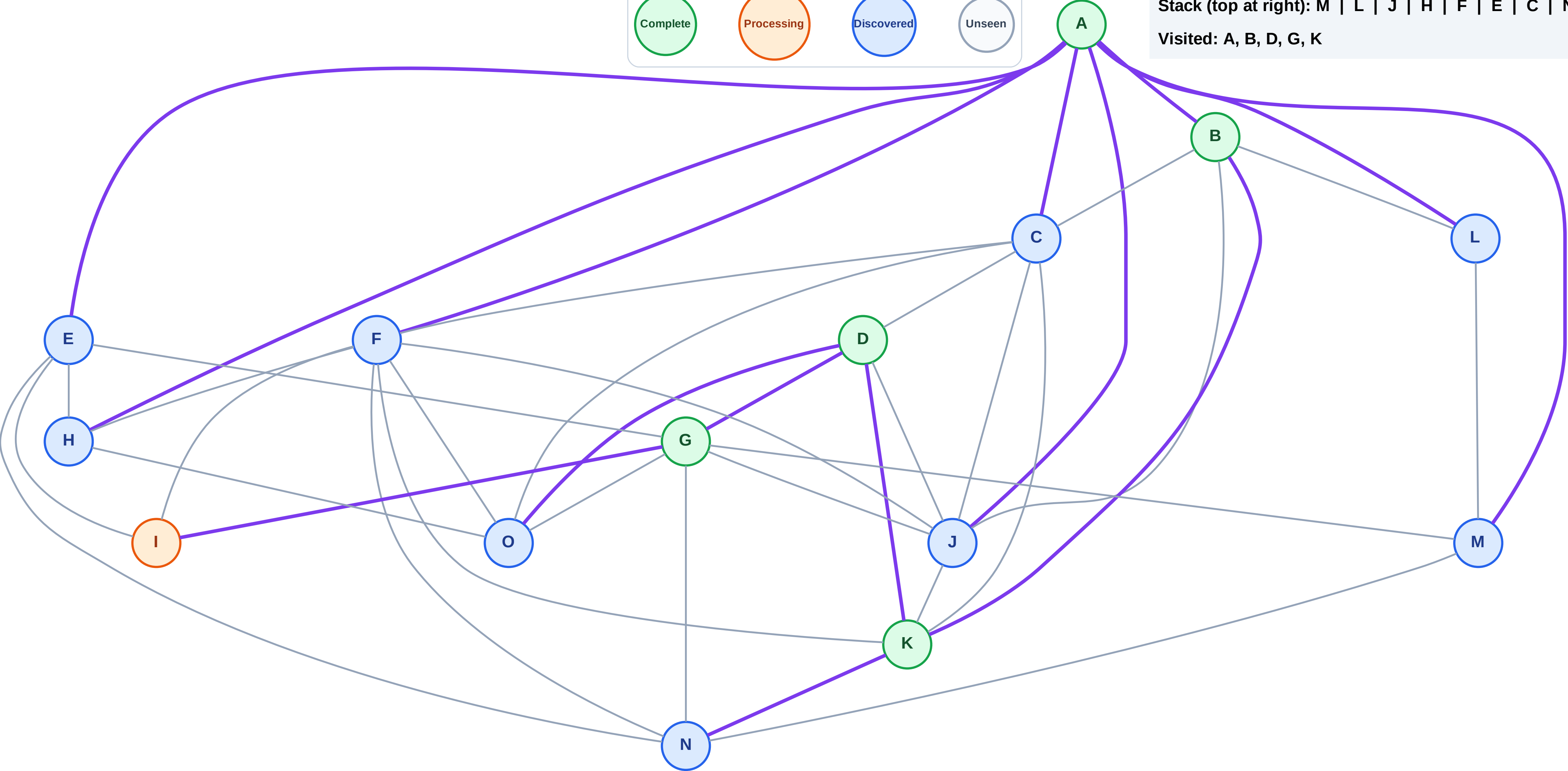
Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | J | H | F | E | C | N | O
Visited: A, B, D, G, K



DFS Traversal

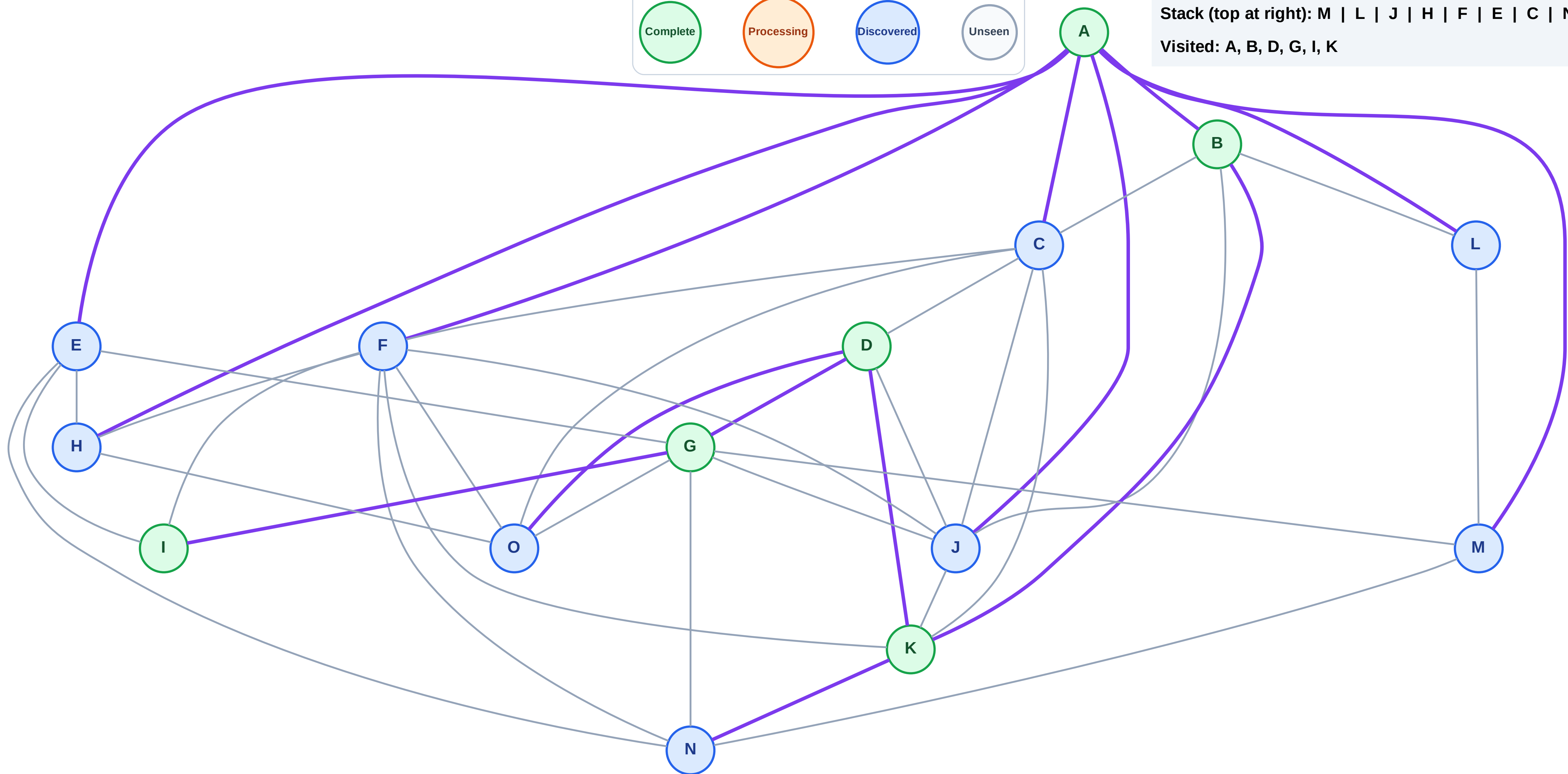
Step 13: No new neighbors from I

Legend



Stack (top at right): M | L | J | H | F | E | C | N | O

Visited: A, B, D, G, I, K



DFS Traversal

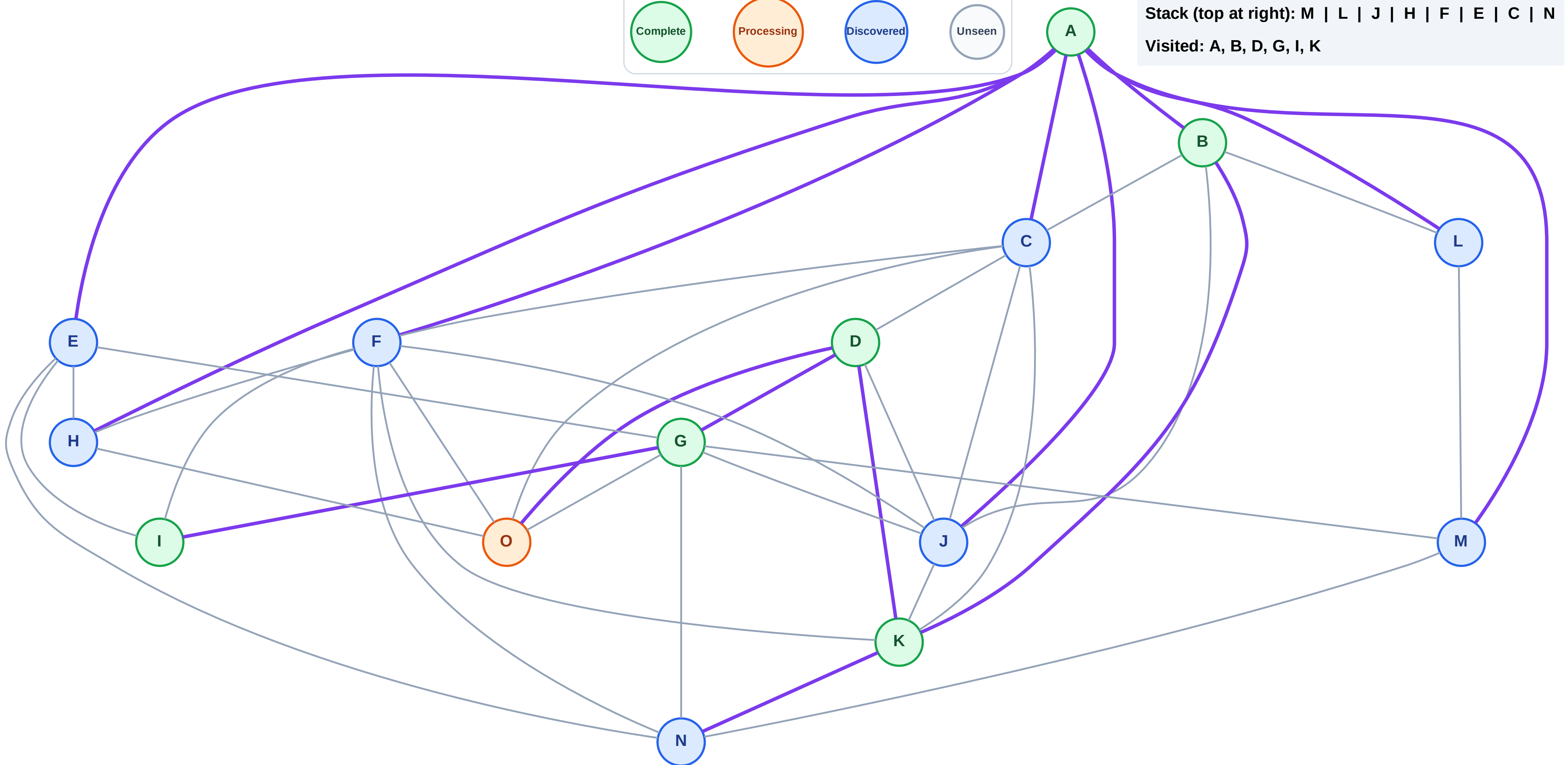
Step 14: Process node O

Legend



Stack (top at right): M | L | J | H | F | E | C | N

Visited: A, B, D, G, I, K



DFS Traversal

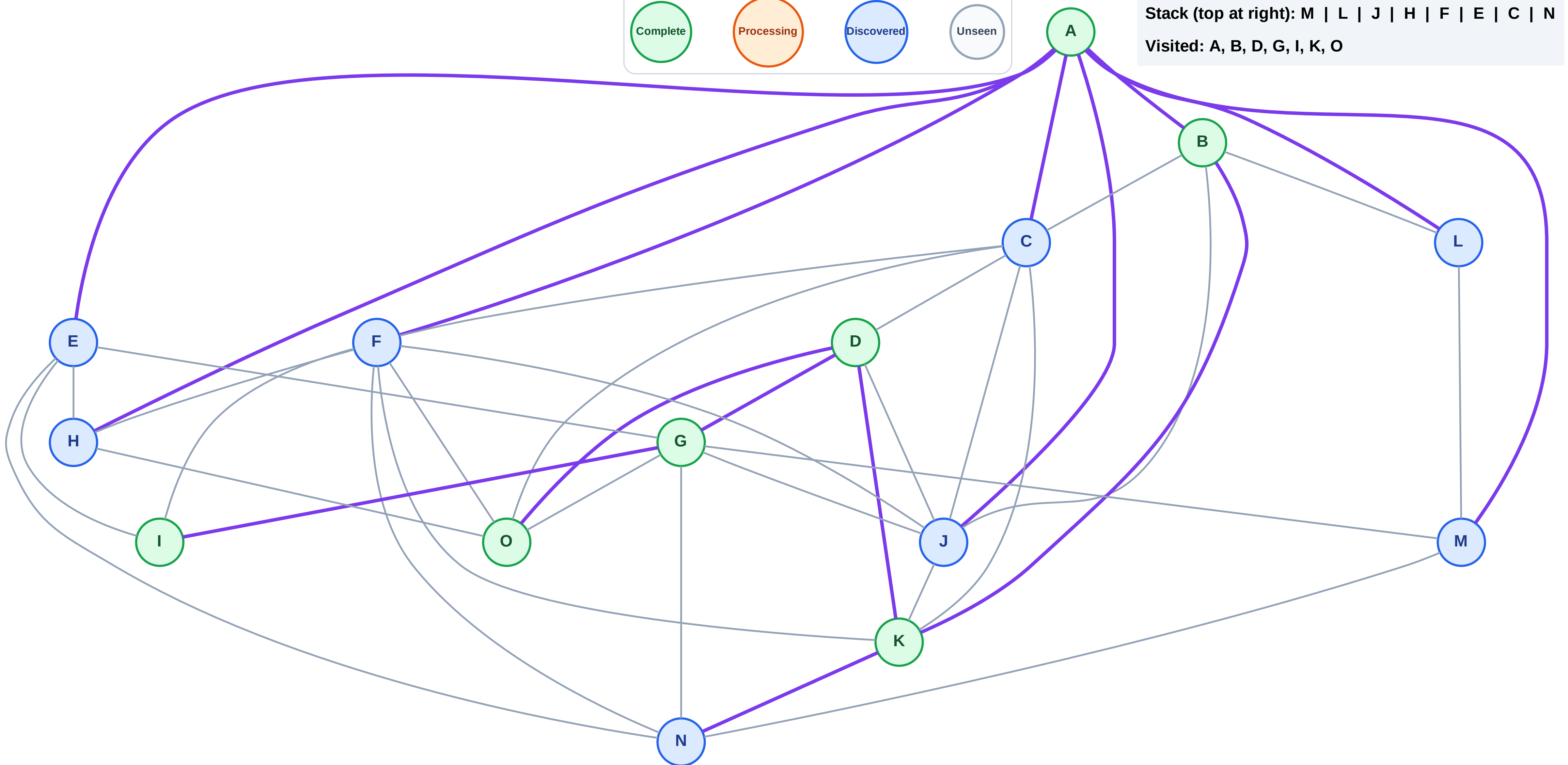
Step 15: No new neighbors from O

Legend



Stack (top at right): M | L | J | H | F | E | C | N

Visited: A, B, D, G, I, K, O



DFS Traversal

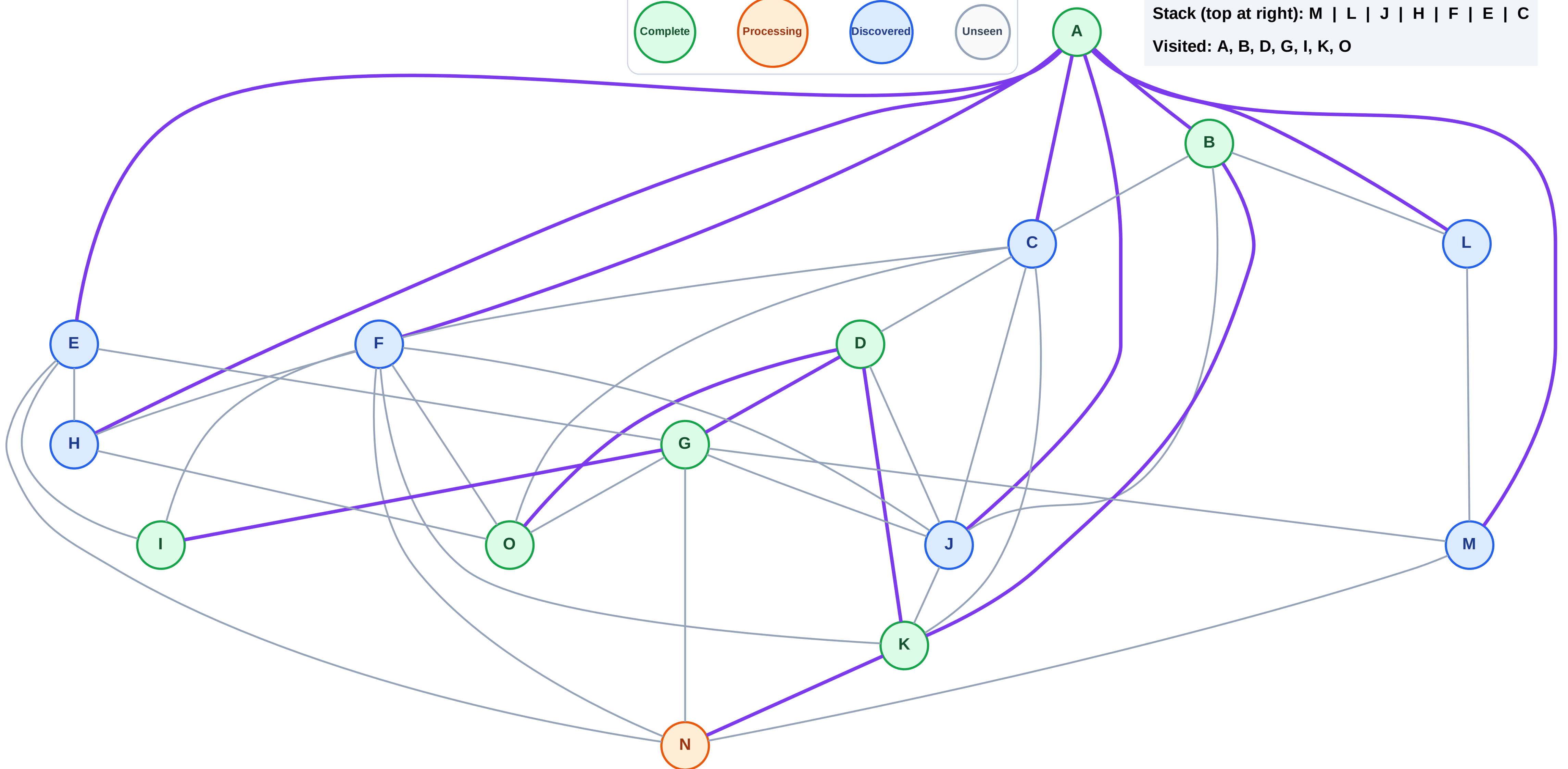
Step 16: Process node N

Legend



Stack (top at right): M | L | J | H | F | E | C

Visited: A, B, D, G, I, K, O

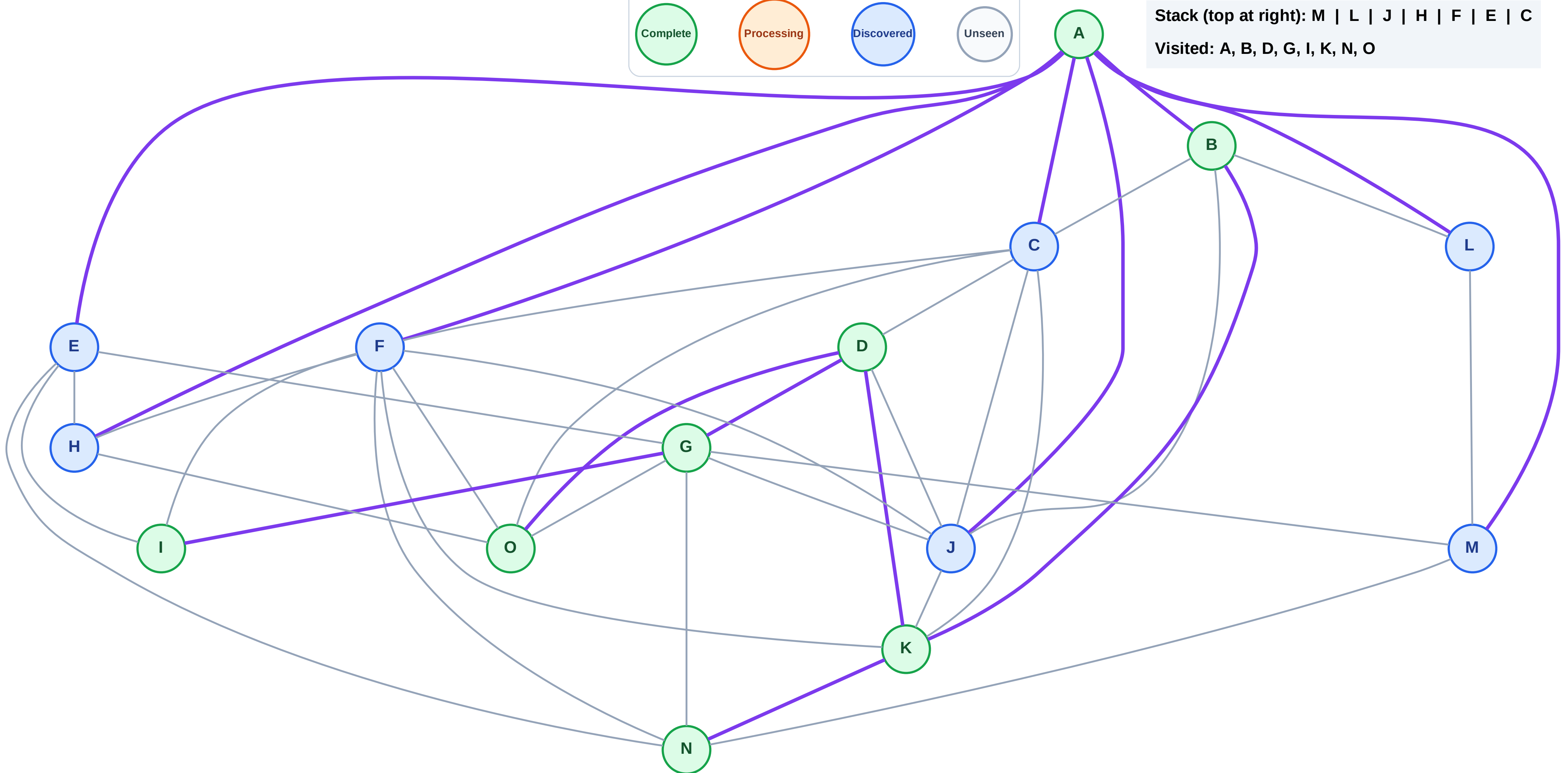


Legend

- Complete
- Processing
- Discovered
- Unseen

Stack (top at right): M | L | J | H | F | E | C

Visited: A, B, D, G, I, K, N, O



DFS Traversal

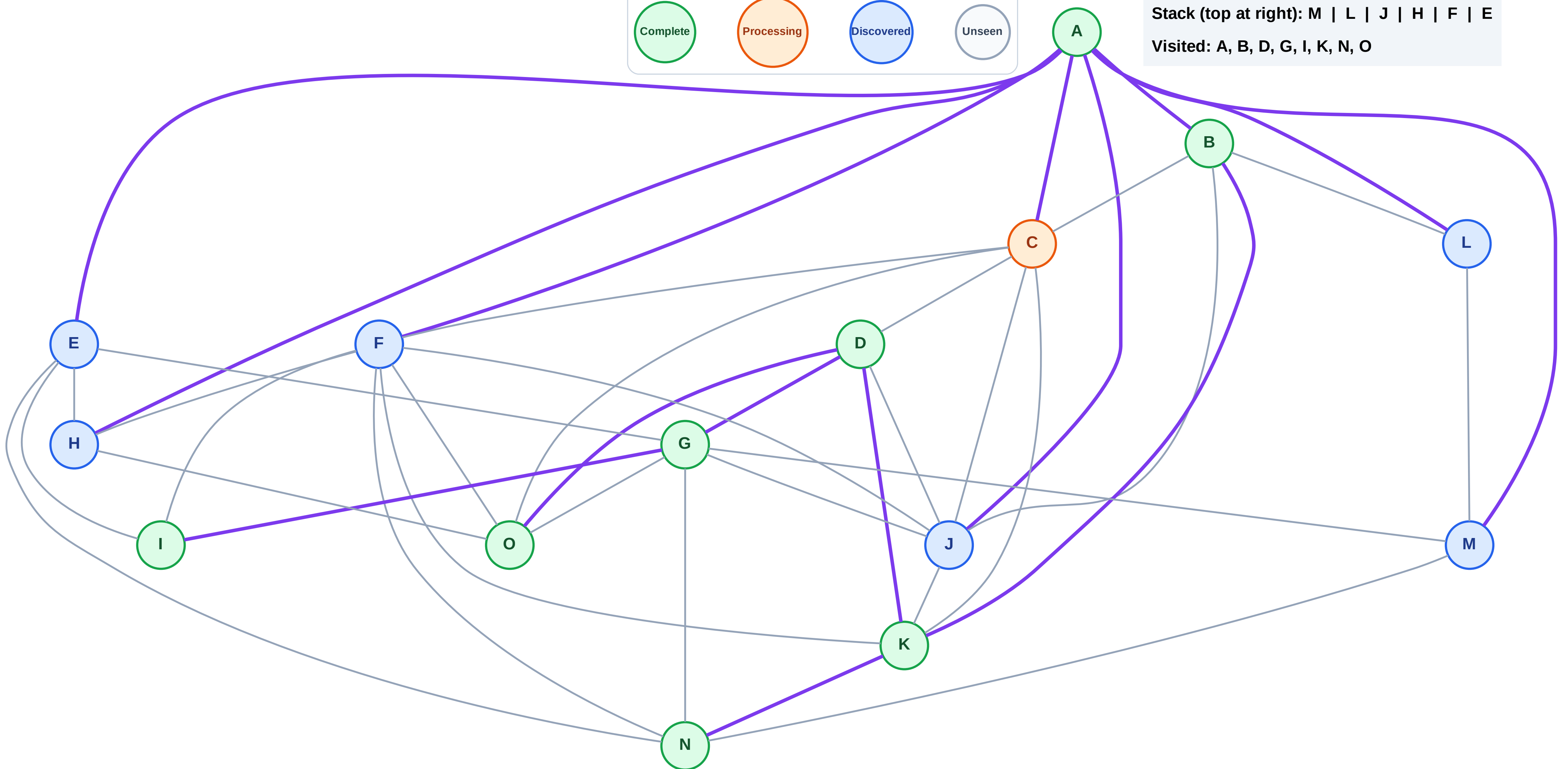
Step 18: Process node C

Legend



Stack (top at right): M | L | J | H | F | E

Visited: A, B, D, G, I, K, N, O



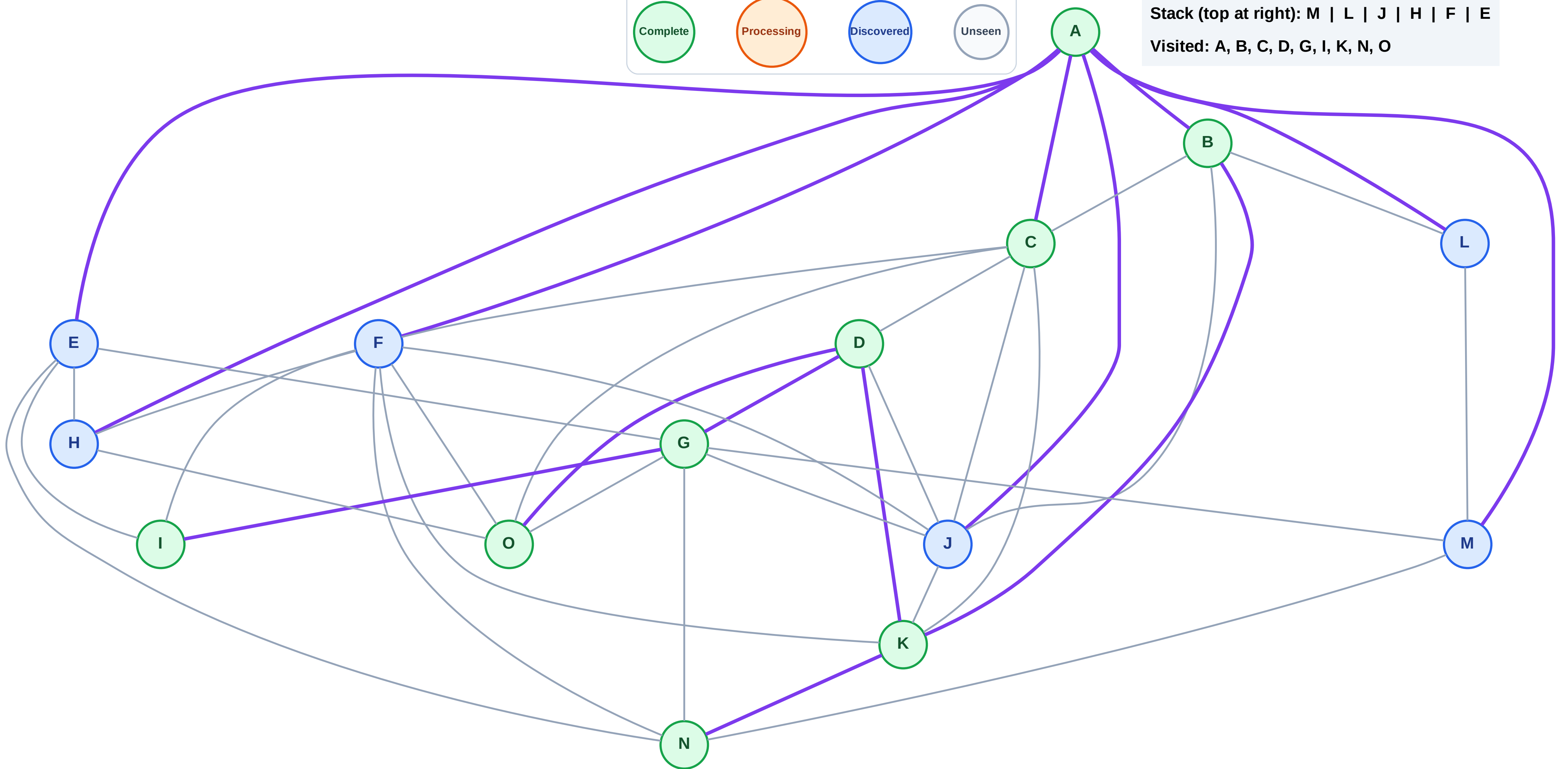
DFS Traversal

Step 19: No new neighbors from C

Legend



Stack (top at right): M | L | J | H | F | E
Visited: A, B, C, D, G, I, K, N, O



DFS Traversal

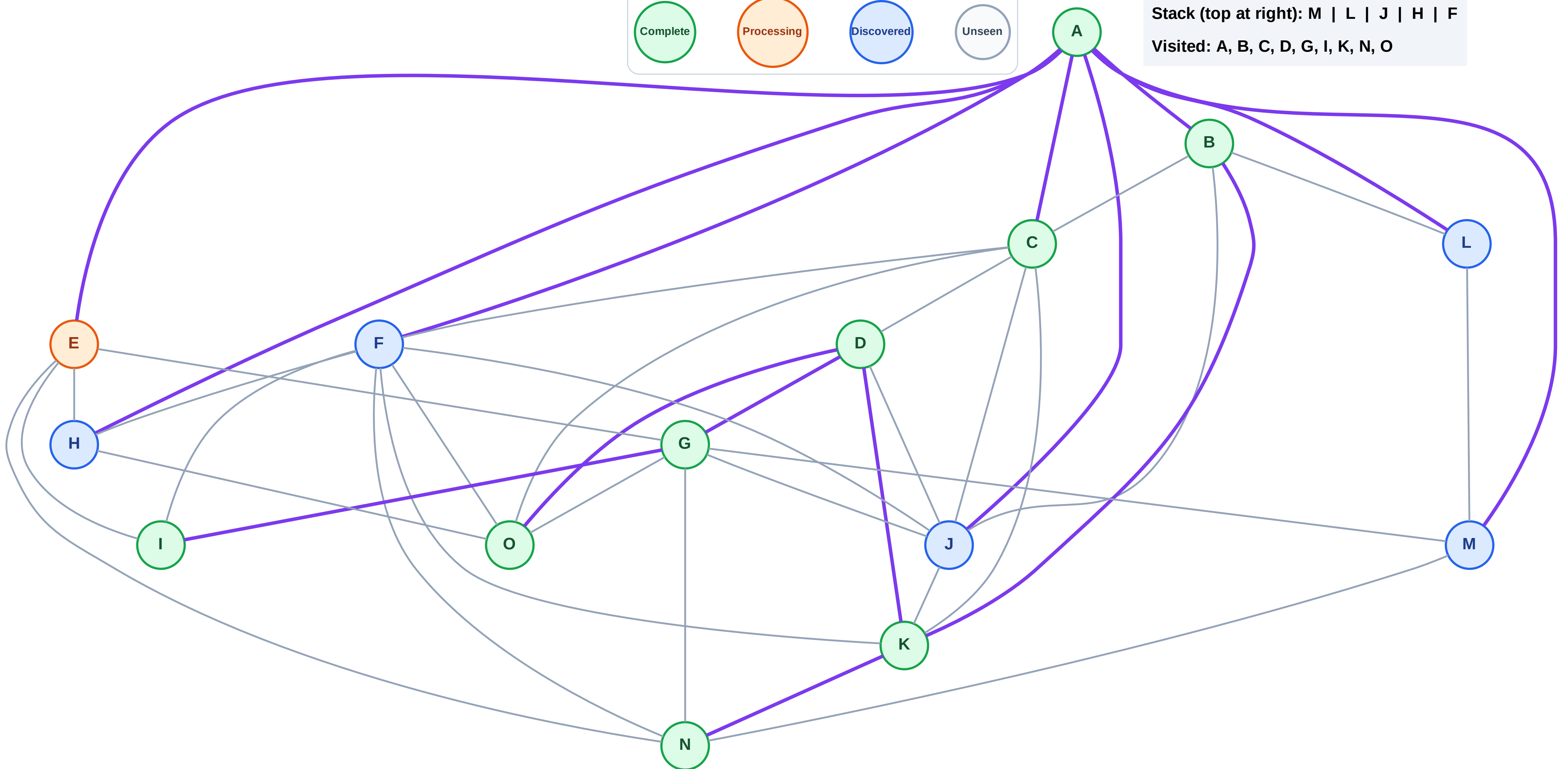
Step 20: Process node E

Legend



Stack (top at right): M | L | J | H | F

Visited: A, B, C, D, G, I, K, N, O

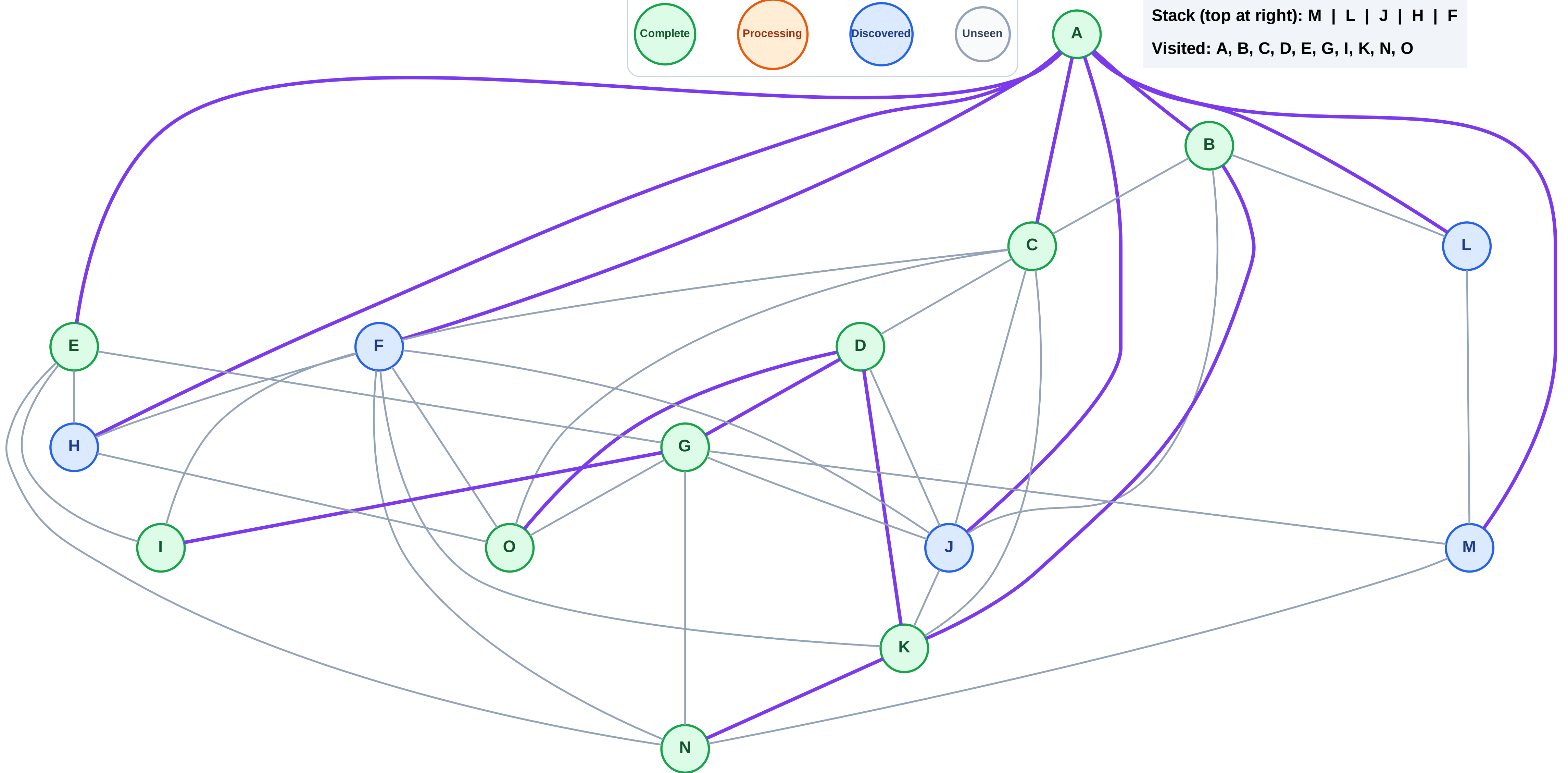


Legend

The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Stack (top at right): M | L | J | H | F

Visited: A, B, C, D, E, G, I, K, N, O



DFS Traversal

Step 22: Process node F

Legend

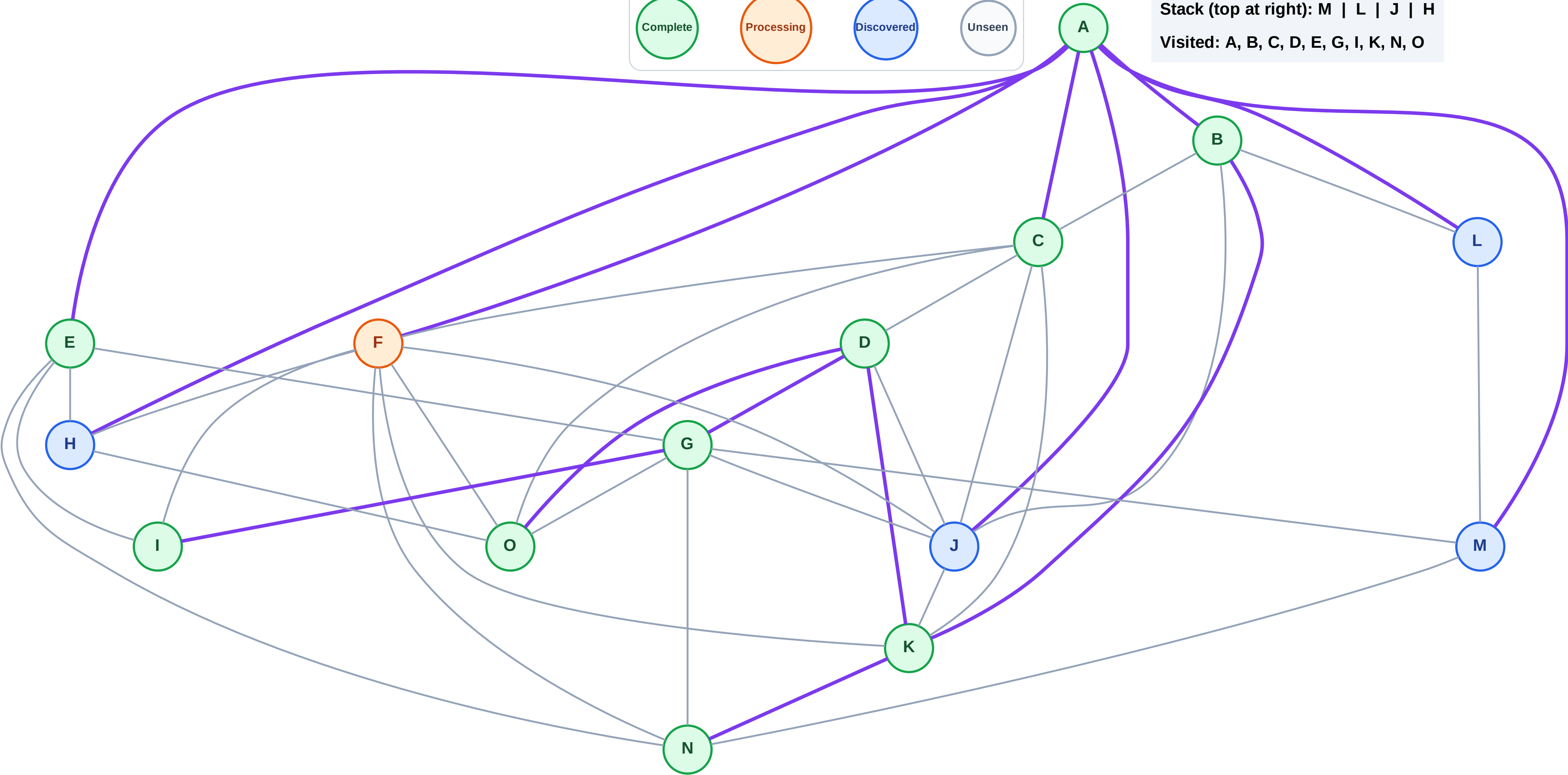
Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | J | H
Visited: A, B, C, D, E, G, I, K, N, O



DFS Traversal

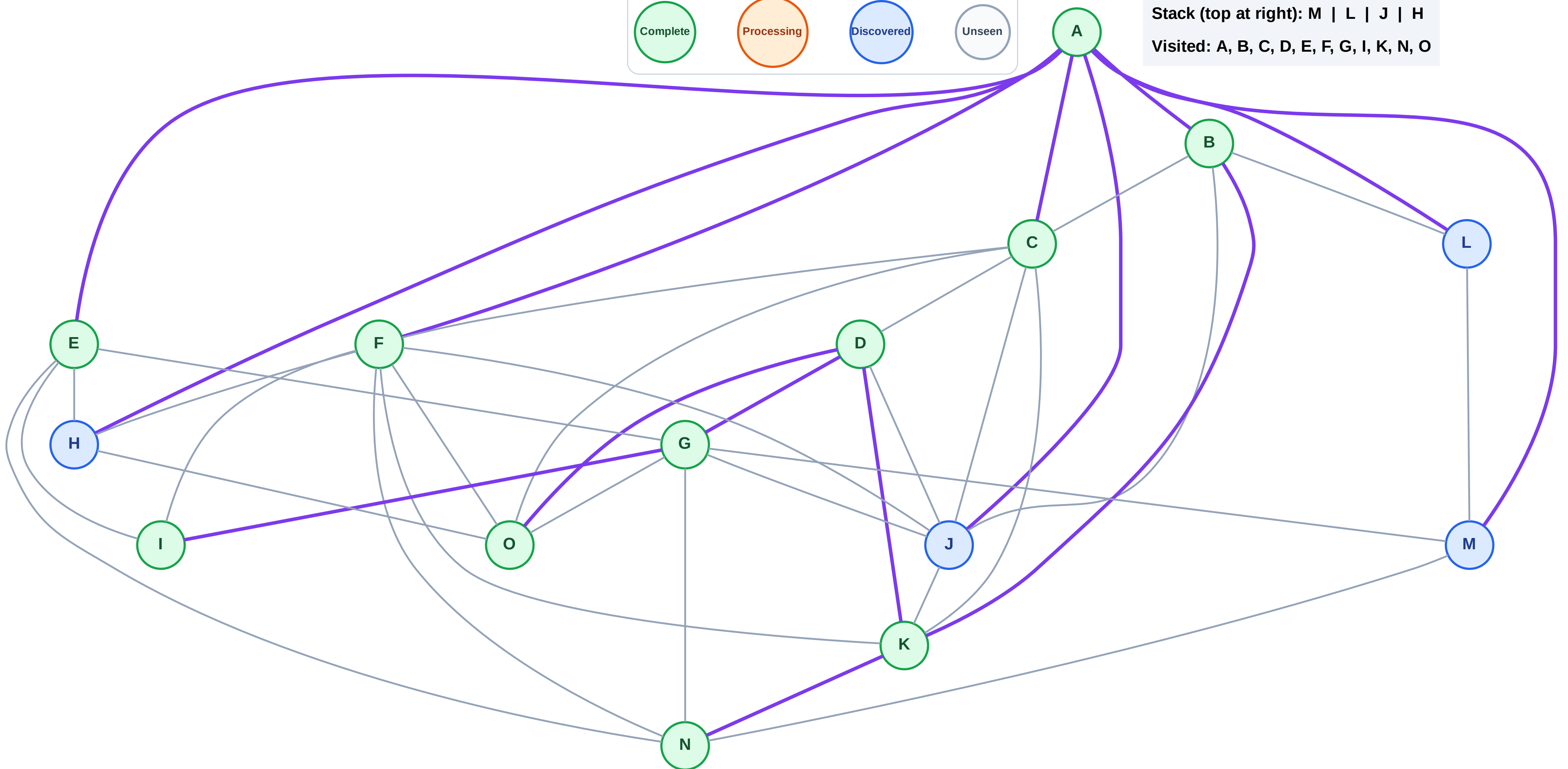
Step 23: No new neighbors from F

Legend



Stack (top at right): M | L | J | H

Visited: A, B, C, D, E, F, G, I, K, N, O



DFS Traversal

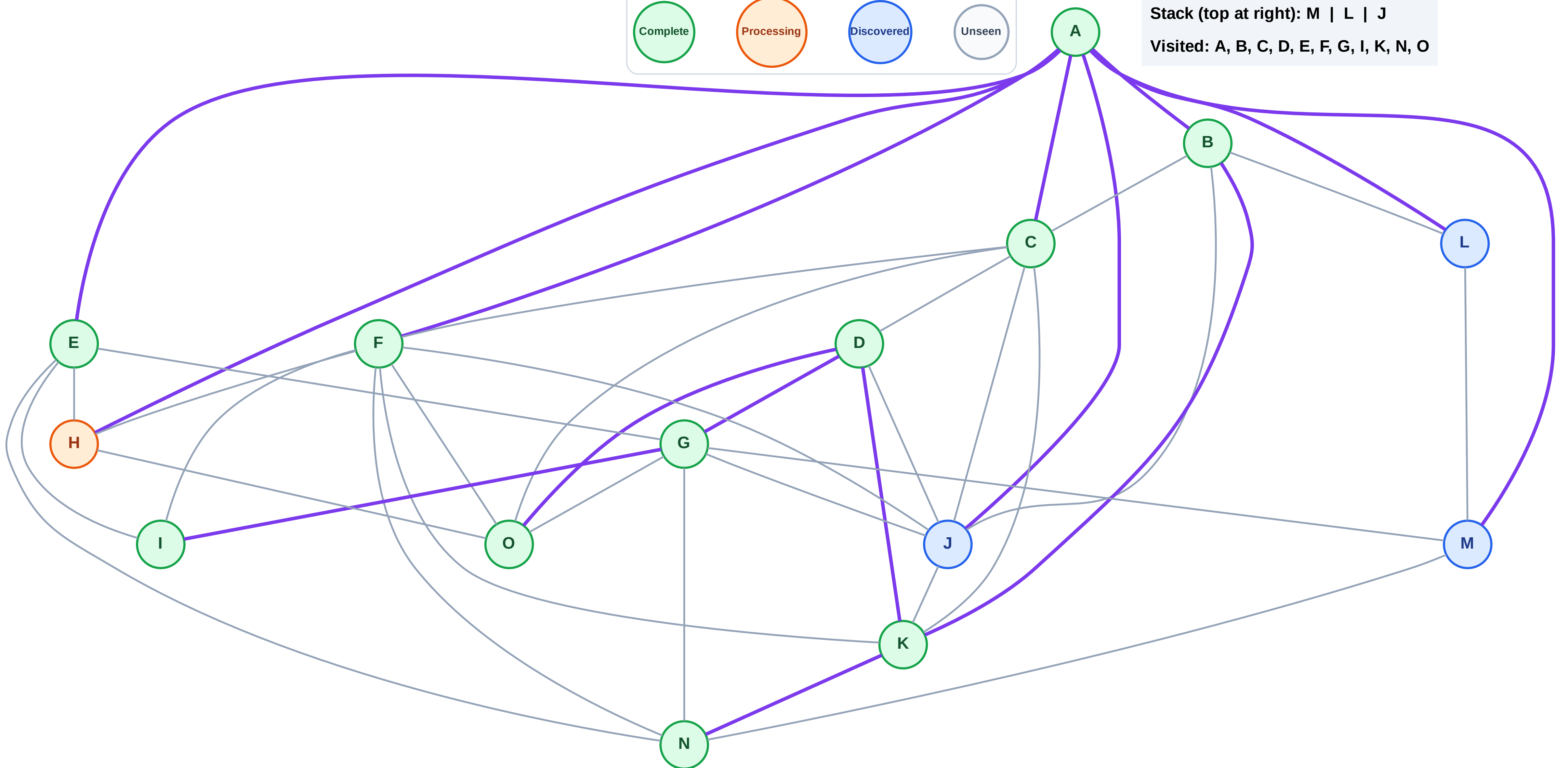
Step 24: Process node H

Legend



Stack (top at right): M | L | J

Visited: A, B, C, D, E, F, G, I, K, N, O



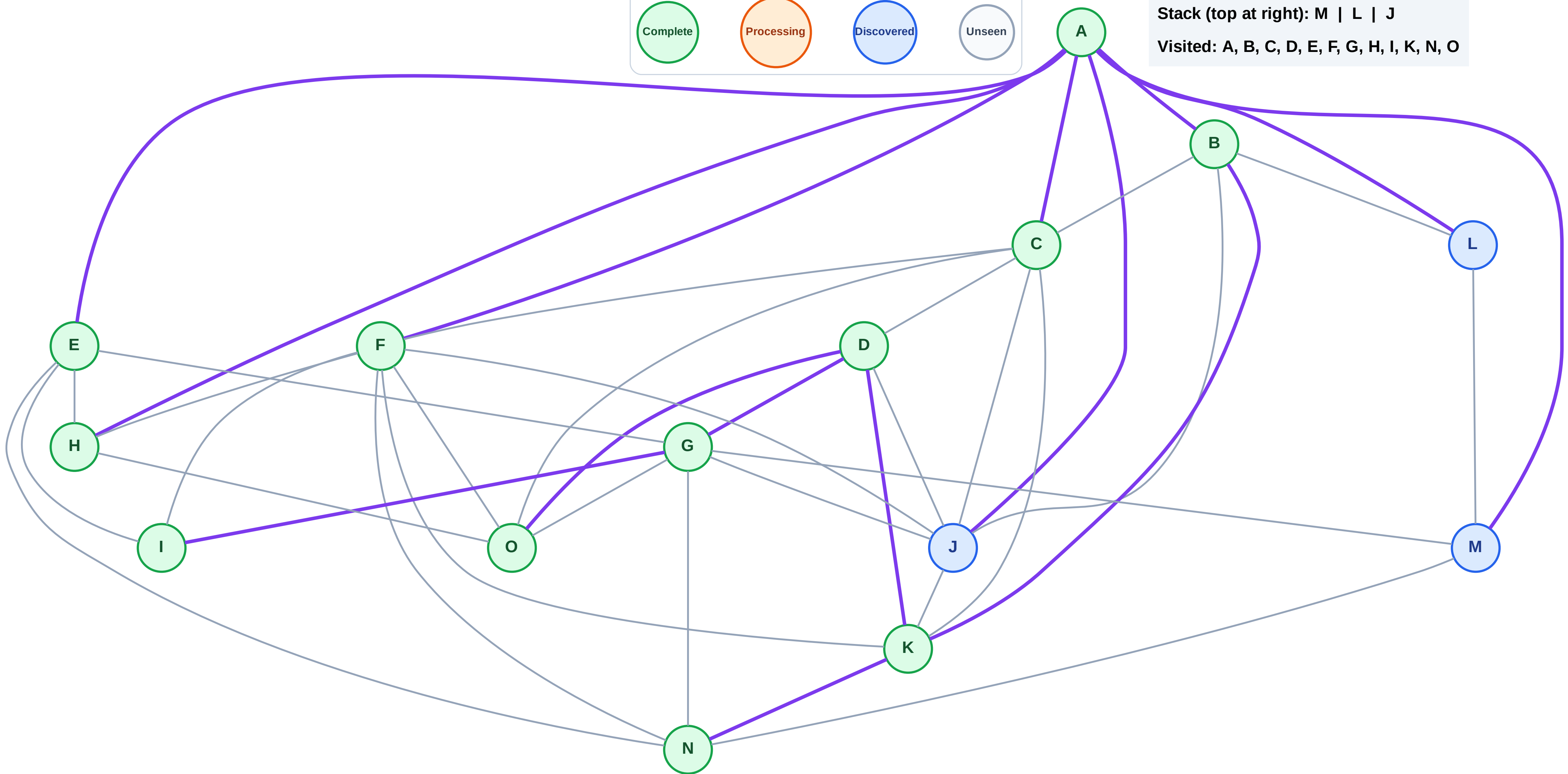
DFS Traversal

Step 25: No new neighbors from H

Legend



Stack (top at right): M | L | J
Visited: A, B, C, D, E, F, G, H, I, K, N, O



DFS Traversal

Step 26: Process node J

Legend

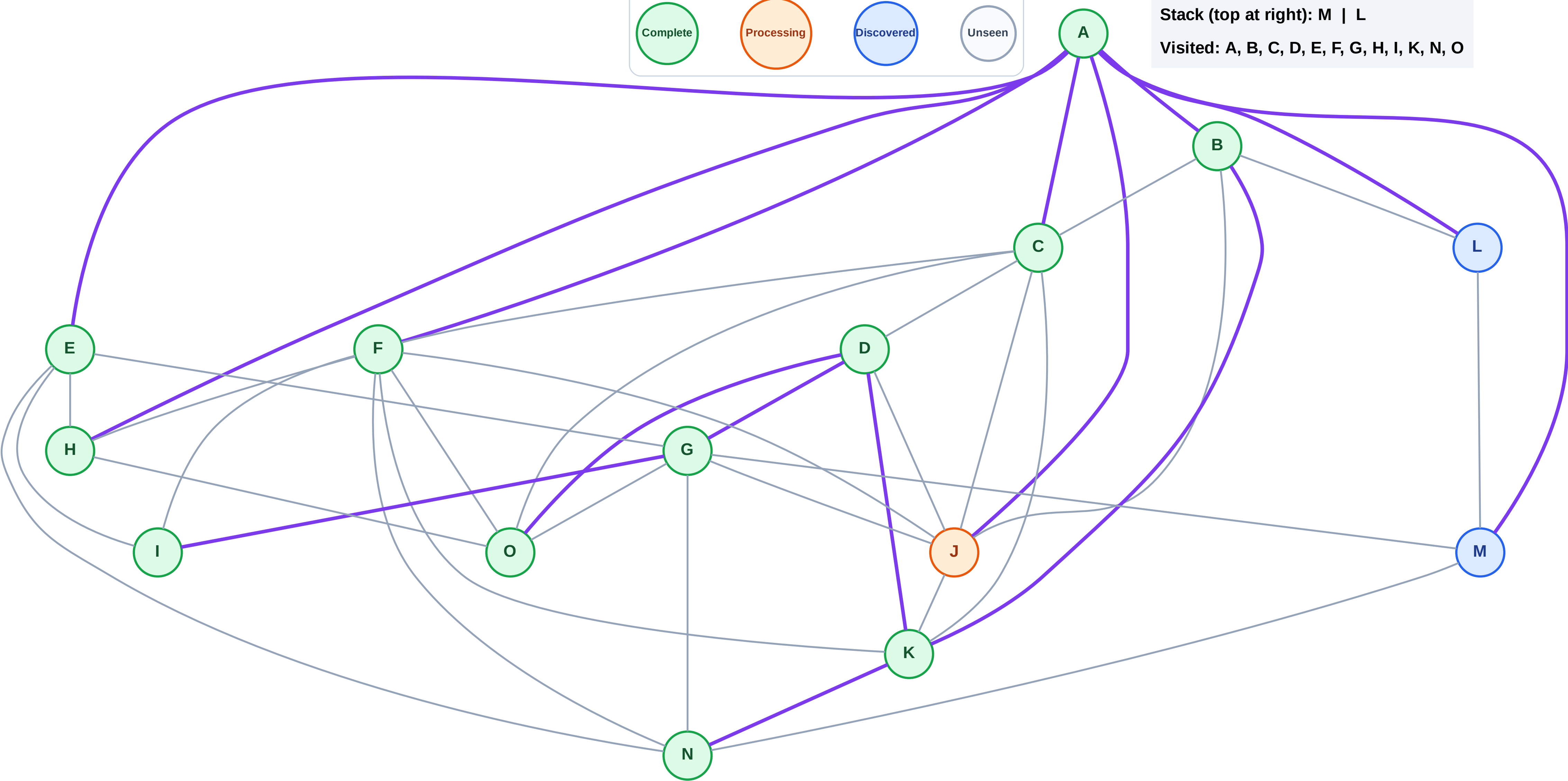
Complete

Processing

Discovered

Unseen

Stack (top at right): M | L
Visited: A, B, C, D, E, F, G, H, I, K, N, O



DFS Traversal

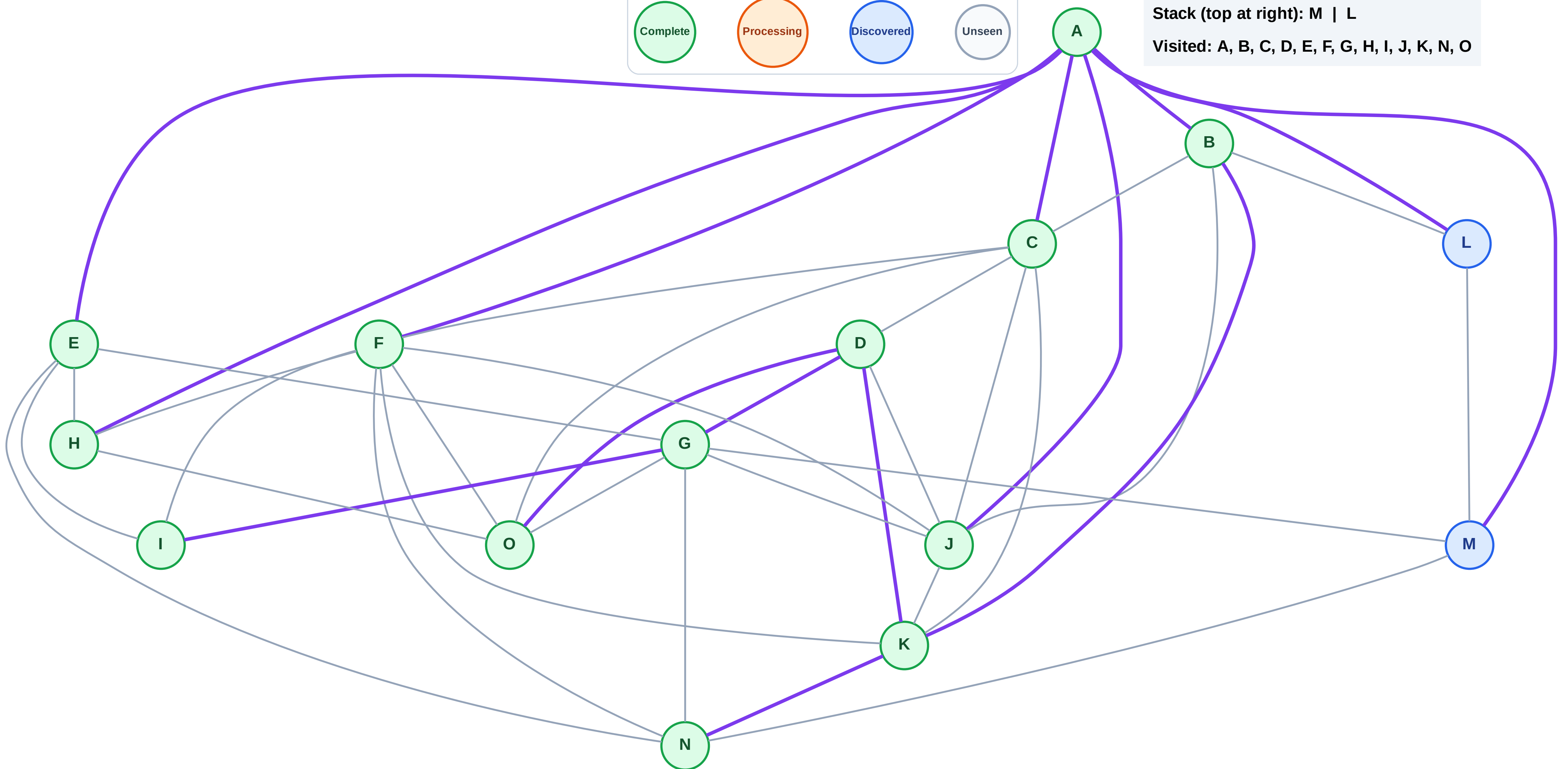
Step 27: No new neighbors from J

Legend



Stack (top at right): M | L

Visited: A, B, C, D, E, F, G, H, I, J, K, N, O



DFS Traversal

Step 28: Process node L

Legend

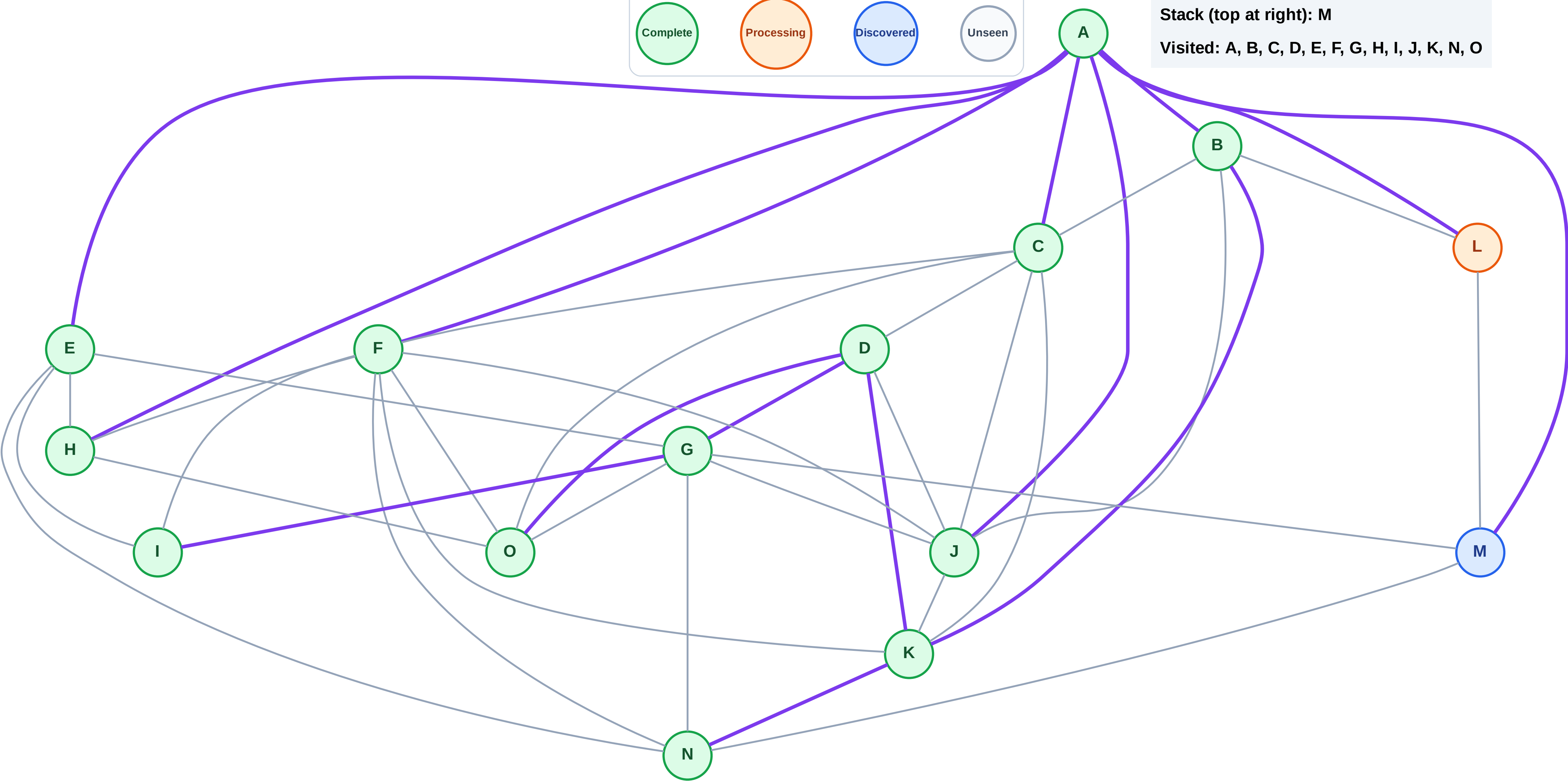
Complete

Processing

Discovered

Unseen

Stack (top at right): M
Visited: A, B, C, D, E, F, G, H, I, J, K, N, O



DFS Traversal

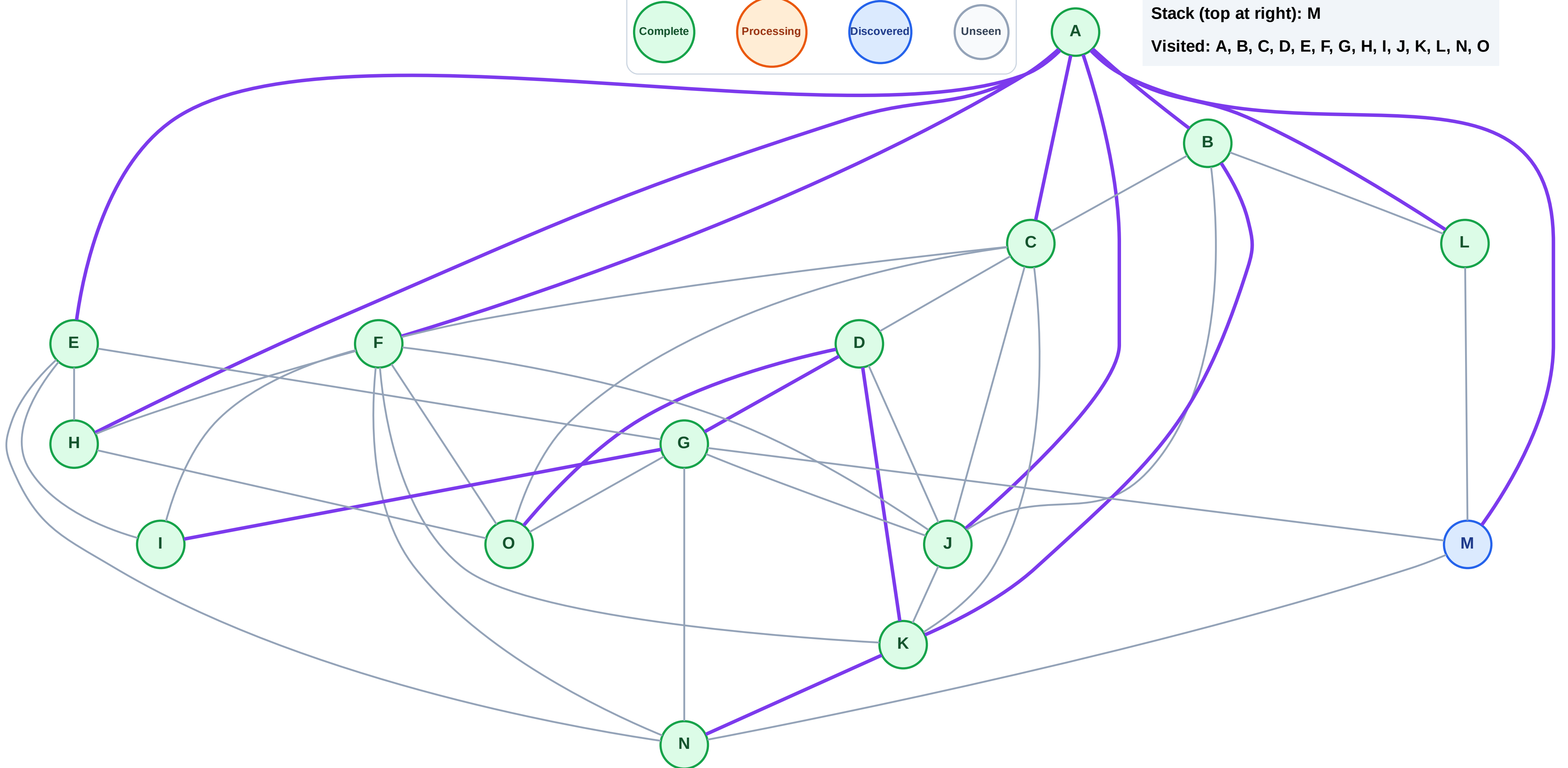
Step 29: No new neighbors from L

Legend



Stack (top at right): M

Visited: A, B, C, D, E, F, G, H, I, J, K, L, N, O



DFS Traversal

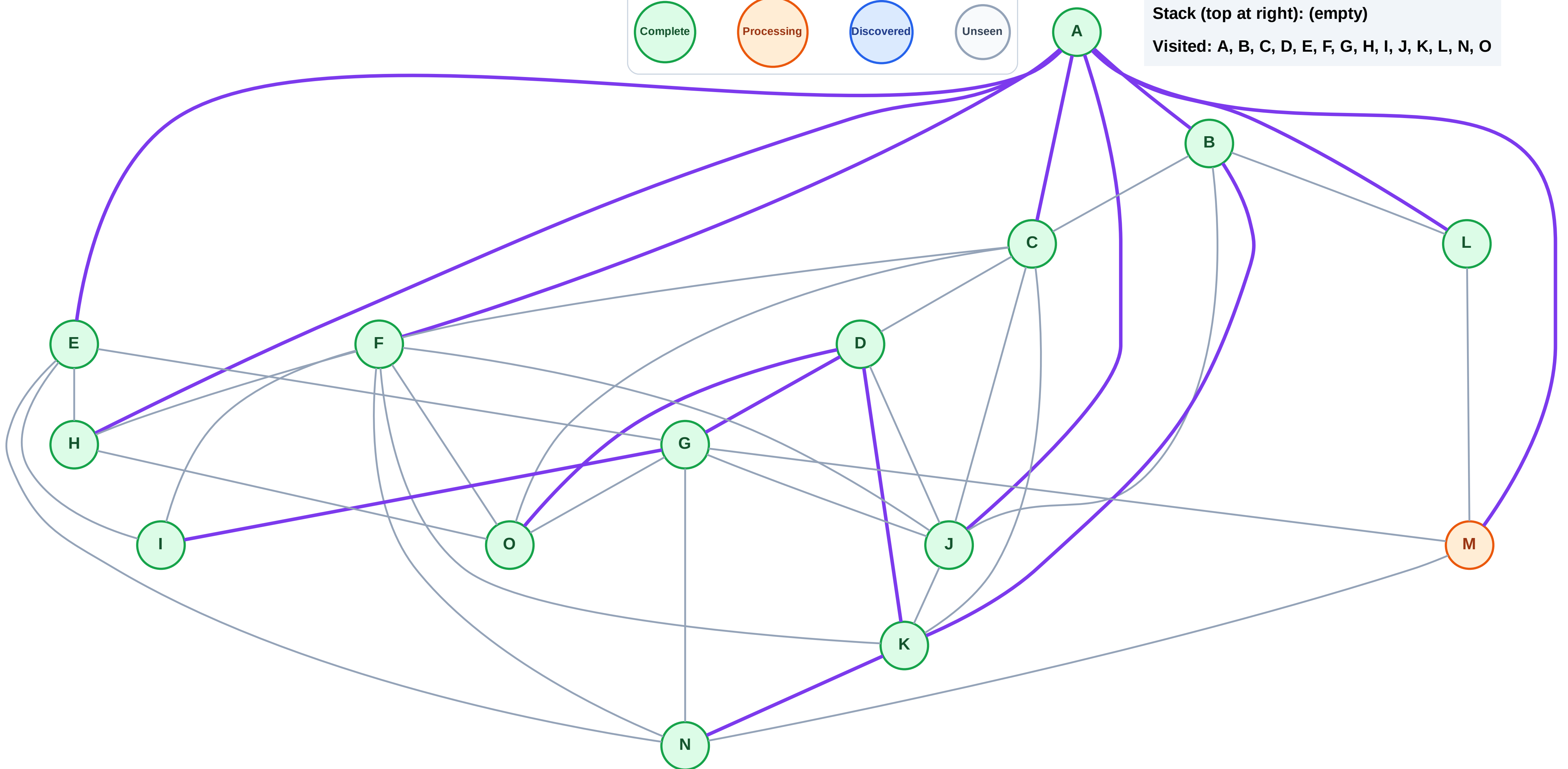
Step 30: Process node M

Legend



Stack (top at right): (empty)

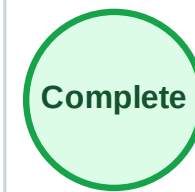
Visited: A, B, C, D, E, F, G, H, I, J, K, L, N, O



DFS Traversal

Step 31: No new neighbors from M

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

